

The World Economy over the Very Long Run

A History in Data and Debates



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Preface

This is a history of the world economy over roughly five thousand years — from the first riverine cities to the container ship — written from an unusual vantage point. Most accounts of the long-run world economy are told, implicitly, from the North Atlantic looking outward: Europe rises, the world responds. This book stands instead on the shore of the **Indian Ocean** and looks the other way. From there a different shape appears. For most of recorded history the centre of gravity of the world economy sat in **Asia** — visible in the direction precious metal flowed and in where the world's goods were actually made — and Europe's two centuries of dominance look less like the natural order of things than like a real, transformative, but **bounded** interruption.

That is a hypothesis, and the book's method is to **test it**, period by period, against the evidence — and to be honest about where the evidence runs thin or the scholars disagree. Hence the subtitle: *a history in data and debates*.

How the book works

Every chapter asks the same **four questions** of its period — about the *direction* of integration, the *channels* that carried it, the *modes* (goods, technology, people, capital) that did the integrating, and the changing *role of Europe*. Those four questions are the reader's toolkit; they are introduced in full in the Introduction.

Every chapter also has the same recurring furniture, designed so the book can be read as a narrative or mined as a course text:

- a **Data exhibit** — one empirical centrepiece, with its provenance and its caveats;
- a **Debate** — the live scholarly controversy, presented as genuinely open;
- running **callouts** that *follow the money* and ask *how we know*;
- a **source endnote after every paragraph** — pointing you to the evidence behind the claim and to where to *read more*, so the prose stays clean but nothing is unsourced;
- a short “**At the research frontier**” review of recent quantitative (cliometric) work on the period — the live edge of the scholarship, not just the classics;
- and end-of-chapter **questions for consideration** (3–5, essay/exam style), a reproducible **data exercise**, and **further reading** — with a bank of **cross-cutting questions** at the end of the book.

The narrative between that furniture is free to breathe — some periods leave abundant data, others almost none — but the scaffolding is constant, so you always know where you are.

Open access

This book is published open-access online and as a downloadable PDF. Its data exercises are **reproducible**: the figures and tables are generated from datasets you can download and re-run yourself. Where a number is disputed, the book says so and shows you the dispute, rather than picking a side without saying so.

Introduction: The vantage, the framework, the evidence

! Preliminary version — for trial in 2026/27 academic year. Content, figures and citations may still change.

A survey of the world economy, and a hypothesis about it.

The view from the centre

In the spring of 2026 the India Meteorological Department published its outlook for the southwest monsoon. NOAA had put the chance of an El Nino developing over May to July at 82 per cent; IMD put it higher and gave a 60 per cent chance that the season rainfall would come in deficient, below 90 per cent of the long-period average. That average was 87 centimetres, across four months that deliver over two thirds of all India's rain. By early September, newspapers were reporting that there was a monsoon crisis, because of El Nino, and that it was worsening. Media coverage drew a straight line from the original phenomenon, El Nino, through the weak monsoon and a poor harvest to the prospect of hunger and potentially famine. The relationship between El Nino and a weak monsoon is real and it is physical. It is also loose: a correlation of about minus 0.5 to minus 0.6, weaker since the 1980s. In 1997, the Pacific ran one of the strongest El Ninos ever recorded and the monsoon came in normal. A relationship can be genuinely causal, well understood in its mechanism, and still fail to predict any particular year.¹

The later steps — not just that a warm Pacific makes a dry season more likely but that a dry season makes a smaller harvest more likely and that in turn a smaller harvest makes hunger more likely — go from physical systems to human ones. Historically, and in the present day, the causal chain runs through granaries, roads, prices, armies and the rules about who has claims on the food produced and thus who may eat. A very long time series from Egypt highlights the importance of understanding how human and physical systems interact. From 622 CE, the height of the Nile flood was measured every year at Roda, because the flood set the land tax: a high flood meant a wide inundation, a large sown area and a full treasury. In measuring a tax base, thirteen centuries of annual hydrology is the by-product. The two lowest floods in the whole series, in 967 and 1200, both coincide with famines. This highlights the importance of physical systems. But the worst subsistence crisis in medieval Egyptian history was 1065–1072, during the Fatimid period, the *shidda al-uzma* or the great calamity. Across those years, the gauge reads only about half a standard deviation below normal. However, what followed was famine, civil war, the collapse of the military, and the breakdown of the system that moved grain from where it grew to where it was eaten.²

¹**Sources:** the seasonal-forecast structure follows the India Meteorological Department's long-period-average convention, under which a season at or below about 90 per cent of the long-period average is classed as deficient; the June-to-September monsoon supplies over two-thirds of India's annual rainfall. **Read more:** the module's Topic 0 slides carry the current year's figures, which change annually and are deliberately not printed here.

²**Sources:** the Roda Nilometer series, ed. Toussoun (1925), digitised by Koutsoyiannis et al. (2013); on the Fatimid crisis and its non-climatic character, the discussion in Chapter 3. **Read more:** Koutsoyiannis (2013), doi:10.1080/02626667.2013.804626.

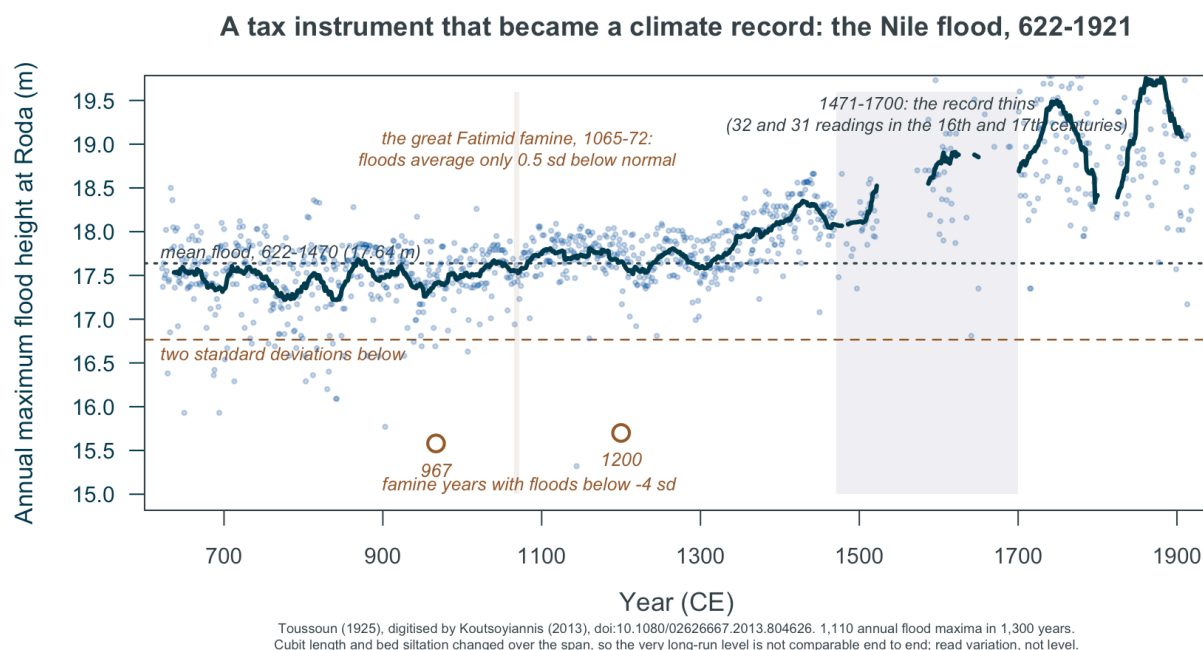


Figure 1. The Roda Nilometer at Cairo: annual maximum Nile flood height, 622–1921 CE. The two lowest floods in the record, in 967 and 1200, are both famine years, which is what a climate explanation predicts. The shaded band marks the great Fatimid famine of 1065–72, the largest subsistence crisis in medieval Egypt, through which the floods ran only about half a standard deviation below normal. Cubit length and the river bed both changed over the span, so read variation rather than level. Data: Toussoun (1925), digitised by Koutsoyiannis (2013).

Two systems, and the cost of forgetting either

That is the tension the whole book works inside. Most economics teaching presents the economy as a purely human system, with the physical world kept off the page for simplicity. Across almost all of the five and a half millennia covered here, that will not suffice. What a society could produce was set by how much land it could reach, how reliably the rain came, what the soil would bear and how many people the disease environment left standing. For the Bronze Age those four things are most of the economy itself. And it is important not to push too far that way, because the opposite error can do just as much damage. Geography and climate do not settle outcomes on their own, and a history that lets them do so has taken the people out of it: Egypt in 1065 had the same river it had in 1064. But physical shocks are still economic events and belong in the account: a volcanic eruption that lowers a Nile flood is a fiscal event, and a pandemic that removes a third of a labour force changes the supply of labour for a century. Both physical and human systems are running at every point in this book and it is important to not let one appear to do all the work.

By and large, geography can explain levels but, to explain timing, we need to have something that varies over time. As Chapter 1 shows, with temperatures largely constant and fluctuating only within a narrow band, aspects of the physical environment such as coastlines have been largely constant for the entire span covered, even if important exceptions exist. Mountain ranges, prevailing winds or natural harbours can, in other words, say why a place was persistently rich, or persistently isolated, or persistently expensive to reach. If these do not change from one century to the next, however, they cannot explain in full why something happened in one century rather than the next: a constant cannot explain a variable. Climate does vary, which is what makes it admissible where geography is not, and which is also why it has to be dated precisely enough. That is a demanding level and, at the current scientific frontier, it is one that is rarely passed. This highlights another tension, in

this instance between the healthy scepticism a well-trained social scientist should have about causal claims – correlation is not causation – and the inherent limits of what the available evidence can offer.

Standing on the Indian Ocean

Most long-run histories of the world economy are written from an Atlantic perspective, looking outward with the sometimes-accidental implication that it is Europe (and later European settlers in North America) that was the active party and the rest of the world a stage on which European energy plays out. Perhaps the simplest version of world economic history is just that: the world is poor and largely static until, some time between 1500 CE and 1750 CE, Europe stirs and modern prosperity radiates from a single source. In a history of the world economy over the last one or three centuries, this is the full story. For a history of the last millennium it is perhaps closer to the denouement.

Consider the world economy, however, over its full span. Imagine it not from London or New York but from a beach on the Malabar coast or the deck of a dhow running before the monsoon from Aden to Gujarat. For most of recorded history the largest and richest economies, the most commercially developed, were in Asia: Han and Song China, the Abbasid caliphate, Mughal India. The world's most coveted stores of value, silver and gold, typically flowed toward them rather than away. Europe spent long centuries as a periphery, exporting slaves, furs and raw metal, importing manufactures and ideas, and paying for what it wanted in bullion it could barely produce itself. The period in which Europe genuinely ran the world economy is, in truth, short: concentrated in the two centuries after 1750, and even at its height it rested on the Asian periphery it governed.

The spine: the claim we are testing

The goal of this book is to interrogate the following argument:

Viewed from the Indian Ocean, the world economy spent most of its history centred on Asia — visible in the direction bullion flowed and in where the world's goods were made. Integration advanced and retreated in waves shaped by climate, disease, technology and money, along durable maritime channels and periodically resurgent overland ones. Europe's dominance was real and transformative but bounded — concentrated in two centuries after 1750, built partly on the Asian centrality it overturned, and resting at its height on the Indian Ocean periphery it governed. Its timing and depth remain debated, and the East's aggregate return — measurable but partial, achieved within a Western-built order — can be seen as the long arc closing.

This spine treats European dominance as an interruption or anomaly, rather than a destination or punchline. It is one bounded by four qualifications:

1. The interruption was consequential, stretching across two transformative centuries;
2. Its timing and depth are genuinely open, and the debate is reviewed in Chapter 7;
3. The return east has been partial, being aggregate weight rather than a restoration of per-capita parity; and
4. It is system-dependent, since the Asian re-emergence happened inside a US-anchored order.

The framework

What makes a place productive

The book relies throughout on one core economic framework as an organising heuristic: the production function. It is used as a scaffold, rather than a model to be solved or estimated. It serves a role each topic, bringing us back to which of the four inputs were doing the work of integration (or disintegration). At its core, what a society can produce depends on four things: its useful knowledge, its physical capital, its people, and its land:

$$Y = f(A, K, L, N)$$

A is technology and useful knowledge, of which machines are the smallest part. A is also technique, crop and animal, navigation, and the ability to write a contract that a stranger will honour. Most of what raised output before the nineteenth century arrived in that form: a wheat variety that ripened a fortnight earlier, a heavier plough and the draught animal that could pull it, a lateen sail that let a hull work into the wind. Seeds and animals are technologies in the strict sense, because they let a society get more out of the same land, and land is the term that was fixed. Institutions live here too, as does trade: a country short of land can import land-intensive goods; a country short of silver can monetise somebody else's. Like any other technology, voluntary exchange raises what a given endowment will yield.

K is physical capital: ships, warehouses, irrigation works, tools, pack animals, stock in hand. Capital is non-consumed output: something produced and set aside to raise what could be produced later. A merchant ship built with a year of carpenters' work is an embodiment of choosing to defer consumption today for future consumption. We distinguish here between (real) capital assets, credit and money. Credit is a relationship, in which one party holds a claim, another carries the liability, and purchasing power has moved across time; we know that Mesopotamia had interest-bearing debt by about 3000 BCE, two and a half millennia before the first known coins. Money, on the other hand, is a function. It tells us about what an object (or a claim) *does*: quote a price, settle a debt, hold value between now and later. Silver by weight does it, and so do cowries, coins, banknotes and a bank deposit. Capital is distinct from both credit and money: it is a thing somebody built and can now use. Whether a ship was paid for with borrowed or saved silver is irrelevant to its impact. The categories overlap — cattle are capital and a store of wealth — but it is worth distinguishing between real and monetary aspects. Given the central role of precious metals, it is helpful to consider how they relate to all three concepts. Silver in the ore body is N, while once refined and cast into an ingot, it is output, Y, and wealth. Promised across time it sits inside a credit relationship. Used to quote a price or settle an obligation it is performing a monetary function. Refined silver was always an economic good; in a great many societies it was also, at the same time, doing monetary work. Coinage standardised a monetary object and certified its content, but it did not invent money. Silver remains perhaps the best single thing to follow across five and a half thousand years, as something that sheds light on the features of the world economy.

L is human capital, labour and people. As described below, population growth (growth in L) is a core measure of success, displaced only in the last topics. But it is also central to understanding changes in integration: voluntary migration, slavery and indenture are ways in which the ratio of L to other endowments changes. It also means human capital: the skills, training and literacy carried in the people themselves. Moving beyond simple headcounts, two workforces of the same size are not the same if one can navigate by the stars, keep a double-entry account, cast bronze to a standard or read a contract. Skill is the part of L that behaves most like A.

N is natural capital: land, energy and resources — the stock of productive assets a society did not build and cannot make more of. In many economics courses, N is folded into K, usually motivated by model tractability (or potentially because land is a small share of costs in some modern settings). Over the full sweep of human history, however, land is fundamental and central. Energy came from land, as food for people and animals and as wood for heat; materials came from land; and the binding constraint on almost every pre-modern economy was how much land it could reach. Putting N as equal to A, K and L is how climate, energy and disease come back into the account as inputs rather than as background. Foregrounding N allows us to highlight also what happens when a resource is *concentrated* in such a way that its holder can earn a surplus. Those surpluses then become subjects of conflict: geology often sits underneath sources of power. In the modern period, this is visible in oilfields and the coalfields before them. But it can be seen across every topic: from Banda nutmeg and Potosi silver to Baltic timber and amber, and in geographic bottlenecks such as the Straits of Malacca and Suez. Much economic history can be viewed as a fight over N, with the armed trading companies of Chapter 6 the clearest case: they are organisations built to hold chokepoints and secure monopoly access to sources of economic rents.³

At the end of every chapter, there is a review of the elements of the production function: which of Y, A, K or L is leading integration or disintegration? And which of A, K, L and N is a binding constraint? The two lists differ because they answer different questions. Integration is carried by what can cross a distance, and N cannot: a coalfield or a monsoon stays where it is, so it reaches another economy only embodied in the goods it produced. That is why output belongs on the first list and land on the second, and it is the same logic Heckscher and Ohlin built on, in which the traded good rather than the endowment is what moves. N is the fixed factor here, in the role labour plays in the simplest growth models. We can highlight here two distinct phases. With N roughly fixed and A improving slowly for most of human history, any gain in output was absorbed by more people rather than by richer people. This is, in effect, Malthus restated. Almost every flourishing economy in the first six chapters flourishes by getting bigger — by increasing its ratio of L to N — not by getting richer in per-person terms. The second is the escape: as useful knowledge becomes cumulative and fossil-fuelled, A surges. The N constraint relaxes at the same time, partly because coal substitutes for land as a source of energy and partly because colonial acreage substitutes for domestic acreage as a source of food and fibre. This is when A grew too fast for L to keep up and, instead, output per person started to rise well above subsistence levels.⁴

The scale of that change, and the limits of what we know about it, are both visible in the growth record. Over the eighteen centuries to 1820, world population multiplied by about 4.6 and income per head by perhaps 1.4; over the two centuries that followed, population multiplied by 7.3 and income per head by 14.8. Measured per generation, population growth after 1820 ran roughly thirteen times faster than before it and income growth at least eighty times faster, and the gap between those two multiples is the escape stated arithmetically. The caveat belongs in the same breath, however, because the income figure is far weaker than the population one. There is no century-by-century estimate of world income per head before 1500, and the pre-1820 rate is a single implied number stretched across the whole span. It is also in large part an assumption, since Maddison assigned a subsistence floor wherever nothing was known. Gregory Clark's review makes the point sharply: had that floor been set at seven hundred dollars rather than four, there would be no measured growth at all between 1 CE and 1820. What can be defended, then, is that income per head grew by at most about forty per cent over eighteen centuries and possibly not at all, which makes the eighty-fold comparison a floor rather than an estimate.⁵

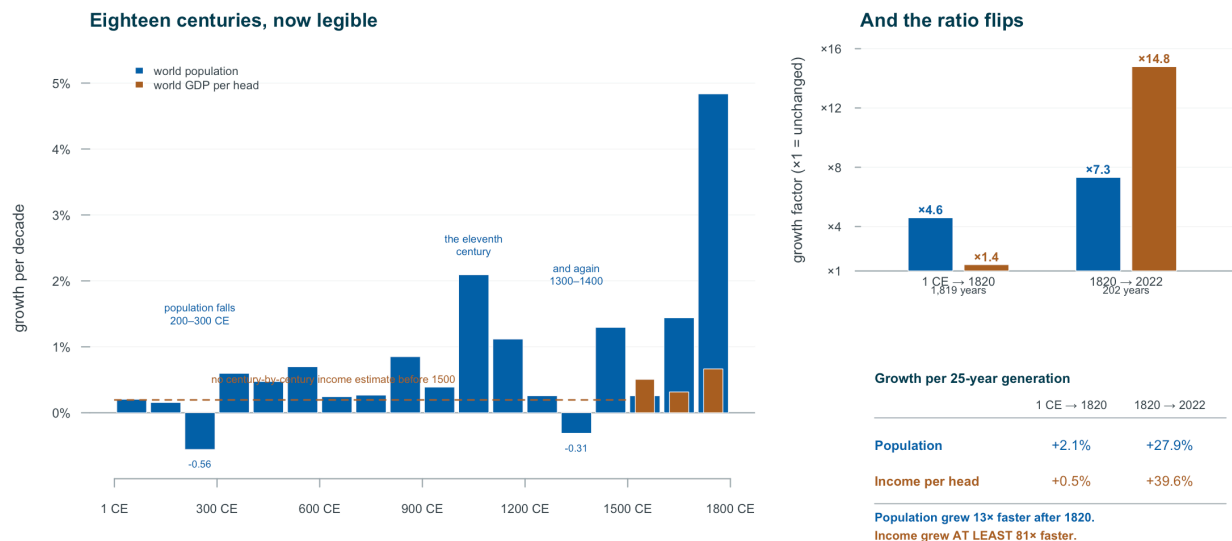
³**Sources:** the framework follows the module's own governing-framework document; on chokepoints and armed trade, Chapter 6. **Read more:** Findlay and O'Rourke, *Power and Plenty* (2007), for the same argument made at book length.

⁴**Sources:** Wrigley, *Energy and the English Industrial Revolution* (2010), on the organic-economy constraint; Pomeranz, *The Great Divergence* (2000), on ghost acreage. **Read more:** Chapter 7, where the framework is why the chapter can treat coal and colonies as two versions of the same relief rather than as two unrelated advantages.

⁵**Sources:** population from HYDE 3.3, doi:10.24416/UU01-AEZZIT; income per head from Maddison (2010) world

After 1820 population grew 13 times faster — and income at least 80 times faster

For eighteen centuries population crept up and income per head barely moved. Left: growth per decade, by century, to 1800. Right: each era in total. It took eighteen centuries to multiply world population by 4.6, and 200 years to multiply it by 7.3 again.



Fourteen of the twenty centuries have no world income estimate at all: the dashed line is the 1 CE–1820 average, not a measurement of any century. So the 80× is a FLOOR, not an estimate: the 13× for population rests on real counts, but a smaller assumed pre-1820 income growth makes 80 larger without limit. Clark: “had Maddison assumed subsistence was \$700, there would be no growth from 1 AD to 1820” — in which case the income multiple is unbounded. Population: HYDE 3.3. Income per head: Maddison (2010) and MPD2023, growth factors chained rather than levels converted.

Figure 2. World population and world income per head, by century to 1800 and by era. For eighteen centuries population crept up and income per head barely moved; after 1820 both accelerate, and the ratio between them reverses. The dashed brown line is not a measurement of any century: fourteen of the twenty centuries have no world income estimate at all, so it is the single implied 1 CE–1820 rate drawn across the whole span. Population: HYDE 3.3. Income per head: Maddison (2010) and the Maddison Project Database 2023, growth factors chained rather than levels converted.

That Malthusian regularity is strong rather than universal, and the exceptions are where the interest lies. The best-documented is Roman Egypt, where the wage, price and rent record has been rebuilt from the surviving papyri. Measured in hectolitres of wheat a year — a unit that no currency debasement can get into — the real wage rises from about ten and a half around 75 CE to about fourteen by the middle of the second century, and then falls back to about ten after the Antonine plague. That is intensive growth, in a pre-modern economy, sustained across roughly a century. It also shows what it takes to see such a thing at all: fifty-two observations across two hundred and fifty years, arriving on daily, monthly and annual bases, from the one province that combined a paper habit with a dry climate. The conclusion to draw is not that the Malthusian account fails, but that it is a strong regularity rather than a law, and that the evidence needed to test it survives almost nowhere.⁶

If the production function says what a society could produce, demand decides whether it does and control of N helps determine who keeps the gains. Capacity is the supply side, and it is what the production function describes. Demand is whether that capacity is used at all, which is not automatic: European households had to acquire a taste for sweetened tea before Asian and Atlantic production

benchmarks and the Maddison Project Database 2023, Bolt and van Zanden (2024), *Journal of Economic Surveys* 39(2), 631–671, doi:10.1111/joes.12618, with growth factors chained rather than levels converted; Clark’s review of Maddison’s subsistence floor, and the observation that a higher floor removes the pre-1820 growth entirely. **Read more:** Chapter 7.

⁶**Sources:** Harper, “People, plagues, and prices in the Roman world: the evidence from Egypt”, *Journal of Economic History* 76 (2016), 803–839, doi:10.1017/s0022050716000826, and the underlying *Database of Prices, Wages, and Rents for Roman Egypt*, DARMC Scholarly Data Series 2016–5, Harvard University. Used with Kyle Harper’s permission, granted 11 September 2026 on condition that both the article and the repository are cited. **Read more:** Chapter 2, which takes the case up and follows it past 165 CE.

had a market to grow into, and Chinese demand for silver as a tax medium is why American mines mattered to a country that never traded with the Americas. Power is who captures the output once it exists, which is where the bottleneck-holder, the armed company and the colonial state come in.

What we mean by a world economy

We should say what the object is before spending eleven chapters locating its centre, and this is harder than it sounds, because most books in this field use the term and never define it. The definition this book works to is deliberately modest. A world economy is a set of regions whose production and prices respond to one another across distances that none of them controls. The test is responsiveness at a distance, and the crucial word is *respond*: what happens in one place has to alter what is produced, or what things cost, somewhere far enough away that no single authority governs both ends. That is a weaker requirement than a world market, which would need something close to a single price, and a different one from an empire, which is a political object. It is also a stronger requirement than contact, since a diplomatic gift or a shipwrecked sailor establishes contact without establishing anything economic.

The object of interest, then, is the connected part of world output: goods produced for exchange, and above all for exchange over distances long enough that neither end could dictate the terms. Before about 1800 that connected fraction has to be read through population, for reasons set out below, and read with an eye on connectivity rather than head-count alone. Responsiveness itself comes in degrees, and the section on connection below sets that range out. The definition is deliberately weak on institutions, because they change out of all recognition across five and a half thousand years, from temple redistribution to chartered companies to container shipping, and a definition anchored to any of them would exclude most of the period. Responsiveness is the one thing that can in principle be measured at every point on the arc, even where the instrument changes completely, and it sets a bar each period has to clear rather than assume. That is why Chapter 1's verdict on the centre of gravity is *not applicable* rather than *no*: in the third millennium BCE the question does not yet have a subject.

Where the centre of gravity sat, and how sure we are

The spine is a claim about where the weight of the world economy sat, and a claim like that is only as good as its definition of weight. Weight here is output: the centre of gravity is the point world economic activity balances around, and the quantity being balanced is production, with territory, armies and head-counts entering only in so far as they move it. But not all output counts equally. What the spine is about is the connected economy, meaning output that trade enabled, or that was produced for trade, and above all for long-distance trade. A subsistence farmer's grain is output, and in aggregate it dwarfs everything else in every chapter before the nineteenth century, but it never moved and it cannot move a centre of gravity. So the weighting favours the goods that travelled, the workshops that made them and the ports that handled them, which is why a chapter can call Rome the largest single economy of its age and still put the centre of gravity east of it. The two statements measure different things and both are true.

That choice needs defending, because there is a good objection to it. The connected economy was small, a few per cent of the output of the economies doing it in the periods this book opens with, so weighting it and disregarding the subsistence bulk locates the centre wherever the tradeable goods were made, which was largely Asia. On that reading Asia cannot lose. The reply is that the two measures answer different questions. Total output answers where most production was happening, which before 1800 is very nearly where most people lived, and it says nothing about who could set a price. Connected output answers whose goods other people reorganised themselves to obtain, and

on whose terms, which has testable consequences: it predicts where metal accumulated, observable in hoards and ice cores, and it predicts who could refuse a deal. Chapter 2 catches the second one happening, when Indian merchants stopped accepting Roman silver at the moment its content fell. Two honesties follow. For much of the arc the two measures agree, so the choice does less work than the objection implies, and where they come apart the book says which one it is using. The measure also treats the ability to set terms as the thing worth measuring, which is a position, and a reader who prefers aggregate production will read the same evidence and reach a different verdict.

Before about 1800, output is read through population, and the circularity has to be said out loud. In a Malthusian world more food meant more people rather than richer people, so a region's share of world output tracked its share of world population fairly closely, which makes population a defensible proxy. But most historical output estimates for the pre-modern world were themselves built from population figures and assumed income levels, so when the two series agree before 1800, part of that agreement is a fact about the world and part of it is an artefact of how the numbers were made. Bullion, likewise, indicates without defining. The direction metal flowed is this book's running tracer, because metal moved toward the workshop, but what a persistent flow reveals directly is where the metal was relatively more valuable and how exchange between the two ends was being settled. It is evidence about monetary demand and about trade imbalances before it is evidence about size.

Every verdict is scored, on a four-point scale plus a null. *High* means several instruments with different biases give one answer. *Moderate* means the reading is supported, but one instrument is carrying it. *Low* means it is a reasonable reading and a different one would not be unreasonable. *Not applicable* means the question has no meaning in this period yet, which is an honest answer rather than a dodge, and is the entry for Chapter 1. Direction and magnitude are scored separately, because they are separate claims and the evidence for them is usually of different quality. Chapter 2 is the standing example: that Roman metal moved east is about as secure as anything in ancient economic history, and how much of it moved is contested by two orders of magnitude. An answer that gives one confidence level for both has run two claims together.

Applied across all ten verdicts, that scale turns up an asymmetry. Of the nine periods where the question has a subject, seven score high on direction and five score low on scale, so we can usually say which way the weight sat and which way it was moving, and rarely say by how much. That is a property of the evidence rather than of this book: the instruments that establish direction, such as the way metal flowed or what a place exported against what it imported, are robust to being wrong about quantities by a factor of two. It is why Appendix C scores the two separately, and it means a reader who wants numbers will be more often disappointed than one who wants an argument. Two things follow for how the verdicts are made. For nine chapters the centre has to be inferred, from city sizes, trade composition, the direction of bullion and shares of world manufacturing; only in Chapter 10 can it also be computed, using Danny Quah's distance-weighted average location of world output, which weights all output equally where this book weights connected output and so answers a slightly different question. And there is one switch, announced in advance: Chapters 1 to 6 score aggregate weight, and from Chapter 7 the measure that matters becomes income per head. That switch *is* the Great Divergence, so the early and late verdicts are not marks on the same ruler, and Appendix C keeps them in one table precisely so that the seam is visible.

The bullion tracer, and why it flips

Before modern capital markets, metal moved toward the workshop. People who made things the rest of the world wanted, Indian cotton and Chinese silk and porcelain, were paid in silver and gold, which pooled where production was most valued. So in antiquity Roman silver drained east to India (Chapter 2), and in the early-modern centuries New World silver flowed east to China (Chapter 5).

The tracer even works in reverse where it should: the Abbasid core paid silver north to its slave-and-fur periphery (Chapter 3), mirroring Rome. That reversal matters, because a tracer that only ever points one way is a rule rather than an instrument. One complication arrives before the pattern does, because the earliest traceable silver flow in this book runs the wrong way. In the third and second millennia BCE the metal moves west to east into Mesopotamia rather than east toward a workshop, since at that date the workshop in the later sense does not exist and silver is moving toward demand rather than toward production. That is a reason to state what the tracer assumes rather than to abandon it: it works when one region is making things another region wants badly enough to pay in metal for, and before that condition holds, metal moves for other reasons entirely.

After the nineteenth century the tracer flips, and what flips is what the metal is doing. Under the classical gold standard, gold is not merely accepted by weight at the far end of a voyage; it anchors a settlement system that reaches across borders, and claims denominated in it circulate on the credit of institutions rather than on assay. Follow it then and it leads to the financial core, London and afterwards New York, rather than to the workshop. Note who underwrote that core: India's colonial drain helped finance Britain's payments deficits (Chapter 8), sourced in the periphery and intermediated by the centre.

Most of the economy was never priced

For almost all of the period this book covers, most of what was produced was never bought or sold. Output sits in three layers. The bottom and by far the largest is subsistence and obligation: what a household grew and ate, plus what it handed over as rent, tribute, tithe or labour dues. That second category has no clean modern counterpart, and it matters, because a peasant handing over a third of the harvest is transferring real output without any price being formed anywhere. Above that sits regional exchange, in local markets, at prices that were real but seldom recorded. Above that, thinnest of all, sits long-distance trade. Our evidence comes overwhelmingly from the top layer, because that is the layer that generated records: customs registers, merchant letters, tax farms, port dues. So when a source gives us a number, the first question is always what the number is a number of.

Confuse the layers and it goes badly. Chapter 2 has to ask how large the Roman eastern trade was, and the published estimates of the annual value of Rome's Indian Ocean imports run from about 920 million sesterces to about 17,640 million, where the high end is roughly the size of the most-cited estimate of the entire Roman economy. Nobody involved is careless: the estimates differ because each takes the tax assessment of one documented cargo and multiplies it by an assumed number of ships, and the assumptions differ. But notice what has happened. A number drawn entirely from the thin top layer, the customs record, is being compared with a number for all three layers at once, and the comparison only works if we know what fraction of Roman output ever passed through a market at all. It did not, and we do not.

The four questions, and where they meet the framework

The book asks the same four questions of every period. They organise every chapter, and they are the method the book works with.

! The four motivating questions

- **Q1 — Direction.** Is the system integrating or disintegrating? Two refinements the evidence forces on the simple cycle: integration breeds fragility, since the links that carry

goods carry shocks; and disintegration has winners, since a fracturing core can loosen its grip on a periphery.

- **Q2 — Channels.** Which routes carry the integrating weight? Three run through this book: the maritime Indian Ocean, the desert-and-oasis corridor, and the steppe. The ocean is the durable artery; the overland channels had one true climax, the Mongol Pax of Chapter 4.
- **Q3 — Modes.** What crosses the connection? The answer is read off the framework, input by input. The real flows are output itself, useful knowledge, people and capital assets — Y, A, L and K. Alongside them run monetary and financial flows: settlement assets, and claims and liabilities. Bullion can sit on both sides at once, which is why its direction serves as the book's running tracer of the centre of gravity.
- **Q4 — Europe.** What is Europe's role: peripheral, entrant, dominant, one of several cores? And how does it change? The book asks this in every period and answers it from the evidence.

Q4 is the odd one out, and readers are entitled to ask why Europe gets a question to itself. The answer is not that Europe is more important than the other actors, since for most of this book it is plainly less so. It is that Europe is where the prior sits: we have absorbed a version of the story in which European development is the main sequence and everything else is background. The way to test that prior is to make it a question, with the possible answers laid out in advance and none of them privileged. Asked that way, the answer changes across the ten chapters in a pattern that is genuinely interesting, and in Chapter 1 it changes twice, since Bronze Age Europe turns out to be a resource periphery supplying tin to an Asian-centred system, which is a much stranger finding than either "central" or "absent".

The four questions and the production function are two axes of a single piece of apparatus. The production function is vertical, asking what makes a *place* productive; the four questions are horizontal, asking how places *connect*; and they meet at N. That gives a definition of integration with some economics in it. To integrate is to relax a binding constraint by drawing on somebody else's inputs: importing land-intensive goods for want of land, monetising foreign silver for want of good mines, importing labour that cannot be raised at home, borrowing technology that was invented elsewhere. Q3's four flows map onto the framework without matching it one for one. Technology is A, people are L, capital is K, and goods are N embodied, in the sense that a cargo of grain or tin is somebody else's land arriving in usable form. The mapping is close enough to be useful and loose enough to be worth checking in each period rather than assuming.

Connection and integration

The framework asks what makes a single place productive. The rest of the book asks how places reach each other, and that is a separate question, with its own instruments and its own characteristic errors.

From contact to market integration

Connection between two economies comes in degrees, and four will do for the whole book. Contact is the weakest: a diplomatic gift arrives, or a shipwrecked sailor, or a single bead, and nothing about either economy has changed. Diffusion is the arrival of something that stays, such as a crop, an animal, a technique or a design, and a transfer that happens once can outweigh a trade that happens every year, since neither the chicken out of Southeast Asia nor the potato into Europe ever needed

a second voyage. Sustained connection means the flow repeats: ships come back, a route acquires a season and a set of prices, and both ends start to plan around it. Market integration is the strongest and the rarest, with production and prices in one place responding to conditions in another closely enough that a shock at one end shows up at the other.

Where along that range the phrase “world economy” begins to apply is a live argument. Dennis Flynn and Arturo Giraldez date globalisation to 1571 and the Manila galleon, on the criterion that every populated continent was then in continuous and consequential exchange. Kevin O’Rourke and Jeffrey Williamson answer that contact without converging prices is not integration, which puts the date after 1820. Two definitions, two and a half centuries apart, argued across three papers in the same journal. This book runs both instruments and says in each period which one it is using, because taking either one strictly destroys the subject: on the price test the first six chapters have nothing to measure, and on the diffusion test every stray bead becomes evidence of a world economy.⁷

The two yardsticks

The book measures two different things and keeps them apart, and half the disagreements in this field are two people scoring on different instruments without noticing. Integration is the question set out above, measured on whichever of the two instruments the period allows: diffusion where there are no prices, price convergence once there are, with the switch around Chapter 3. Success is the other yardstick, and it has two settings. Quantity, or extensive growth, means more people, bigger cities, larger total output. Quality, or intensive growth, means sustained income per head, which is the modern escape and, on this timescale, a very recent one. Almost every thriving economy in this book is thriving on the first setting: Han China, Song China, Mughal India and Qing China were all much larger than their contemporaries and none of them richer per person in the way the phrase implies today. An answer that says China was richer than Europe in 1700, without saying whether it means in aggregate or per head, has not yet said anything.

Trade costs, and what price convergence cannot see

The strongest instrument in the book is price convergence, and what it does is narrower than its reputation. A good rarely costs the same in two places. The test is that the gap between the two prices is bounded by the cost of moving the good, and narrows as that cost falls, so a stable price gap between two trading places is the cost of integration, quoted. Trade settles where the gap the market will bear equals the trade cost: freight, plus tolls, plus risk, plus whatever has to be paid for protection. That last term is why integration in this book is as much a political variable as a technological one, and it means a narrowing gap can arise three ways. Demand can rise where the good is bought; supply can rise where it is sold; or the cost of carriage can fall. Only the third is integration, so rising volumes are not evidence of integration. Converging prices are.⁸

The blind spot follows directly, and it is the one that matters in the early chapters, since before roughly 500 BCE we are working almost entirely inside it. Chapter 1 does contain one place where the test can almost be run, and it shows what the instrument feels like when it works. In the Old

⁷**Sources:** O’Rourke and Williamson, “When did globalisation begin?”, *European Review of Economic History* 6 (2002), 23–50, doi:10.1017/S1361491602000023; Flynn and Giraldez, “Path dependence, time lags and the birth of globalisation”, *EREH* 8 (2004), 81–108, doi:10.1017/S1361491604001066; O’Rourke and Williamson, “Once more: when did globalisation begin?”, *EREH* 8 (2004), 109–17, doi:10.1017/S1361491604001078. **Read more:** Chapter 5, where the Manila galleon is the case in point.

⁸**Sources:** Shiue and Keller, “Markets in China and Europe on the eve of the Industrial Revolution”, *American Economic Review* 97 (2007), for the method applied comparatively. **Read more:** Chapter 6 and Chapter 8, where the test does most of its work.

Assyrian trade of the early second millennium BCE, tin bought at Assur cost about 6.7 shekels of silver per hundred shekels of tin, and the same tin sold in Anatolia at about 14.3, a gap of roughly 2.1 times across about a thousand kilometres of donkey caravan. It is not arbitrage, because arbitrage is riskless and simultaneous and this was neither: the merchant financed the goods, waited a hundred days, and bore the risk of losing everything to bandits or a change of ruler. The gap is the cost of carriage, quoted, in a world with no coinage and no market in the modern sense, and it did not close. It persisted for generations, which says the costs were structural rather than a temporary friction somebody was about to compete away.

Integration breeds fragility

The first refinement to Q1 earns a section of its own, because it is the closest thing this book has to a general finding. The links that carry goods carry shocks: a network that moves pepper efficiently moves pathogens efficiently, and a system whose parts have specialised in what they do best is a system in which no part can feed itself when the connections fail. The pattern shows up at every scale. The Black Death is the clearest case, moving along the Mongol roads that had just made overland trade viable for the first time in centuries, and the Late Bronze Age collapse, the Antonine plague and the financial crisis of the 1930s all repeat it. The lesson is not that integration is bad, which would be a strange thing to conclude from a book about it. It is that integration and fragility are the same property observed at different moments, and that a period's prosperity and its vulnerability are usually built out of the same links. When a chapter reports a system at its peak, the next question is what that peak has made it unable to survive.

The evidence, and what it will bear

Across the periods covered in this book, we have to use different types of instruments. But the nature of any given instrument decides what can be claimed: a chapter using ice cores can say things a chapter resting on customs registers cannot, and vice versa. In this section, we take the production function framework and outline what each input, each stock of wealth, has for its own evidence base, as well as the potential traps in each.

Reading N: land, climate and resources

The evidence for N is mostly physical and mostly not written by anybody. Ice cores from Greenland and the Alps record atmospheric deposition year by year, so a layer of lead in the ice is a by-product of somebody smelting silver out of galena a continent away. Speleothems, which is to say stalagmites, record rainfall in their oxygen isotopes. Lake sediments, pollen cores, marine cores off river deltas and tree rings all give climate at various resolutions, and ore bodies have chemical fingerprints, so a metal object can sometimes be traced to the ground it came out of. The characteristic failure is dating, and the second is scale. A stalagmite might be dated to within a century, which sounds precise until the question is whether a drought preceded a civilisation's collapse or followed it. The cleanest instance in this book is the drying around 2200 BCE, where three independent archives agree on the direction and their error bands are each as wide as the three-century gap they are being asked to bridge, so none of them can order the climate event against the settlement change it supposedly caused. Chapter 1 runs that argument out against the Harappan de-urbanisation, which post-dates the climate event by some three hundred years. The scale problem is related, since a core measures its own catchment, not the world.

Reading L: people

Population is reconstructed from settlement survey, from cemetery demography, from skeletal and stature evidence, and increasingly from ancient DNA. Isotopes in tooth enamel give something different again: strontium and oxygen laid down in childhood record where a person grew up, and enamel never remodels, so the molars carry an ordered record of where somebody was at particular ages. The majority of early population figures in this book are from an area multiplied by an assumed number of people per unit of area; the assumed coefficient does most of the work. Change the coefficient and the answer changes proportionally; change which line on the ground counts as the edge of the settlement and it changes by more than that. Appendix D sets out two worked cases: an urbanisation rate for Sumer where two careful scholars differ by a factor of three, and a population for Pataliputra where the published estimates span a factor of about thirteen without anybody disputing any evidence at all.

Isotopes and DNA look like two ways of measuring the same thing and are not, because isotopes measure residence and DNA measures descent. A merchant who sailed to Carthage, traded for a season and sailed home is visible to the first instrument and completely invisible to the second, because he left no descendants there; his grandchild, born in Carthage to a local mother, is the exact reverse. This is why the finding that western Mediterranean Punic populations carry almost no Levantine ancestry shows that Phoenician settlers did not dominate the gene pool, and does not show that Phoenicians did not go there. Both report lower bounds, and where they disagree the disagreement is itself a measurement: the ratio of circulation to settlement, which is a real economic quantity we would otherwise have no way to see.⁹

Reading K: capital, money and credit

One heading covers three things the framework keeps apart, and the reason is the archives rather than the economics. Hoards, stray coin finds, die studies and bullion shipment records are evidence about money; loan tablets and partnership contracts are evidence about credit; ship timbers, irrigation surveys and warehouse accounts are evidence about capital itself. In practice they arrive in the same box, because the people who wrote down one of them wrote down the others on the same tablet, and above all they arrive in merchant archives: the letters, loans and accounts of people doing business. The categories still have to be kept apart, because coin is evidence about settlement and says nothing about how many ships there were. The characteristic failure is that archives survive by accident, and disproportionately by catastrophe. Mesopotamia's cuneiform tablets were unbaked clay, and what fired them hard enough to last four thousand years was the burning of the palaces that held them, so we read those economies in detail because they were destroyed. A place that was never sacked, or whose records were kept on palm leaf or paper in a wet climate, is simply harder for us to read, and legibility is a poor proxy for commerce. Coin hoards carry their own version of the same problem, since a hoard dates when somebody decided to bury money and did not come back for it, which is a fact about insecurity as much as about circulation.

A second and subtler difficulty is that coins are recycled: the metal in a coin struck in 200 CE may have been mined in 50 BCE, melted down twice and alloyed with something else on the way, so a chemical analysis of coinage reports what was in the melting pot rather than what came out of the ground. Chapter 2 puts an ice core and a coin series side by side for exactly this reason, because the ice records what was smelted and the coins record what was refined and recycled.

⁹**Sources:** Ringbauer et al., "Punic people were genetically diverse with almost no Levantine ancestors", *Nature* 643 (2025), 139-47, doi:10.1038/s41586-025-08913-3. **Read more:** Chapter 2 and Chapter 3, where both instruments are used.

Reading A: technology and technique

A is read from the objects themselves, and from the sciences that interrogate them: archaeometallurgy, lead and tin isotope provenancing, residue analysis, and radiocarbon dating of the contexts things are found in. Its characteristic failure runs the opposite way to the others, and it is a cheerful one, because better dating destroys more claims than it creates. When a technique improves, the usual result is that something long believed turns out to rest on a specimen that is intrusive, or younger than its context, or misidentified. The cleanest example is the chicken. Domestic chickens were for decades said to originate in the Indus Valley, and a systematic re-dating programme found that every European Neolithic and Bronze Age specimen tested was intrusive, moving the earliest secure domestic chickens to Thailand and about a thousand years later. Nothing was fabricated and nobody was careless; the instrument got sharper and a claim fell over. That is how to read the confident sentences in older books, including good ones. A claim about origins established before radiocarbon calibration, or before a genome could be sequenced from a tooth, has not been refuted so much as never tested, and the pattern of the last twenty years is that most such claims do not survive.

What each proxy can and cannot support

The limits are worth setting out in one place, because most over-claiming in this field is one of these four moves.

Table 1. What the standard proxies will and will not bear.

Proxy	What it supports	What it does not
Diffusion of goods	Contact, and the geographic range of a network	Volume, frequency, price, or who profited
Direction of carriage	Relative shipping cost; which end wanted the other's goods	How the surplus was divided between the two ends
Population	Aggregate scale, under a Malthusian assumption	Income per head, ever
Settlement counts	An upper bound on decline	The decline itself, since the record-keeping collapses too

The fourth row is the one most often missed. When a state falls apart, the number of attested settlements drops, but the thing that produced the attestations, an administration writing things down, fell apart at the same time. So a fall in recorded settlements is a composite of real decline and lost visibility, and it gives a ceiling on the first rather than a measurement of it.

Absence of data is not absence of activity

What has survived reflects some combination of conventions about record-keeping and the accidents of climate and preservation, not economic weight. Organic records rot quickly in the humid monsoon tropics; they may survive better in cool dry Europe and arid Egypt, so the surviving record systematically under-documents the ocean world this book is written around. Because the direction of that bias is known, we should treat the figures we have as lower bounds: trade was at least as large as the record shows. But it does not license filling the gap with a conclusion: "the evidence is biased against parts of Asia" is true, but this does not automatically mean that "Asian trade was much larger".

Distinguishing the map of evidence from the map of production

One commodity demonstrates the problem: black pepper, which is south Indian. Peppercorns that have been recovered from the Roman period survived in one of three ways: desiccation, in the hyper-arid Egyptian desert (at Berenike and Myos Hormos), waterlogged in anaerobic mud in places like Germany, Britain and France, or mineralised in cesspit fill in Campania. But where pepper was actually grown, none of those three conditions holds. So until very recently, essentially none of the finds were from the producing region. In this case, it is easy to read the map as one of trade and distribution, not one of production but the lesson may well hold for other products as well. In short, across this whole book, and especially for earlier periods, the map of the evidence is not the map of the world economy; rather, it is the map of preservation conditions. A set of products with very poor chances of preservation are textiles. Goods made from cotton, silk and wool rot in every climate that grew them.¹⁰

The oldest annual series in the world are tax records

The longest continuous annual measurements we have of anything physical reflect human priorities: The Nilometer measure of the Nile flood runs annually (with some gaps) from 622 CE because flood height set the Egyptian land tax. Similarly, data on the Beaune grape harvest run from 1354 because the town council had to pay the pickers on a date. Data on Kyoto cherry blossoms or Lake Suwa's freeze similarly have motivations related to human institutions. But because they were measuring a tax base or a wage bill, the geography of the pre-modern climate record is closer to a map of fiscal and institutional capacity. Further, it degrades when the polity degrades. The Nilometer holds between seventy-seven and a hundred readings per century up to 1470, and then thirty-two in the sixteenth century and thirty-one in the seventeenth. The density of a series is itself a datum about the state that produced it. And what the series tells us needs care: the apparent upward drift after about 1300 reflects the silting up of the river bed, rather than a wetter Nile.¹¹

Grading claims we read

Because the evidence is so uneven, it will be necessary to grade individual claims, as well as verdicts. We can consider a claim to be: **secure** when a named source has been read and says it; **probable** when it takes one step of inference, or rests on a synthesis rather than the original study; and **contested** when specialists disagree in print. And it is **single-source** when one authority stands behind it and nobody has independently confirmed it; that is not a criticism so much as a warning to the reader. Related, agreement between two documents that share an ancestor is not evidence of the strength of a claim, which is why a claim has to be traced back to the source that first measured or observed the thing. In preparing Chapter 2, this book compiled a register of claims about early historic South Asia. On the forty-five entries whose subject is genuinely South Asian, fewer than one in six came out secure and nearly one in three contested. There is value in that ratio: it marks the limits of what can be known about the Mauryan and post-Mauryan economy at the moment.

An example, from Java, shows another value, intrinsic to the scientific method. In 2023, a team published a study of Gunung Padang in West Java arguing that the hill is actually a construction, begun by at least 14,000 BCE, and "potentially the oldest pyramid in the world". The journal retracted it in March 2024, because the radiocarbon dating "was applied to soil samples that were not associated

¹⁰**Sources:** Cobb, on the recovery contexts of Roman-period pepper, citing Cappers (2006), Livarda (2011), and Robinson and Rowan (2015). **Read more:** Chapter 2.

¹¹**Sources:** Toussoun (1925), digitised by Koutsoyiannis et al. (2013); Kondrashov, Feliks and Ghil (2005) on the series' spectral structure. **Read more:** Chapter 3.

with any artifacts or features that could be reliably interpreted as anthropogenic". The laboratory work was, undoubtedly, careful but what failed was the join between the number and the claim. The age belongs to the soil; the construction event was inferred separately and then had the soil's age attached to it. The authors still disagree with the retraction, so the case is still open.¹²

The instruments improve, and that is the shape of the book

Set against the chapters, the four evidence bases show a progression. For Chapter 1 there are no price series at all. Integration has to be read from the diffusion of objects and standards and the verdict on the centre of gravity is *not applicable*. From Chapter 2 there are coins, quoted prices in a handful of documents, and the first physical output series, so direction can be argued and magnitude cannot. From Chapter 3 to Chapter 6 the merchant archives thicken and price series begin to run continuously in a few places, which is enough to test convergence over decades. From Chapter 7 there are wages, prices and the first constructed national accounts, and the argument shifts from where the weight sat to why it moved. By Chapter 9 and Chapter 10 there are trade statistics collected by states for their own purposes, and finally a centre of gravity that can be computed rather than inferred. Two things follow. The first is that the book's later verdicts are better evidenced than its earlier ones, which is stated in Appendix C chapter by chapter. The second is the one people forget: better evidence does not make the later arguments less contested. Chapter 7 has more data behind it than any chapter before it and contains the book's least settled question. The quantity of evidence and the level of disagreement are close to independent of one another, and assuming otherwise is how a reader ends up trusting a well-measured claim more than it deserves.

One cargo, and what four questions can do with it

The apparatus is easier to follow once it has been run on a real object, and there is a good one to hand. In the late fourteenth century BCE a ship went down off Uluburun, near Kas on the southern Turkish coast, and it was found in 1982. It is the richest single deposit of Bronze Age trade goods ever recovered, and it is a dataset of one. Aboard were about ten tonnes of copper, as 354 four-cornered "oxide" ingots and 121 bun-shaped ones, and roughly a tonne of tin, which is close to the ten-to-one ratio bronze needs, so the cargo is about eleven tonnes of alloy in kit form. With it went about 175 glass ingots, a tonne of terebinth resin in at least 149 Canaanite jars, 135 Cypriot vessels, ebony and ivory, Baltic amber and Mesopotamian bitumen: one hull carrying the products of at least nine or ten cultures. The four questions can then be put to it. Q2, channels, it answers well: this is a maritime cargo moving counter-clockwise around the eastern Mediterranean, and the amber alone establishes a link running to the Baltic. Q3, modes, it answers partly, since the flow is overwhelmingly goods, which is N embodied, and the merchants' weight sets aboard suggest private commercial calculation alongside gift exchange. Q1, direction, it barely touches, because one wreck is a point and direction requires a series. And Q4, Europe, it raises without settling, because where the tin came from is precisely what two laboratories now disagree about.

What it cannot tell us is the more useful half. It cannot say how many such ships sailed, or how often, or what any of it cost, or who profited. It cannot even settle whether this was commerce or diplomacy, which is the question Chapter 1 takes up with the evidence on both sides. And there is a preservation trap sitting inside it. The 135 Cypriot vessels aboard are roughly twice the number

¹²**Sources:** Natawidjaja et al., "Geo-archaeological prospecting of Gunung Padang buried prehistoric pyramid in West Java, Indonesia", *Archaeological Prospection* 31:2, doi:10.1002/arp.1912, retracted at doi:10.1002/arp.1932 on 18 March 2024. **Read more:** Ernenwein, "Retraction and reflection", *ArcheoSciences* 49-1 (2025), 607-10, the editor's own account of the case.

known from the whole Aegean for the entire Late Bronze Age. If one wreck can double the record, then the record was never a sample of the trade. It is a sample of what happened not to be lost.¹³

Using this book

Three ledgers to keep

Ten topics can be ten surveys or one argument, and the difference is whether things are written down along the way. Keep three running lists, one line per chapter. The centre-of-gravity ledger records where the weight is, whether it is moving, and how confident the reading is, on the four-point scale above. The input ledger records which of A, K, L and N is doing the work, and which is binding. The natural-system beat records the climate, energy or disease force in play, and is there to stop the physical world becoming background scenery. Three short rows for each topic, kept as the topics go by, and by Chapter 10 the book's argument will have been assembled by hand, which is a considerably better position to argue with it from than having read it.

Six threads that run the length of the book

Everything above recurs, and it recurs as six threads. Naming them together is a way of noticing what a straight chronology would hide.

1. **Centre of gravity**, with a verdict and a stated confidence in every chapter, collected in Appendix C.
2. **Quantity to quality**, the second yardstick, resolving at Chapter 7.
3. **Follow the silver**, the early-modern form of the bullion tracer. The same American metal met different monetary conditions and had different fates: Asia absorbed and specialised, the Ottomans inflated, West Africa debased and depleted, and Europe turned a monetary loss into armed-trade and financial power. That last fate is the bridge to the divergence.
4. **Measurement of integration**, on instruments that improve from diffusion through bullion flows and prices to a computed centre of gravity.
5. **Forces**: climate, disease, technology and policy, two of them natural.
6. **Data survival and bias**, and whose eyes we are seeing through.

A note on method

Two commitments follow, and they are why the subtitle is *data and debates*.

- **Data**. The book leans on numbers, because numbers are how a claim like the spine gets adjudicated rather than asserted. Figures and tables are reproducible, generated from datasets that can be downloaded and re-run. But the data thin out sharply further back, and they are biased toward the literate, fiscal, European end of the record. A chapter on the Bronze Age and a chapter on the twentieth century are not equally evidenced, and the book says which is which.

¹³**Sources:** Pulak, "The Uluburun shipwreck: an overview", *International Journal of Nautical Archaeology* 27 (1998), 188-224, at p. 219 for the cultures represented and p. 204 for the Cypriot vessels; on the dating, Pulak's own later statement of the end of the fourteenth century and Manning et al. (2009) at 1320 plus or minus 15. **Read more:** Chapter 1.

- **Debates.** Where the scholarship is genuinely contested, and the standing examples are when the Great Divergence began and why, whether interwar disintegration was man-made or inevitable, and when globalisation really started, the book presents the disagreement rather than resolving it out of sight. Disputed figures are flagged and taught rather than silently settled.

The complaint this book is built on is not its own invention. David Northrup observes that textbooks “almost invariably” break their coverage at 1500, and Merry Wiesner-Hanks, introducing the *Cambridge World History*, names the problem in her own preface while conceding that four of that series’ nine books are given to 1400 and after. Northrup pushes further back, in a form that reaches this book’s own starting line: the deeper exclusion is the convention of beginning with Sumer and with writing, which removes between ninety-five and ninety-nine per cent of the human past before the story starts. There are honest defences of that imbalance. David Christian argues the demographic case, that perhaps eighty per cent of all human lives were lived between the Palaeolithic and 1750, and the counter is that a historian can work only with evidence, which accumulates toward the present. Michael Lang, in the same volume, goes after this book’s method rather than its coverage, arguing that periodising by chronology smuggles in assumptions about what changes and when.¹⁴

This idea of this book is to test the claim made here. This means that it is possible to read the evidence assembled here and reach a different verdict. The book aims to highlight the places where disagreement is most likely: for example, where an argument rests on a single instrument or the verdicts is uncertain. The derivations in Appendix D show that a headline figure turns out to be a coefficient multiplied by an area and thus the choice of coefficient is critical. A reader who finishes convinced that European dominance was longer, or deeper, or earlier than the spine allows should ask which of those places would have to be attacked to make that case. That is the idea behind setting out the hypothesis: being clear about what it would take to refute it.

i How to read the chapters

Each chapter opens by following one object that carries its period’s story, locates the period on the arc, sets out the cast and how we know, narrates the period on its own terms, then reads it through the four questions, delivers a centre-of-gravity verdict with a confidence level, and hands off to the next. Look out for the recurring Data exhibit and Debate, and the Follow the money and How we know callouts. Three appendices do the heavy lifting the chapters send out to them: Appendix C collects the ten verdicts in one table, Appendix D sets out how the most quotable numbers were actually constructed, and Appendix E gives the live scholarly disputes in full.

¹⁴**Sources:** Northrup, “Globalization in historical perspective”, in *The Cambridge World History*, vol. 1 (2015), pp. 110-31, at pp. 111-12 and 123; Wiesner-Hanks, preface to vol. 1, pp. xvii and xix; Christian, “Introduction: the constructions of a global past”, vol. 1, pp. 22-23; Lang, “Evolution, rupture, and periodization”, vol. 1, pp. 84-109, at pp. 96-97. **Read more:** Sachsenmaier, “The evolution of world histories”, vol. 1, pp. 56-83.

Part I: 3500 BCE to 500 CE

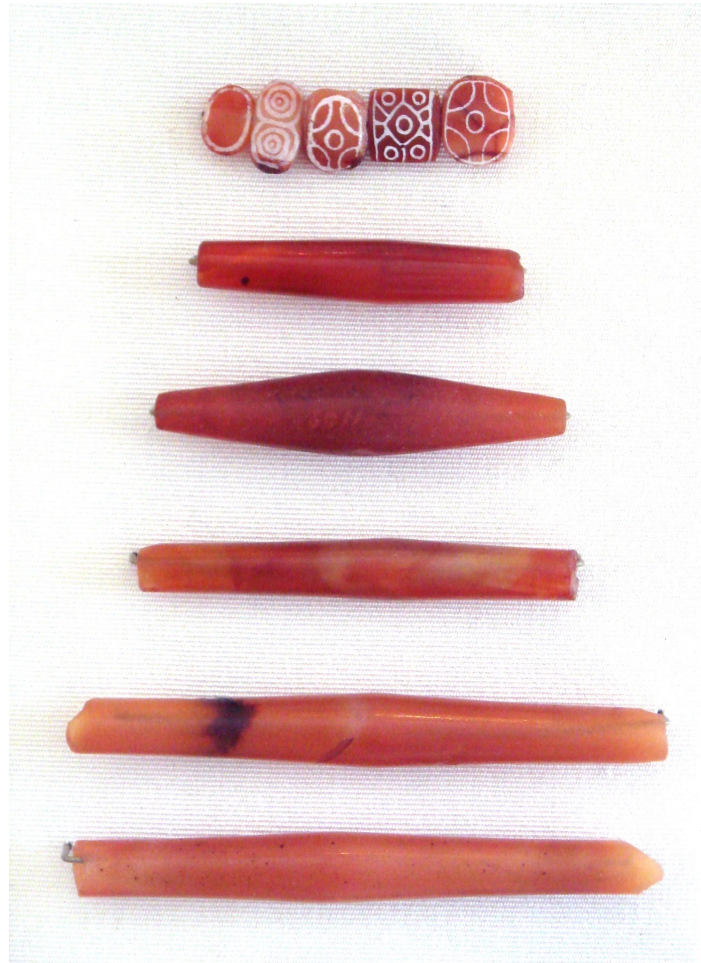


Figure 3. Etched carnelian beads of the Indus civilisation, excavated far from home at Susa in Elam — the emblematic long-distance trade good of a world of many loosely linked cores and no single centre. Photo: ALFGRN, Wikimedia Commons, CC BY-SA 2.0.

1. Deep structures: a world without a centre

! Preliminary draft — under review. Published for review; content, figures and citations may still change.

Before prices, the only measure of integration is the diffusion of goods and ideas.

“Ships from Meluhha, ships from Magan, and ships from Dilmun he made tie up alongside the quay of Agade.” — inscription of Sargon of Akkad, c.2300 BCE

Follow one thing

Follow a bead. Sometime around 2500 BCE an artisan in one of the Indus cities took a piece of hard orange carnelian, drilled it lengthwise with a slow stone drill, and etched a white pattern into its surface by a technique that was, for the moment, a Harappan monopoly. The bead travelled. It went down the Indus, along the coast of the Arabian Sea, up the Persian Gulf through the entrepôts of Magan and Dilmun, and ended in a grave at Ur, some two thousand kilometres from the hand that made it, and a good deal further by the route it actually took. No contract survives to say who carried it, what was paid, or at what price. The bead itself is the evidence: that two distant economies had touched, that goods and standards moved between them, and that an exchange existed at all. This is how the whole chapter has to be read, because in a world before coins and price-lists, integration is visible only in things.

The bead carried one more clue, in the direction it travelled. Indus goods reached Sumer, and Meluhhan ships docked in Sumerian ports, while much less Mesopotamian material has been found in the Indus. In a period with no prices to compare, an asymmetry like that is the crudest available sign of who held the initiative. It is also a treacherous one, and this chapter takes it apart later, beginning with the observation that a carnelian bead is exactly the kind of thing that survives to be counted. But it does place the Indus, in its first appearance in this book, among the most active traders of a world that did not yet have a centre.¹

Where we are on the arc

This is the first chapter, and it is the one place in the book where the central question cannot yet be asked. The book exists to test a claim, that the weight of the world economy sat in Asia for far longer than we tend to assume, period by period against the evidence. But “where did the weight sit?” only means something once there is a single connected economy whose weight can be located, and across these three thousand years there was not one. There were several loosely linked riverine cores, trading at the edges, rising and falling on their own clocks. So this chapter returns no verdict on the spine. It builds the ground the spine will later stand on, laying down the geography — the

¹**Sources:** Dalrymple (Loc 1163, 1168) on the Meluhha trade and the goods reaching Ur; Anthony (loc 4821) on carnelian moving west. **Read more:** Dalrymple, *The Golden Road* (2024).

1. Deep structures: a world without a centre

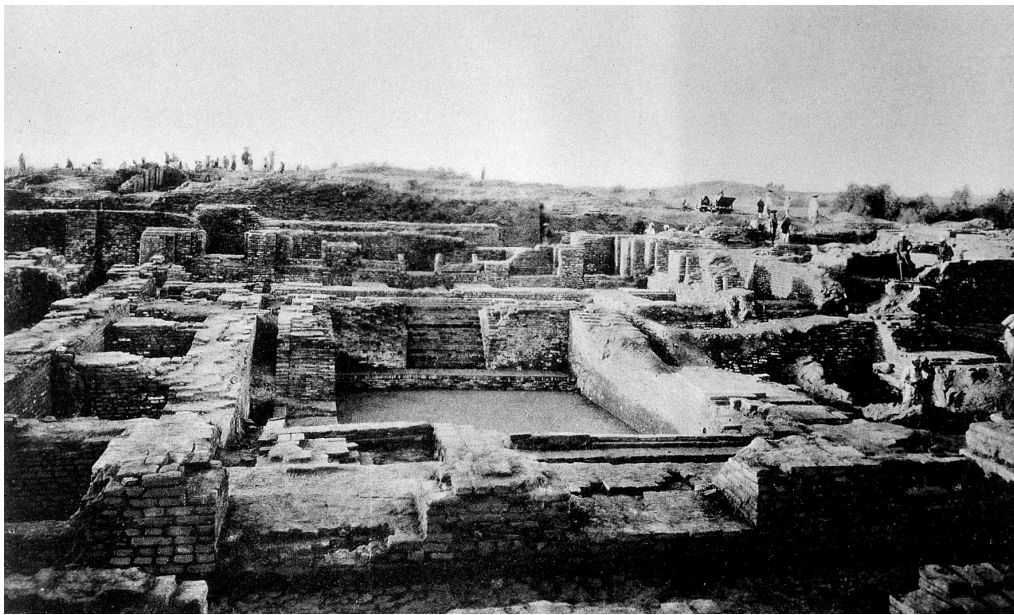


Figure 1.1. The Great Bath at Mohenjo-daro — the planned monumentality of the Indus cities, built without the palaces or royal tombs of contemporary Mesopotamia and Egypt. Photo: Wellcome Collection, CC BY 4.0.

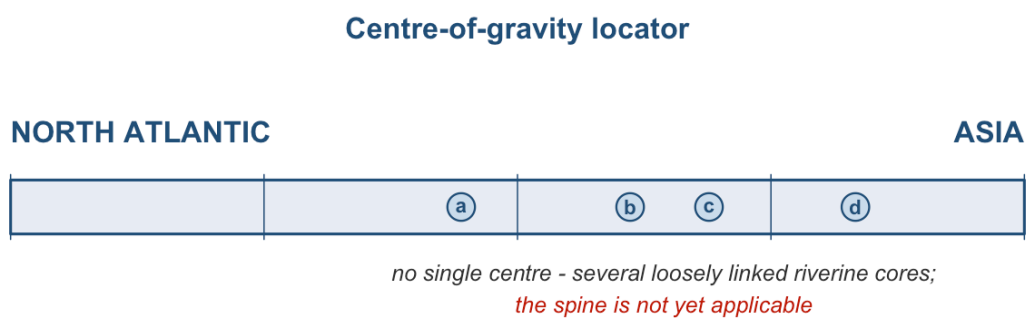


Figure 1.2. Centre-of-gravity locator: in c.3500–500 BCE there was no single centre to locate — only several loosely linked riverine cores, keyed west to east: (a) Egypt, (b) Mesopotamia/Uruk, (c) the Indus (Harappa), and (d) the forming East Asian core. The Near East (Uruk) was the most connected node and the Indus the most active long-distance trader; the spine is not yet applicable, and the chapter's contribution is structural (stylised; not to scale). Author's schematic.

monsoon and the three great channels — that will eventually make the Indian Ocean the natural integrator.²

1.1. The stage and the cast

The stage was built by geology and weather. A continental collision some fifty million years ago raised the Himalayas, and the mountains organised the climate of half the world: they created the monsoon, the great reversing wind that blows from the south-west in the northern summer and swings back from the north-east in winter, and they fed the river basins — the Indus, the Ganges,

²**Sources:** Cunliffe (Loc 158) on the channels frame; Pearson (Ch.1) on the monsoon deep structure. **Read more:** Cunliffe, *By Steppe, Desert, and Ocean* (2015).

The stage and the three channels (c.3500-500 BCE)

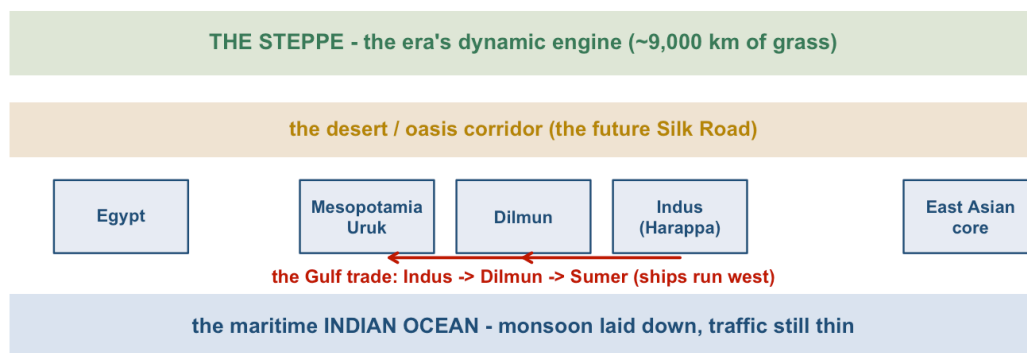


Figure 1.3. Orientation: the stage and the three channels. The riverine cores — Mesopotamia/Uruk, the Indus (Harappa), Egypt and the forming East Asian core — linked at the edges by the maritime Indian Ocean (still waiting), the desert/oasis route, and the Eurasian steppe (the era's dynamic engine), with the Gulf trade running Indus → Dilmun → Sumer (stylised; not to scale). Author's schematic.

and by analogy the Yellow and Yangtze — where the first surplus-producing societies grew up. The monsoon's deepest gift was reliability. Because the wind reversed on a known schedule, a sailor could ride out on one season's breath and return on the other, which made predictable round-trip voyaging possible across the Indian Ocean in a way the Atlantic and Pacific could not match. That reliability is what the geographer Michael Pearson called the ocean's "deep structure", and it is the master variable of this whole book. But it set possibilities rather than outcomes. Three corridors were born in these millennia and would carry goods, people and ideas for the next four thousand years: the maritime Indian Ocean, the desert-and-oasis route later called the Silk Road, and the great Eurasian steppe. Around them sat the cast, the riverine cores that could generate a tradable surplus.³

💡 Dramatis personae

The economic actors of c.3500–500 BCE, profiled east to west. There is no centre yet, so these are loosely linked cores rather than partners in one system. **India appears in every chapter at the fullest depth** — here as the Indus civilisation, the period's most active long-distance trader on the evidence that survives, a qualification the chapter goes on to make good. The steppe comes last as the era's dynamic engine rather than as a periphery.

Six profiles follow, and they can be skipped without losing the thread, because what each establishes is carried forward here. Four findings carry forward into the narrative. The first is that a large economy does not need a visible sovereign. The Harappans left no royal tombs, no palace quarter, no royal iconography and no depictions of warfare, and they built more urban public infrastructure per head than any of their contemporaries — house-by-house wells, bathing platforms, covered drains — with nobody's name attached to any of it, across a territory the size of western Europe in which only five of more than a thousand known sites count as urban. The second is that the state can arrive before the market rather than after it. Western Zhou China ran on obligation between aristocratic lineages,

³**Sources:** Cunliffe (Loc 158) on the steppe corridor and the three channels; Pearson (Ch.1) on the monsoon deep structure. **Read more:** Pearson, *The Indian Ocean* (2003).

1. Deep structures: a world without a centre

with markets in any ordinary sense absent, and it was the Warring States contest that ground out conscription, censuses, uniform weights and deliberate grain-price smoothing. The instinct is to assume markets came first and states arrived later to regulate them; here the sequence runs the other way.

The third finding is that writing follows the money, and does so differently in different hands. Mesopotamian cuneiform began as accounting and stayed institutional for a very long time, producing a scribal class; Shang China's oracle bones were made to record divination rather than transactions; and Egypt expanded writing for a fiscal reason, then found, in Brian Muhs's account, that documented and enforceable titles lowered the cost of a private transaction relative to going through the institution, so that the state's own revenue interest pushed it into underwriting the exchange that competed with its redistribution. The Phoenician alphabet later in this chapter completes the set by putting writing in a merchant's hands. The fourth is a warning the Introduction sets up and this period makes concrete. Mesopotamia kept its accounts on clay, which survives when a city burns, while early China kept its on bamboo and wood slips, which do not begin to survive until about the fifth century BCE, so a comparison between a bureaucratic, account-keeping Mesopotamia and a ritual, divining China may substantially be a comparison between clay and bamboo. Dilmun sharpens the same point from the other side, having run complex repeated long-distance transactions in contact with three literate civilisations without any Dilmun script ever being identified.

One comparison cuts across the six. Three societies of roughly the same era each generated a large agricultural surplus, and each put it somewhere different. Uruk put it into the city itself and into the apparatus for running one: walls, workshops, ration bowls turned out in the millions, and the accounting that became writing. The Harappans put it into infrastructure that nobody signed, with the seals that recorded claims on resources sitting in ordinary households rather than in palaces. Egypt put it into monuments to named individuals, spread along a river. None of these is a market economy, and the differences between them are not differences of wealth. They are differences in who could command the surplus and what they wanted it for, which is a question this book asks in every later chapter with better data and rarely a clearer contrast.

The forming East Asian core

The Yellow and Yangtze valleys were taking shape as an agrarian world in their own right while the Gulf trade ran in the west, but they sat largely outside it. Wild rice was being gathered along the Yangtze from about 8000 BCE, cultivated from about 6000, and fully domesticated in the lower Yangtze only around 4000 to 3000 BCE; by 3000 BCE farming had spread across much of what would become the East Asian core, and the first Chinese state emerged around 1800 BCE. This was a populous, river-fed agrarian base of the same broad type as Mesopotamia or the Indus, but it grew up on its own metabolism rather than as a node in the western relay. What marks it out for this period is how late the western bronze-and-chariot package reached it. The earliest known chariots come from Sintashta graves in the Urals, before 2000 BCE, and the type was in the Near East by the nineteenth century BCE, yet it did not appear in China until about 1200 BCE, arriving by way of the steppe rather than the Gulf. The same was broadly true of the metallurgy that defined the Bronze Age elsewhere.⁴

Two states then carried the core through the rest of this chapter's span. The Shang (c.1600–1046 BCE) were northern China's first securely documented state, and they cast bronze at something close to industrial scale: foundries at Zhengzhou and Anyang, ritual vessels surviving in the thousands, the largest weighing over 800 kilograms. Their expansion followed commodities, north-east toward the Bohai salt and south toward metal, with jade, ivory, cowries and turquoise moving on top of that. They also left the oracle bones, the earliest Chinese writing, and the contrast with Mesopotamia is worth pausing on: cuneiform began as accounting, while the oracle bones are records of divination. The two earliest writing systems in this book

were made to preserve different things, which shapes what each civilisation can now be asked about.⁵

The Zhou who followed show state capacity arriving well before anything an economist would call a market. Western Zhou (1045–771 BCE) was a patrimonial arrangement in von Glahn’s reading: the king shared authority with noble houses and lived off rights over land and labour, a domain state rather than a tax state, its three ministries named for Lands, Construction and Horses. Exchange ran between aristocratic lineages, and markets in any ordinary sense were absent. Eastern Zhou (770–256 BCE), and the Warring States contest that ran on to Qin’s unification in 221, then ground capacity out of endemic war — conscription, censuses, uniform weights, deliberate grain-price smoothing by about 400 BCE, primogeniture abolished in the 350s — while iron lifted farm output and coinage pulled interregional trade and specialisation, iron in Hann, salt in Qi, gold in Chu. The sequence is the counter-intuitive part. The instinct is to assume markets came first and states arrived later to regulate them; here the state came first, by a wide margin.⁶

A warning has to travel with all of this, and it is sharper than a general complaint about thin evidence, because the bias is in the writing medium. Mesopotamia kept its accounts on clay, which survives when a city burns; early China kept its on bamboo and wood slips, which do not begin to survive until about the fifth century BCE. What we have from Shang China is therefore inscriptions meant for display, above all the oracle bones, and Wang Haicheng’s point is that the vast majority of bones cracked for divination were never inscribed at all, while everyday administrative documents were almost certainly written on perishable materials and are gone. The graph for a bundle of writing slips appears in the divination texts themselves, so the Shang were keeping records on slips; we cannot read them. A comparison between a bureaucratic, account-keeping Mesopotamia and a ritual, divining China may therefore be substantially a comparison between clay and bamboo. That does not make the western record false, but it does mean the eastern core is under-documented for this era rather than demonstrably marginal.⁷

Trade profile

- **Main exports** — little evidence of long-distance export into the western system this early; the core’s surplus was consumed and circulated locally.
- **Main imports** — the bronze-and-chariot technology package, arriving overland from the steppe around 1200 BCE rather than by sea.
- **Export markets** — none securely attested in the Gulf or Mesopotamian record for this period.
- **Import sources** — the Eurasian steppe corridor to the north and west, the one channel that reached it.⁸

⁵**Sources:** Fuller, Qin et al., “The domestication process and domestication rate in rice”, *Science* 323 (2009), and Kingwell-Banham, Petrie and Fuller in *The Cambridge World History*, vol. 2, for the staged chronology; Anthony (loc 7344, 6171); Cunliffe (loc 1266, 2001). **Note on a date corrected:** an earlier draft gave 9,500–7,500 BCE for rice, following Mithen (2003); that is the Diaotonghuan-era dating and has been superseded. **Read more:** Fuller, Qin et al. (2009).

⁶**Sources:** Liu & Chen (2012) on Shang bronze casting at Zhengzhou and Anyang; Anthony (loc 6171) on the chariot reaching China c.1200 BCE. **Note on a figure clarified:** the very large surviving vessel (the Houmuwu ding, over 800 kg) is Shang and from Anyang; the often-quoted figure of some twelve thousand inscribed vessels is Zhou-era, and the two should not be merged. **Read more:** Liu & Chen, *The Archaeology of China* (2012).

⁷**Sources:** von Glahn (2016) on the Western Zhou as a patrimonial domain state and on Warring-States fiscal and administrative capacity; Scott (Page 631) on early state formation. **Read more:** von Glahn, *The Economic History of China* (2016).

⁸**Sources:** Wang Haicheng, “Writing and the city in early China”, in *The Cambridge World History*, vol. 3 (2015), pp. 131–57, on perishable media, the survival threshold of about the fifth century BCE, display writing, and the uninscribed divination bones; Mithen (Loc 7937) and Scott (Page 100) on the general preservation asymmetry. **Read more:** Wang (2015).

⁸**Sources:** Anthony (loc 6171); Mithen (Loc 7945). **Read more:** Mithen, *After the Ice* (2003).

💡 The Indus civilisation — the agency-holder

The Indus, or Harappan, civilisation grew up along the Indus and its tributaries from roughly 3000 BCE, when its first cities appeared at much the same moment as the Early Dynastic cities of Sumer, and it reached its mature urban phase between about 2600 and 1900 BCE. Its base was agrarian, like every economy in this chapter. What set the Harappans apart was how much of what they built on it had been *designed*: Mohenjo-daro and Harappa were laid out to a plan, with gridded streets, fired-brick housing and elaborate covered drains, at a degree of urban engineering with no contemporary equal. Mohenjo-daro covered more than 200 hectares, within reach of Uruk's 250, and Harappan urban sites ran from about 50 hectares to over 200. The distinguishing feature was deliberation rather than size. The settlement pattern is the stranger fact. Mature Harappan material turns up at more than a thousand sites spread over more than a million square kilometres, and of those only five count as urban: Harappa, Mohenjo-daro, Ganweriwala, Rakhigarhi and Dholavira, with the rest typically around five hectares. Those five stand far apart, 280 kilometres from Mohenjo-daro to Ganweriwala and 600 from Harappa to Mohenjo-daro, across a territory the size of western Europe, where Mesopotamia in the same centuries was a crowded landscape of neighbouring cities within a few days' walk of one another. Whatever held the Indus together was not proximity.

Two of the things this economy stood on were domesticated locally, which matters for a region routinely written up as a recipient of other people's innovations: zebu cattle at or around Mehrgarh, and the earliest evidence anywhere for Indian tree cotton, on which a textile tradition rests that later chapters follow for four thousand years.⁹ Two features of the internal economy stand out, and the second is genuinely puzzling. The first was standardisation, and it went further than the phrase usually implies. Harappan cubical stone weights double in a binary sequence up to a standard of about 13.7 grams and then continue in decimal multiples, the largest at Mohenjo-daro reaching almost eleven kilograms — a system, not a convention, and one exported as far as the Gulf. Fired bricks hold a 1:2:4 ratio across the whole settled zone, and that last uniformity is where the state instinct should be resisted, since Kenoyer's explanation is a class of brickmakers whose craft tradition spread from settlement to settlement rather than a building code. Alongside the weights ran a corpus of inscribed seals, and the two together are the apparatus of an economy in which goods were measured, marked and exchanged to common conventions over long distances.¹⁰

The second feature is what the cities lack, and the usual shorthand overstates it. There are no royal tombs, no monumental religious architecture, no royal iconography and no depictions of warfare. Two entries on that list need care: temples are not securely absent, since Kenoyer allows that House AI at Mohenjo-daro may have been one, and no depictions of warfare is not the same as no warfare, because skeletal remains from Harappa carry clear evidence of interpersonal violence. What is absent is a palace *quarter*, the largest buildings being scattered across the mounds rather than gathered into a royal precinct, and the structures long called granaries have gone too, having produced no charred grain, no storage vessels and no sealings. What these cities were reticent about was the display of personal power, which is not the same as being socially flat: Possehl, who denies this was a state, still writes of upper ranks and a social pyramid, and remarks that the Great Bath was not the playground of the Indus *hoi polloi*.¹¹

How such a civilisation was governed, and how its surplus was gathered without the visible apparatus of monarchy, is the liveliest question in Indus archaeology. Three positions run as a spectrum rather than a quarrel — Kenoyer's rulers of a kind, Possehl's elites without a state, and Adam Green's collective governance by many intersecting institutions — and Appendix E sets them out with the evidence that divides them, which turns largely on where the seals were found. One fact cuts across all three: no Indus seal has ever been found in a burial, which is exactly where Egypt and Mesopotamia put the marks of individual status. What the three

agree on is enough to carry the economic argument. There were no kings, no royal display and no depictions of warfare; there was extensive investment in urban public goods, from house-by-house wells to bathing platforms and covered drains; and there was a standardisation apparatus reaching from villages to the Gulf. A society can build all of that without anyone's name on it, and that matters beyond the Indus.

The chapters that follow spend a great deal of time on states that taxed, borrowed, minted and spent, and it is easy to slide from that into assuming an economy needs a visible sovereign before it can build anything. The Harappans built more urban infrastructure per head than any of their contemporaries and left no king's name attached to any of it.¹² With no prices and no ledgers, the integration the Harappans achieved is read from the things themselves: Harappan seals, cubical weights and above all etched carnelian beads turn up in Mesopotamian contexts, and carnelian from the Indus borderlands was moving north through the Caucasus by the late fourth millennium. Externally the Harappans ran the period's great long-distance trade, the Gulf route that linked the Indus — the Meluhha of the cuneiform records — to Mesopotamia through the intermediary ports of Magan, on the Oman coast, and Dilmun, the entrepot on Bahrain. Two qualifications matter more than the inventory. Few ships sailed the whole way: the traffic moved in stages, and in the Ur III period Mesopotamian boats went only as far as Magan, and later only to Dilmun, even though most of what they collected there had started in India, so Dilmun grew rich precisely because the route was broken in the middle. And the westward flow was not the whole story.

Meluhhan boats put in at Sargon's quay at Akkad, and an Akkadian seal inscription names an interpreter of Meluhha, so there were Harappans in Mesopotamia and enough of them to need translating; but Mesopotamian weights have turned up at Harappa and Mohenjo-daro, and in one Harappan cemetery a young woman was buried wrapped in a reed shroud inside a wooden coffin, a rite unknown in the Indus world and familiar from Ur and Kish. Somebody from Mesopotamia lived, and died, in India.¹³ What survives is a changing asymmetry rather than a clean one. Through the mature phase the initiative ran west: Harappan goods, standards and people are far more visible in Mesopotamia than the reverse, and no Mesopotamian commercial text records trade with Meluhha at all, the references being royal and tributary. Later the balance shifted, once Harappan ports on the Makran coast came within reach of sailors who did not have to cross open-ocean monsoon water, and the carrying passed increasingly to Mesopotamians and Dilmunites. Agency followed the sailing conditions and the position of the ports rather than belonging to anyone, and who carried the goods is in any case a fact about shipping capability and voyage risk rather than about who captured the gains. Two cautions therefore govern the asymmetry. The first is that the gains turn on each side's alternatives, and Mesopotamia had the deeper pockets, so the safer proposition, and the one the rest of this book uses, is that whoever could reach the other party on their own terms was better placed to take the larger share.

The second is about what each side sent. Indus exports were carnelian beads, seals, ivory and stone weights: hard, distinctive, durable objects that survive four thousand years and are identifiable at a glance as Harappan, where Mesopotamia sent back silver, tin, wool, oil and grain, which were consumed, remelted, rotted, or survive without being identifiable as Mesopotamian at all. An exchange of exactly equal value in both directions would leave a lopsided record, tilted the way this one is tilted. That does not dissolve the asymmetry, since the Harappan presence includes people and standards and not only goods, but the record cannot carry the weight the strong version of the claim puts on it.¹⁴

The urban phase ended around 1900 BCE: the great cities were given up, the weights-and-seals apparatus fell out of use, and the population shifted east toward the Ganges. The decline was broadly coincident with the weakening of the monsoon at the 4.2 ka event, and the temptation to read one as the cause of the other should be resisted, starting with the arithmetic. The 4.2 ka

event is around 2200 BCE and the de-urbanisation around 1900, three centuries apart, which is a long time for a cause to take effect. The palaeoclimate record shows no drought of the right timing and depth, and both standard authorities decline the climate explanation anyway, Possehl for an internal ideological failure and Kenoyer for flooding routed through a crisis of legitimacy. Both attribute the drying of the Ghaggar-Hakra to tectonic change rather than monsoon failure, which detaches the most-cited piece of hydrological evidence from the climate story altogether. Appendix E sets the four readings against each other. On all of them, “de-urbanisation” fits better than “collapse”.¹⁵ **Trade profile**

- **Main exports** — carnelian and etched-carnelian beads, ivory, hardwoods (teak), pearls, and lapis lazuli carried on and re-exported from further north. Cotton is deliberately left off this list: the Indus had the world’s oldest cotton tradition, but no cuneiform record places cotton among the Meluhha goods and no textile survives to show it moving west.
- **Main imports** — silver, tin, wool, and oil and grain.
- **Export markets** — Mesopotamia (Sumer and Akkad), reached through the Gulf intermediaries Magan (Oman) and Dilmun (Bahrain).
- **Import sources** — the same Gulf route in reverse, with Dilmun and Magan as the entrepôts through which Mesopotamian silver and the metals of the wider Near East passed back to the Indus.¹⁶

💡 Dilmun — the Gulf entrepot

Dilmun, centred on Bahrain, was the junction of the Gulf trade that linked the Indus world to Mesopotamia, at its height between about 2600 and 1700 BCE. The early third-millennium Dilmun of the cuneiform texts most likely lay in eastern Arabia, around Tarut, while Bahrain's own urban sites — Qal'at al-Bahrain, Saar, Diraz — come in from the middle of the third millennium and Bahrain becomes the centre later still, so the identification shifts across the period. Goods moved up the Gulf in stages: the Indus supplied carnelian, ivory, hardwoods, lapis and etched beads; Magan, on the Oman side, supplied copper; and Dilmun sat in the middle, gathering these and passing them on into Sumer. Its wealth came from controlling a chokepoint rather than from making anything, taking a cut on goods it never produced, which is the bottleneck-and-toll pattern the module returns to again and again: whoever controls the narrow passage between a supplier and a hungry market can levy the difference. Few vessels ran the whole

- ¹⁶**Sources:** Roy (p.20) on the Indus settlements beginning c.3000 BCE, contemporary with Early Dynastic Sumer; Mohan (Loc 531) on the end of the urban era c.1900 BCE; Kingwell-Banham, Petrie and Fuller, "Early agriculture in South Asia", in *The Cambridge World History*, vol. 2 (2015), pp. 261–309, on zebu domestication at Mehrgarh and on the tree-cotton evidence, which rests on Moulherat et al., "First evidence of cotton at Neolithic Mehrgarh", *Journal of Archaeological Science* 29 (2002); on the redating, Mutin et al., "New radiocarbon dates of human tooth enamel reveal a late appearance of farming life in the Indus Valley", *Scientific Reports* 15 (2025), 11345, doi:10.1038/s41598-025-92621-5, whose co-authors include one of the original excavators; Sinopoli, "Ancient South Asian cities in their regions", in *The Cambridge World History*, vol. 3 (2015), pp. 319–42, at pp. 322, 324–25 and 330, for the urban site count, the 50-to-200-plus hectare range, Mohenjo-daro's extent, the intercity distances and the five-hectare norm for non-urban sites. Older syntheses give appreciably smaller figures for the great cities — Chakrabarti (1979), pp. 210–11, via Allchin (1995), p. 202, puts Harappa and Mohenjo-daro at about 85 hectares each — but Sinopoli's are the current figures and are used here. **Note on a date corrected:** Mehrgarh's dating has moved sharply, to an aceramic Neolithic of 5250 to 4650 BCE rather than a start around 7000 or 8000, on tooth enamel rather than the root-contaminated charcoal the old chronology rested on; the cotton grave moves with it. Nothing about the local-domestication argument depends on the older dates, but a great deal of popular writing about Mehrgarh does. **Read more:** Kingwell-Banham, Petrie and Fuller (2015).
- ¹⁶**Sources:** Kenoyer, *Ancient Cities of the Indus Valley Civilization* (1998), on the weight sequence, the brick ratio and the brickmakers' craft tradition. Cited by chapter where the held edition lacks pagination. **Read more:** Kenoyer (1998).
- ¹⁶**Sources:** Possehl, *The Indus Civilization: A Contemporary Perspective* (2002), on Mohenjo-daro's largest buildings, the social pyramid and the Great Bath, and on the rejection of the granaries; Kenoyer (1998) on House AI. **Read more:** Possehl (2002).
- ¹⁶**Sources:** Kenoyer (1998) for rulers and the weight system; Possehl (2002) for the no-state-but-elites position and the reading of seal findspots; Green, "Of Revenue Without Rulers: Public Goods in the Egalitarian Cities of the Indus Civilization", *Frontiers in Political Science* 4 (2022), open access, for collective governance, the Mohenjo-daro seal count after Mackay and the household distribution after Franke-Vogt, and the Harappa gateway weights after Kenoyer and Miller (2007); Kenoyer (1998) on the absence of seals from burials. Green's comparative framework is Blanton and Fargher's, in which societies drawing on internal revenue such as agrarian taxation tend to produce more public goods and wider political participation than those living off external resources. **Read more:** Green (2022), open access.
- ¹⁶**Sources:** Anthony (loc 4821, 4896) on carnelian from western Pakistan moving north via the Maikop conduit c.3700–3400 BCE; Ratnagar, *Encounters: The Westerly Trade of the Harappa Civilization* (1981), on transshipment and the staged character of the route, the dragoman inscription (after Gelb 1968), the Mesopotamian weights, and the Cemetery R 37 burial (after Wheeler 1947); Dalrymple (Loc 1163, 1168) on the goods reaching Ur. Note Edzard's caution, reported by Ratnagar, that not every textual mention of Dilmun or Meluhha need refer to the people of those lands. **Read more:** Ratnagar, *Encounters* (1981).
- ¹⁶**Sources:** Ratnagar, *Encounters* (1981), for the absence of Meluhha from Mesopotamian commercial texts, the tributary character of the Akkadian-period references, and the argument that trade handling passed to Mesopotamians and Dilmunites once the Makran ports were in use; Roy (p.27) on the long-run pattern of ships carrying goods to and from India. **Read more:** Ratnagar, *Encounters* (1981).
- ¹⁶**Sources:** Possehl (2002), for the rejection of environmental causation and the ideological reading; Kenoyer (1998), for flooding mediated by a crisis of legitimacy, and both for the tectonic account of the Sarasvati; Williams (loc 1194, 1197) on reduced river flow c.4500–4200 years ago and Mithen (loc 9959) on c.2000 BCE drought, both representing the climate reading these authorities argue against; Staubwasser and Gupta et al. for the abrupt-forcing reading, Giosan et al. for gradual hydrological change, and Possehl for multi-causal social transformation. **Read more:** Possehl (2002).
- ¹⁶**Sources:** Dalrymple (Loc 1163, 1168) on the Meluhha trade goods and the Gulf route; Roy (p.20, p.27); Mithen (Loc 9058) on the Indus cotton tradition. **Read more:** Dalrymple, *The Golden Road* (2024).



Figure 1.4. An Indus seal from Mohenjo-daro. Standardised seals and cubical weights — recurring from the Indus to Mesopotamia — are how a long-distance economy is read in a world before prices. Photo: Huzi30, Wikimedia Commons, CC BY-SA 4.0.

route, so the middle of the chain was where the goods changed hands and the margin was taken, and that is why the entrepot mattered more than its own production ever warranted.¹⁷ Two things about Dilmun cut against what an entrepot is supposed to look like. The first is that it was not culturally derivative. Harriet Crawford's summary is that Dilmun's material culture owed "remarkably little either to its immediate neighbours, or to the trading partners at either end of the trade routes", and the seals make the point concretely: Dilmun used round stamp seals with its own iconography, Mesopotamia used cylinder seals, the Indus used square stamps. Three partners in one trading system, three different ways of sealing a transaction. The second is stranger and bears on an argument this chapter has been building. Dilmun was in sustained contact with three literate civilisations — Mesopotamia, Susa and the Indus — and ran exactly the kind of complex, repeated, long-distance transactions that writing is supposed to have emerged to keep track of. Yet no Dilmun script has been identified. Crawford finds it hard to believe that such canny operators managed without one, and allows that the need may have been met in other ways. Either way the case is a useful check: if writing came out of the need to track transactions, here is a society with the transactions and, so far as anyone can show, without the writing.¹⁸ **Trade profile**

- **Main exports** — re-exported goods rather than home produce: Indus carnelian, ivory and hardwoods; Magan copper passed upstream into Sumer.
- **Main imports** — silver, wool, oil and grain coming down from Mesopotamia, plus the goods it held in transit.
- **Export markets** — the Mesopotamian cities at the head of the Gulf, above all the Sumerian demand-core.
- **Import sources** — Meluhha (the Indus) and Magan (Oman) below it on the route.¹⁹

The Indus: the agency-holder that ran the Gulf trade

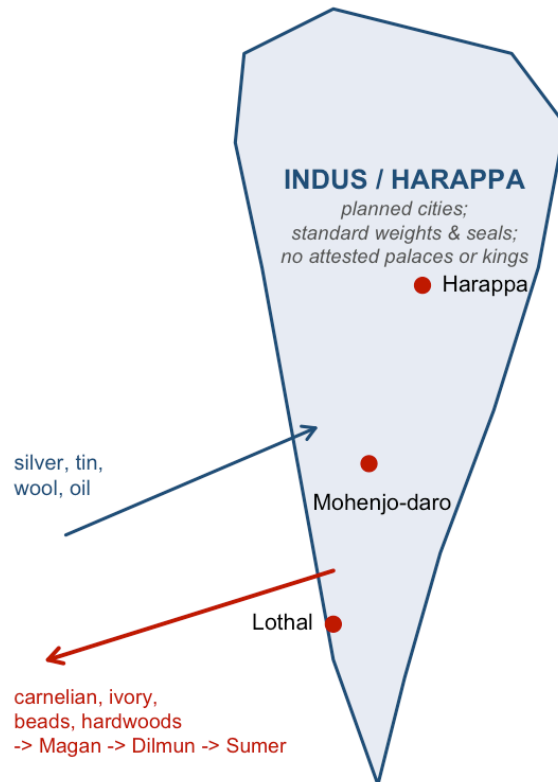


Figure 1.5. The Indus civilisation as the agency-holder: the planned cities of Mohenjo-daro and Harappa on the Indus, running the Gulf trade west to Sumer through Magan and Dilmun, with carnelian and beads moving out and silver and tin coming back (stylised; not to scale). Author's schematic.

💡 Mesopotamia and Uruk — the demand core

Mesopotamia was the period's most commercially central node, and Uruk was its showpiece. The scale of what happened there is easier to grasp as a rate than as a size. Through the proto-urban centuries, settlements in the region seldom passed 20 hectares, and counters and stamp seals were enough to keep track of what they produced. Then, by the end of the Late Uruk

¹⁹ **Sources:** Pearson (Ch.3); Frahm (Page 46); Kriwaczek (Page 49); Ratnagar, *Encounters* (1981), on transshipment and the staged route. **Read more:** Pearson, *The Indian Ocean* (2003).

¹⁹ **Sources:** Crawford, *Dilmun and its Gulf Neighbours* (Cambridge University Press, 1998), on the distinctiveness of Dilmun's material culture, the contrast between round, cylinder and square seals across the three partners, and the absence of an identified Dilmun script despite contact with three literate neighbours; the argument she invokes about writing emerging to track economic transactions is Nissen's, used elsewhere in this chapter for Uruk. **Read more:** Crawford (1998).

¹⁹ **Sources:** Pearson (Ch.3); Frahm (Page 46). **Read more:** Pearson, *The Indian Ocean* (2003).

period, Uruk expanded to at least 250 hectares, some 2.5 square kilometres, holding perhaps 40,000 to 50,000 people, with Nissen putting the city at that size around 3300 BCE. That is more than a twelvefold increase in built area, and the world had not seen anything like it. It rested on an organised hinterland whose size is routinely overstated, since the radii in circulation are limits on early states in general rather than measurements of Uruk, and the tax share usually quoted is an inference from agrarian history at large rather than a figure from any Sumerian record. What can be said is that river transport extended the catchment for bulk goods enormously, which is why the alluvium's cities sat on watercourses. Two cautions. These are reconstructions, and the specialists are more reticent than the synthesis: Emberling opens his own chapter with cities that "may have housed 50,000 people or more" and then declines to say what population *makes* a settlement a city at all.

And the southern alluvium was not the only place cities were forming, since Tell Brak, 750 kilometres north, had reached about 130 hectares before 3500 BCE, so any account making southern irrigation the *cause* of Mesopotamian urbanism has a northern problem.²⁰ Growth on that scale had to be administered, and the administrative technology is what survives. Beveled-rim bowls, mould-made in the millions and mostly holding the same amount, imply a standard unit of measure and, on the usual reading, the issue of rations; the cylinder seal answered the same problem of keeping track of a large and stratified population. Writing came out of that effort, and the earliest tablets did not represent spoken language at all, being in Hans Nissen's phrase barely more than an extension of the techniques of accounting. That writing was invented to do bookkeeping and only later learned to carry literature is one of the more useful things this period has to say about how economies build the tools they need.²¹

The clearest window onto how that economy worked was textile manufacture. Uruk's state workshops are credited with as many as 9,000 women, on the order of a fifth of the whole city, spinning and weaving wool into cloth that the temple and palace then disbursed or traded. That figure needs handling with care, since it comes from Scott, who writes a few paragraphs later that reading such evidence back from the Ur III archives to the Uruk period "is highly questionable", and the archaic Uruk tablets are largely undeciphered administrative accounts that will not support a headcount of that precision. An establishment of that size, fed and housed from central stores and producing on central account, looks like a command economy, in the redistributive pattern Karl Polanyi placed at the heart of the early Near East. Whether that was the whole story is the live debate, because the same society also ran private loans, partnerships and price-sensitive long-distance ventures. Two cautions belong with that. The first is chronological: the archives documenting administered redistribution most fully are not Uruk's but those of the Third Dynasty of Ur, around 2100 BCE, more than a millennium later, and Emberling treats the great collection-and-redistribution centres as an innovation of that empire rather than a feature of the first cities. The second is that the strongest version of the redistributive case has already been walked back, on the evidence of the very archive it was built from. Appendix E has both.²²

Money and credit were already mature, well before any coin was struck. Silver and barley served as units of account and means of payment, so prices existed even though coinage did not. Interest-bearing debt is securely attested from the later third millennium and may reach back to around 3000 BCE, and the rates were later codified under Hammurabi at 20 per cent on silver and 33 1/3 per cent on barley. Debt could spiral into bondage, and rulers periodically wiped the slate: in about 2400 BCE Enmetena of Lagash declared *amargi*, cancelling debts, the first recorded use of the word in a political document. Credit belonged to this economy from the start. What arrived later was minted coinage, and the continuous price series that came with it.²³ What made Mesopotamia the demand pole was its geology. The southern alluvium was rich in silt and grain but had almost no metal, building stone or timber of its own, so the materials of an urban civilisation had to be imported, and even Uruk's monumental architecture

leaned on stone hauled some eighty kilometres. An integrated trading sphere, the “Uruk world system”, already linked the Iranian plateau to the eastern Mediterranean between about 3500 and 3100 BCE, then contracted around 3000 BCE as the long-distance networks thinned and an era of warring city-states began. The luxuries it carried travelled astonishing distances: lapis lazuli from Badakhshan in Afghanistan moved roughly 2,500 kilometres to reach Mesopotamia, and onward to Egypt. The resource-poor core generated the demand that pulled goods across half a continent.

The mechanics of that import trade are best seen in the early second-millennium tin caravans between Assur and the Anatolian colony at Kanesh, where donkey trains of fifty or more animals carried tin and textiles roughly 950 kilometres on a journey of about fifty days, exchanging them for silver. Merchants reckoned gross mark-ups near 100 per cent on the metal and 200 per cent on the cloth — mark-ups rather than profits, since the fifty-day journey, the fodder, the drivers, the tolls, the agent’s commission and something like a hundred days of tied-up capital all came out of them first. To the south-east a separate maritime arm ran down the Gulf, with Sumer drawing copper from Magan and exotic goods from Meluhha through Dilmun, and one Ur merchant, Ea-nasir, known from the Dilmun copper trade and from the most famous complaint tablet in the ancient world.²⁴ **Trade profile**

- **Main exports** — silver and barley (also serving as money and units of account); woollen textiles from the state workshops; grain and other agricultural surplus of the irrigated alluvium.
- **Main imports** — the materials the alluvium lacked: tin and copper for bronze, building stone and timber, and high-value luxuries (lapis lazuli, carnelian, ivory, hardwoods).
- **Export markets** — Anatolia (textiles and silver through Kanesh); the Gulf trade with Magan and the Indus via the Dilmun entrepot; the wider Uruk-system sphere from the Iranian plateau to the eastern Mediterranean.
- **Import sources** — Anatolia and the eastern highlands (metals, silver); Badakhshan (lapis, ~2,500 km away); Magan/Oman (copper) and Meluhha/the Indus (carnelian, ivory, hardwoods) through Dilmun.²⁵

Even the urbanisation rate is a construction rather than an observation. Adams put the region’s ur-

²⁵ **Sources:** Nissen, “Urbanization and the techniques of communication: the Mesopotamian city of Uruk during the fourth millennium BCE”, in *The Cambridge World History*, vol. 3 (2015), pp. 113–30, on the growth from under 20 to at least 250 hectares and the population estimate; Scott, *Against the Grain* (2017), ch. 4, for Uruk’s 250 hectares, the Postgate/Abu Salabikh hinterland calculation and the one-fifth grain levy, and ch. 6 for the forty-eight-kilometre requisition radius and the transport logic. (Cited by chapter: the held text is an EPUB without pagination. An earlier draft gave a “ring of perhaps twenty competing city-states”; that count appears in neither Scott nor Emberling and has been removed.) Emberling, “Mesopotamian cities and urban process, 3500–1600 BCE”, in *The Cambridge World History*, vol. 3 (2015), pp. 253–78, at p. 253 for the 50,000 figure, p. 254 for his refusal to set a population threshold, and p. 263 on markets, walls and temples developing fully only later. Population figures per hectare are themselves a contested method: see Postgate, “How Many Sumerians per Hectare?”, *Cambridge Archaeological Journal* 4 (1994). **Read more:** Nissen (2015).

²⁵ **Sources:** Nissen (2015), pp. 113–30, on the beveled-rim bowls as mass production with a standard capacity, the cylinder seal, and writing as an extension of accounting; Goetzmann (loc 561) and Frahm (Page 34) on the accounting tablets. **Read more:** Nissen (2015).

²⁵ **Sources:** Scott (Page 1040) on the workshops and the Polanyi reading; Emberling, *CWH* vol. 3 (2015), pp. 253–78, at p. 261 on the redistributive model requiring revision, p. 262 on the temple-city model and the Ba’u archive, p. 263 on markets, walls and temples as late developments, and pp. 272, 275 on collection and redistribution as an Ur III innovation. **Read more:** Emberling (2015).

²⁵ **Sources:** Graeber (Loc 8075, 4560); Goetzmann (loc 592, 836, 838); Kriwaczek (Page 101). **Read more:** Graeber, *Debt: The First 5,000 Years* (2011).

²⁵ **Sources:** Scott (Page 1199); Kriwaczek (Page 40, 49, 72, 215, 216); Frahm (Page 46, 47, 48); Goetzmann (loc 940). **Note on a figure withdrawn:** an earlier draft credited Ea-nasir with fifty-one investors in a single venture; that count is not sourced to a page in the held corpus and has been removed. **Read more:** Kriwaczek, *Babylon* (2010).

²⁵ **Sources:** Frahm (Page 46, 47, 48); Kriwaczek (Page 40, 49, 215); Scott (Page 1199). **Read more:** Kriwaczek, *Babylon* (2010).

1. Deep structures: a world without a centre



Figure 1.6. The Great Ziggurat of Ur, built under the Third Dynasty of Ur around 2100 BCE — the temple-and-palace institution that pulled metal and stone across half a continent. Note the date: the administered redistribution best documented in Mesopotamia belongs to this Ur III period, more than a millennium after Uruk, and Geoff Emberling treats it as an Ur III innovation rather than a feature of early cities. Photo: U.S. DoD (Cherie A. Thurlby), public domain.

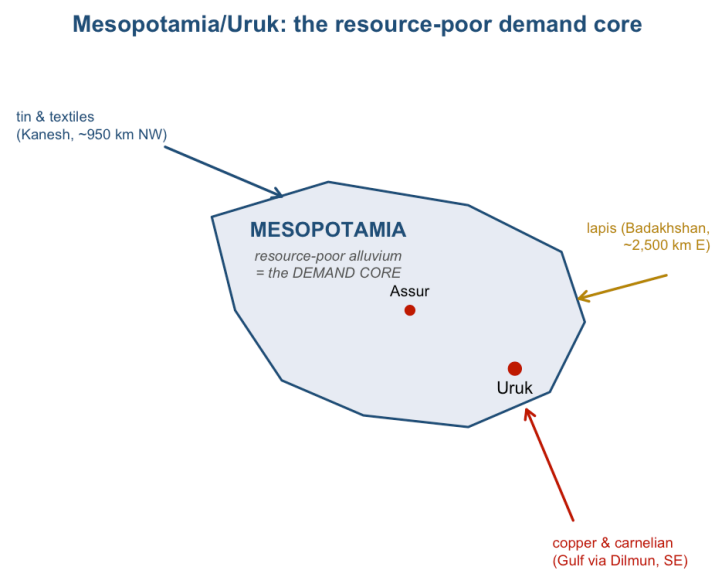


Figure 1.7. Mesopotamia and Uruk as the demand core: a resource-poor alluvium importing tin, copper, lapis and carnelian along the Assur–Kanesh caravan route and down the Gulf through Dilmun, against exports of silver, barley and woollen cloth (stylised; not to scale). Author's schematic.

banised share at as much as 78 per cent across Sumer around 2400 BCE, while his own survey figures for Ur III Umma implied nearer 25; Steinkeller, working from the place-names in the administrative texts rather than from surface survey, put 70 to 80 per cent of Umma's people outside the city. Neither is careless, and the gap between them is mostly a gap between methods, though not wholly, since the two estimates also cover different places and different centuries. The derivation is set out in Appendix D.

💡 Egypt — the Nile apart

Egypt was a riverine core of the first rank, but it stood largely outside the Gulf system that linked the Indus to Mesopotamia. Its wealth came from the Nile: a narrow ribbon of irrigated land whose predictable flood underwrote dense agriculture and a centralised state from unification around 3100 BCE, and through the Old, Middle and New Kingdoms from about 2686, with the two Intermediate Periods as genuine interruptions to that centralisation rather than footnotes to it. Where Mesopotamia depended on uncertain rainfall and rivers, Egypt's annual inundation gave it an agrarian base reliable enough to sustain monumental building and a deep bureaucracy. It traded along its own axes rather than through the Gulf, looking south to Punt for incense, ivory and exotic goods, and east through the Sinai and the Red Sea margins for copper, turquoise and contact with the Levant. Lapis lazuli from Afghanistan reached Egypt by overland relay, a reminder that it was tied into the wider Bronze Age web at the edges, but its trading face pointed down the Nile corridor and across the Sinai land bridge, and the Indus-to-Sumer relay ran without it.

Where Egypt put its surplus is the more interesting question, and the answer is not "into cities". John Baines argues that the values of early Egyptian rulers and elites seem not to have focused on cities at all: power was presented across a landscape rather than concentrated in an urban form, through dispersed mortuary monuments and necropolis sites, movement between them, and a careful interplay of display and concealment. Egyptian cities were constrained by the floodplain and overshadowed by those monuments, and only by the New Kingdom did the society become more city-focused. There is a further reason to keep Egypt in this chapter rather than filing it as a magnificent irrelevance, and it comes from the economics of its record-keeping. Brian Muhs argues that the Egyptian state expanded the use of writing over three millennia for a straightforward fiscal reason, to document and exact taxes more efficiently. But the writing did something its promoters did not intend. Once titles and transfers could be documented and enforced, the cost of making a private transaction fell relative to the cost of going through the institution, and private exchange grew at redistribution's expense; silver, coming into use as a common measure of value, lowered those costs again for high-value dealings, and the state then began taxing transfers in silver, which gave it a direct fiscal interest in documenting and enforcing private property rights still further. That loop runs through the rest of this book. An administrative technology adopted to make extraction work better ended up lowering the transaction costs of the private exchange that competed with extraction, and the state's own revenue interest then pushed it into underwriting the competition.

Set beside Uruk, where writing began as accounting and stayed institutional for a very long time, and beside the Phoenician alphabet later in this chapter, which put writing in a merchant's hands, the point is that the same technology can push an economy in opposite directions depending on who can afford it and what the state wants from it.²⁶ So Egypt is best read as precocious but separate: an early, rich, highly organised state that developed in parallel with the Gulf system rather than as part of it. It mattered enormously in the later eastern-Mediterranean world, and by the fourteenth century BCE its correspondence with Hatti, Babylon and the Aegean shows it deep in the palace-economy network whose collapse around 1177 BCE it barely survived. For the first integration of the Gulf, though, it is a neighbouring core rather than a participant. **Trade profile**

- **Main exports** — Nile agricultural surplus and manufactured goods; gold from the eastern deserts and Nubia.
- **Main imports** — incense and ivory from Punt; copper and turquoise from Sinai; lapis from Afghanistan by overland relay.
- **Export markets** — the Levant and, increasingly, the eastern-Mediterranean palace world.

💡 **Import sources** — Punt to the south, Sinai and the Red Sea to the east, the Levant to the north-east.

The Eurasian steppe was the era's main artery rather than a periphery: an unbroken grass corridor running some 7,000 km from eastern Europe to the Great Wall of China. What turned this ribbon of grass into a power-and-information highway was the horse. A herder on foot could manage about 200 sheep; mounted, the same herder could manage some 500, ranging 50 to 60 kilometres in a working day around them. Read that figure carefully, because it is a herdsman's daily range rather than the pace of a migration. Households moving with their animals covered something closer to 10 to 20 kilometres on a day they moved, and nearer nine or ten averaged across a whole migration, held to the speed of the herd and to the grass it had to stop and eat. What mounted herding bought was reach, and reach is what made the corridor the fastest channel for moving people, animals and news across the continent.²⁷

A cluster of innovations compounded the advantage. The wheel, and then the chariot — the earliest known examples of which come from Sintashta graves in the Urals, before 2000 BCE — spread outward along the grass, reaching the Near East by the nineteenth century BCE and China only around 1200 BCE. Alongside these ran what Andrew Sherratt called a "secondary-products revolution": the use of animals for traction, wool and dairy rather than meat alone. That thesis is contested, since whether it was a single fourth-millennium event or a gradual, patchy adoption is debated, but the underlying shift, animals exploited while alive, was real and it multiplied what a pastoral economy could yield. The corridor was also a vector of migration. The Yamnaya expansion out of the Pontic-Caspian steppe around 3000 BCE has been associated, on present ancient-DNA readings, with a large influx of steppe-derived ancestry into Europe — as much as 90 per cent of the gene pool in Britain at the Bell Beaker transition around 2450 BCE, and about 75 per cent in Corded Ware burials in Germany, rather than a single figure for the continent — a striking figure that should be held cautiously, since the genetics and its demographic interpretation are still moving. The steppe was a channel and an engine at once: it raided the settled cores, traded with them, and at times poured into them.²⁸

Trade profile

- **Main exports** — horses, mounted mobility, and metal-working skill; above all the capacity to move goods, people and force at speed.
- **Main imports** — grain, textiles and finished metalwork drawn from the settled cores.
- **Export markets** — the riverine cores it raided, traded with and migrated into, from Mesopotamia to the forming East Asian core.
- **Import sources** — the same settled cores, supplying the agricultural and manufactured goods the grassland could not produce.

²⁶**Sources:** Baines, "Ancient Egyptian cities: monumentality and performance", in *The Cambridge World History*, vol. 3, on the non-urban focus of early Egyptian elite values; Muhs, *The Ancient Egyptian Economy, 3000-30 BCE* (Cambridge University Press, 2016), whose central argument the second half of this paragraph follows: writing expanded for fiscal reasons, then lowered the transaction costs of private property transfers relative to redistribution, with silver as a unified measure of value lowering them again and giving the state an incentive to document and enforce private titles. Muhs frames the book explicitly in New Institutional Economics terms, and the organisation-versus-market question is Coase's. **Read more:** Muhs (2016).

i How we know

This period leaves no price series at all, and that single absence governs everything the chapter can claim. The test the book uses from Chapter 8 onward — a price gap bounded by the cost of carriage, narrowing as that cost falls — can barely be run here. What the period does leave, in the Assyrian merchant archives, is a handful of exchange ratios at either end of one route, and those turn out to be more useful than they look. Everything else has to be read archaeologically, from the diffusion of goods and shared standards: an Indus seal in a Mesopotamian layer, a set of Harappan weights, a string of etched carnelian beads in a Sumerian grave. Each is a physical trace that two distant economies had touched. None of them says how much, how often, or at what price. And the record is biased the way the introduction warns: organic things rotted in the monsoon tropics and survived in dry Egypt and cool Europe, so what we have systematically under-documents the ocean world this book centres on. *Sources: on reading integration by the diffusion of goods, and the Indus weights, seals and carnelian beads in Mesopotamia, Frahm (Page 46); on the Meluhhan ships and interpreter and the survival bias, Cunliffe (Loc 158). Read more: Cunliffe, By Steppe, Desert, and Ocean (2015).*

1.2. The period on its own terms

Before the cycles, a word about what the base cost. Every economy in this chapter stood on farming, and farming was not obviously a bargain for the people who did it. The skeletal record across the agricultural transition, read at sites worldwide, mostly shows the same direction of travel: shorter stature, more dental caries as fermentable carbohydrates displaced a mixed forager diet, more enamel defects from childhood stress, and heavier loads of infectious disease as people settled at density alongside their animals and their refuse. Sedentism brought contaminated water, vermin, poor sanitation, harvest failure and soil exhaustion, and it narrowed the variety of what was eaten, with social inequality appearing alongside all of it. The finding is not universal, and bioarchaeologists are careful about it, since any one skeletal marker can have several causes and a minority of studies run the other way. But the broad pattern is the physical counterpart of the Malthusian result set out at the end of this chapter: the gains from better technology and better land went into more people rather than into better-fed ones.²⁹

The period ran as a sequence of integrations that rose and then came apart, three times over, rather than as one smooth arc. The three cycles are worth naming at the outset, because the shape returns in every later chapter. An early Uruk sphere reached from the Iranian plateau to the Mediterranean and then contracted. The Gulf trade that the Harappans ran followed it, and ended with their cities. Then the Late Bronze Age system of the eastern Mediterranean, the tightest of the three, broke the hardest. Between the second and the third sits a gap of some three centuries that no maritime cycle fills, and what carried that gap was a technology rather than a trade: the horse, the wheel and the

²⁸ **Sources:** Cunliffe, *By Steppe, Desert, and Ocean* (2015), p. 77, for the 200-to-500 herding gain and the 50-60 km working range, and p. 17 for a rider's daily distance. **Note on a figure clarified:** Cunliffe's sentence makes the herdsman, not the flock, the thing covering 50-60 km; earlier drafts of this chapter read it as a migration rate. Cunliffe gives no source for the number, and it should be treated as an authorial assertion. **Read more:** Cunliffe (2015).

²⁸ **Sources:** on the chariot's spread, Anthony (loc 6171) and Cunliffe (loc 1266); Olalde et al., *Nature* 555 (2018), for the British figure and Haak et al., *Nature* 522 (2015), for Corded Ware Germany; Spinney (Loc 754, 1338); Anthony (loc 5149). An earlier draft gave "up to roughly 90 per cent of Europe's gene pool", which takes the highest regional number and generalises it. **Read more:** Anthony, *The Horse, the Wheel, and Language* (2007).

²⁹ **Sources:** Roberts, "What did agriculture do for us? The bioarchaeology of health and diet", in *The Cambridge World History*, vol. 2 (2015), pp. 93-123, on stature, dental caries, enamel hypoplasia, infectious disease loads and the living conditions of settled communities, and on the studies that do not follow the trend. The econometric counterpart is Ashraf and Galor (2011), discussed at the end of this chapter. **Read more:** Roberts (2015).

1. Deep structures: a world without a centre

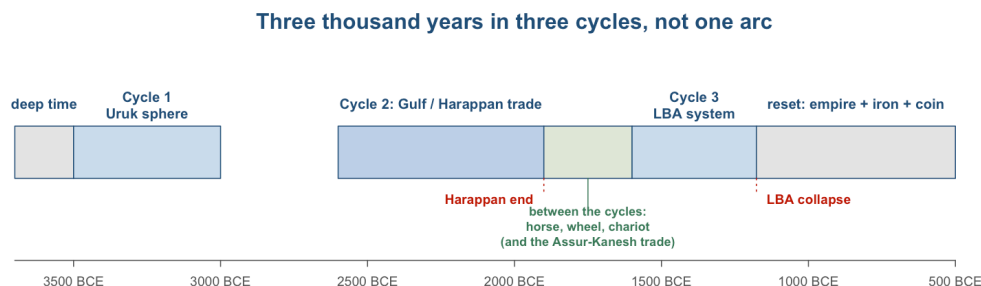


Figure 1.8. Three thousand years in three cycles, not one arc: an early Uruk world system rises and contracts, the Gulf trade the Harappans ran follows and ends with their cities c.1900 BCE, and after a gap that the steppe technologies carry, the Late Bronze Age system rises and collapses c.1177 BCE, before the reset by 500 BCE. Drawn to a linear time axis, so the long interval between cycles two and three is real time rather than spacing. Author's schematic.

chariot, remaking on the steppe what movement cost. The geography underneath all three was set long before any of them, and it left a tension behind. The same monsoon that made predictable round-trip voyaging possible could, when it weakened, help hollow out a civilisation, and both faces appear in the cycles below. The determinism is real and bounded: deep structures shaped what could happen, and people still chose the moves, which is a caution worth holding against the harder geographic readings of Cunliffe and, before him, Diamond.³⁰

Cycle one, the first integration and its contraction, ran from about 3500 to 3000 BCE, and it did not run through the Gulf at all but overland across the northern Near East. From roughly 3500 to 3100 BCE a "Uruk world system" linked the Iranian plateau to the eastern Mediterranean, drawing distant materials toward the Mesopotamian alluvium where the world's first true cities were rising. Uruk itself was already a demand-core of startling size, covering some 250 hectares by around 3300 BCE and holding perhaps forty to fifty thousand people, and such a place pulled goods from far away: lapis lazuli from the mines of Badakhshan in north-eastern Afghanistan travelled roughly 2,500 kilometres to reach Mesopotamia, and the same blue stone reached Egypt at roughly the same period. This was integration in its archaic form, measurable by the diffusion of materials far from their sources rather than by prices. Then it came apart. Around 3000 BCE the Uruk system contracted, the long-distance exchange network thinned, and an era of warring city-states opened in its place. Lapis itself kept coming — the greatest concentration of it outside Afghanistan is the Royal Cemetery of Ur, five centuries after the supposed break — so what changed was the scale and the politics of the traffic rather than its existence. The period's first paired peak-and-fall therefore happened before the Gulf trade that older accounts foreground, which means the integration-then-disintegration rhythm that defines this whole story began one cycle earlier than is often assumed.³¹

Cycle two, the Gulf trade the Harappans led, ran from about 2600 to 1900 BCE, and as the overland system frayed a maritime one took its place. From about 2600 BCE a sea trade linked the Indus cities, Meluhha to the Sumerians, to Mesopotamia through Magan on the Oman peninsula and Dilmun, the entrepot on Bahrain where goods were trans-shipped. The Indus profile above sets out what moved and which way; what matters for the sequence is that agency was not a fixed property of either side. Through the mature phase the initiative ran west, and later, once Harappan ports on the Makran coast could be reached without crossing open-ocean monsoon water, the carrying passed increasingly to Mesopotamians and Dilmunites. The value sat, as it will so often in this book, with

³⁰**Sources:** Cunliffe (Loc 158); Pearson (Ch.1). **Read more:** Cunliffe, *By Steppe, Desert, and Ocean* (2015).

³¹**Sources:** Scott (Page 1199, 805, 1040); Kriwaczek (Page 49, 72); Frahm (Page 34). **Read more:** Scott, *Against the Grain* (2017).

whoever held the chokepoint rather than with producer or consumer: Dilmun took a cut on goods it never made. One qualification on the end of this cycle, because it is usually got wrong. What stopped around 1900 BCE was Indus participation rather than the Gulf system, since Dilmun's commerce with Mesopotamia ran on through the Isin-Larsa and Old Babylonian periods and revived again under the Kassites. The network survived the loss of its eastern terminus by re-sourcing.³²

Cycle two ended when the Harappan cities emptied around 1900 BCE, and the Indus profile above gives that debate in full. Two things belong in the sequence. The 4.2 ka event, invoked as the cause, is around 2200 BCE, three centuries earlier, and the palaeoclimate record does not show the sharp, sustained drought a mono-causal climate story would need. And "collapse" is the wrong word, as the languages show: the cities went, the people did not. Steppe-derived ancestry entered South Asia between about 2000 and 1500 BCE, and the core of the *Rig Veda* is conventionally placed between about 1500 and 1200 BCE. Keep the genetic and the linguistic evidence distinct, because an individual from Rakhigarhi, sequenced in 2019, carried no steppe ancestry at all, so whatever arrived did not arrive early enough to be Harappan. Separately, and from the other direction, a migration out of China after 2000 BCE brought Austroasiatic speech and a new variety of rice in from the east.

Present-day Indian languages still carry that layering — a Dravidian population formed long before, an Indo-Aryan layer arriving after the cities emptied, an Austroasiatic layer from the east — so the movement that the archaeology only gestures at is legible in how the descendants of those people speak. When the material record thins, other records sometimes do not.³³ Between the cycles, the steppe remade what movement cost — but the claim that no trading system occupies the interval will not survive a look at this chapter's own evidence. The Assyrian tin trade between Assur and Kanesh ran from about 1950 to 1750 BCE, squarely inside it, and it is the best-documented commercial network anywhere in these three thousand years: the price ratios, the caravan logistics, the credit and the merchant families all come from there, and the Mari archives and the Old Babylonian Gulf trade sit in the same window. What the interval lacks is a *maritime* cycle of the kind that brackets it, and the Assyrian material is vivid because a merchant quarter's private papers happened to survive, which is a fact about archives rather than about the economy. Read the three cycles, then, as three horizons of surviving evidence that coincide with three real expansions, rather than as an account of when trade did and did not happen.

What the interval does carry, and the bracketing cycles do not, is a change in what movement cost. The steppe profile above sets out the technology; the point to take forward is its pace and its character. The domestic horse lineage that spread across Eurasia expands only from about 2200 BCE, which makes horse and chariot near-contemporaries, and the earliest known chariots, from Sintashta graves east of the Urals before 2000 BCE, reached the Near East within a century or two and China only around 1200. Nine hundred years for one technology to cross a continent is the diffusion speed of this world, and it is the baseline against which every later chapter's should be read. What riding changed was range rather than speed, and the difference is where the usual story goes wrong. Daily rates barely moved: a household travelling with its herds made nine to twenty kilometres on a day it moved, a party on foot eighteen to twenty-five, and a rider travelling light thirty to forty-five. What changed by an order of magnitude was reach: Anthony's Yamnaya herders moved within an annual orbit of under 50 kilometres, while mounted raiding parties struck hundreds of kilometres out and drove livestock home. The economy's orbit stayed small; the reach of force and information did not, and the two never rejoin.

³²**Sources:** Kriwaczek (Page 49); Scott (Page 1551). **Read more:** Ratnagar, *Encounters* (1981).

³³**Sources:** Narasimhan et al., "The formation of human populations in South and Central Asia", *Science* 365 (2019), for the 2000–1500 BCE window on steppe-derived ancestry; Shinde et al., *Cell* 179 (2019), for the Rakhigarhi genome; Mohan (Loc 531, 649, 3292) on the end of the urban phase, the influx, and the Austroasiatic migration and rice after 2000 BCE; Williams (Loc 1194) on reduced river flow. **Note on dates corrected:** an earlier draft placed the Arya at 1700 BCE and the *Rig Veda*'s composition with them, both following Mohan; neither sits inside the consensus range, and this is contested and politically charged ground on which a textbook should be more careful than usual, not less. **Read more:** Narasimhan et al. (2019).

1. Deep structures: a world without a centre

The spectacular courier speeds this period is often credited with — the Persian road, the Roman post, the Mongol relay — belong to later societies that built stations and kept remounts at them.³⁴ Two distinctions inside the word “nomad” matter here, and the second is a warning. Ingold’s is between living off a dead animal and living off a live one: milk, blood and wool are a renewable yield where meat is terminal, and the move to secondary products is what turns a herd from a larder into an income. The second is against reading any of this as environmental destiny. Specialist pastoralism tends to dominate where other farming cannot succeed, but the forest steppes of southern Russia and northern Kazakhstan grow wheat without irrigation today and were held by herders from the Eneolithic to the twentieth century anyway. Farming was possible there, and herders kept the ground for two thousand years regardless. That bears directly on this chapter’s closing debate about how much weight geography should carry. The consequences of horsemanship ran outward from there: bigger herds needed more land, land meant conflict, conflict meant alliances, and alliances were bought with elite goods, so long-distance exchange was partly a by-product of horsemanship. These tools also powered the Indo-European expansions that reshaped the genetic and linguistic map of Eurasia, though the migration-and-linguistics detail is contested terrain and best treated lightly.³⁵

Cycle three, the Late Bronze Age, was already an age of globalisation. By the fourteenth century BCE the eastern Mediterranean and the Near East ran as one working system: Egypt, Hatti, Babylonia, Assyria, Mittani, Mycenaean Greece, Cyprus — the Alashiya of the records, a metal-rich island famous for its copper — and the Syrian port of Ugarit were all in sustained contact with each other, and their post survives. The Amarna letters, some 350 of them from a corpus of 382 tablets, written between about 1365 and 1335 BCE, carry correspondence reaching from Nubia to the Aegean to Iran, and they are written in Akkadian, the diplomatic lingua franca of the day, which almost none of the senders spoke at home. A shared script, a shared protocol and a shared honorific vocabulary, with kings addressing one another as “my brother”, are together the strongest single indicator of system-hood the period offers. Specialisation was visible in what moved: Cypriot copper, Nubian gold, Afghan lapis, Baltic amber in Mycenaean graves.

Hittite iron belongs on a different list, being rare, prestigious, and known largely from a single royal letter about a dagger blade rather than traded as a commodity. The system had rules, too. Its conflicts were wars of readjustment rather than annihilation, and its kings married each other’s daughters, complained in writing about late consignments of gold, and went on trading. What that correspondence describes is trade wearing court dress: royal “gifts” carried explicit reciprocity, stated weights and complaints about quality, and the line between gift-exchange and commerce is one worth refusing to draw too sharply. Cline calls the result a golden age of internationalism, and puts the

³⁴**Sources:** Librado et al., “The origins and spread of domestic horses from the Western Eurasian steppes”, *Nature* 598 (2021), for the DOM2 lineage and its expansion from c.2200 BCE, and Outram et al., *Science* 323 (2009), for Botai horses belonging to the Przewalski rather than the domestic line; Cunliffe (Loc 158, 909, 1266); Anthony (Loc 6171). **Note on a date corrected:** earlier drafts gave domestication at c.4800 BCE, which is Anthony’s 2007 position and has since been substantially revised, by Anthony’s own acknowledgement. **Note on figures corrected:** earlier drafts gave three daily travel rates (ten to thirty kilometres on foot, fifty to sixty driving herds, a hundred to a hundred and fifty riding light) and cited Anthony among the sources. None of the three appears anywhere in his book. The rates given above replace them: households with herds 10-20 km on a moving day and 9-11 km/day of net advance across a migration (Kartaeva, *bilig* 90, 2019, on Kazakh seasonal migration — 800-1,000 km in 90-110 days with some seventy halts; Tawfiq, “Asayer”, *Encyclopaedia Iranica* II/7, giving a Bala Geriva move of 90 km as a ten-day trek; Barth, *Nomads of South Persia*, 1961; and the American cattle drives in McCoy 1874, which run 16-24 on a moving day but 13-14 door-to-door), foot travel 18-25 (Herodotus 5.53 at 27.8 on the Royal Road; Clausewitz’s one-third rest-day discount), and a light rider 30-45 (US *Cavalry Drill Regulations*, 1916, pp. 387-88, at 40 km per marching day). The 100-150 figure is a two-to-four-day forced march or a relay rate, not a sustained one. Denis Sinor, “Horse and Pasture in Inner Asian History”, *Oriens Extremus* 19 (1972), sets out the grazing arithmetic that caps herd movement. Herd-size gain from mounted herding: Cunliffe (2015), p. 77. **Read more:** Librado et al. (2021).

³⁵**Sources:** Outram, “Pastoralism”, in *The Cambridge World History*, vol. 2 (2015), pp. 161-85, for Ingold’s four modes, the primary and secondary products distinction, and the forest-steppe counter-case that warns against simple environmental determinism; Cunliffe (Loc 909, 1266) and Spinney (Loc 754) on the wider consequences. **Read more:** Outram (2015).

globalised economy that defined it at three hundred years and more.³⁶ The physical counterpart of that correspondence is a single hull, and the Introduction has already run the book's four questions across it. The Uluburun ship went down off the Anatolian coast in the late fourteenth century BCE, conventionally around 1300 and on some recent readings nearer 1320, carrying roughly ten tonnes of copper and an estimated tonne of tin — close to the ten-to-one ratio bronze requires, so the cargo was a bronze industry in kit form — along with glass ingots, terebinth resin, ebony, ivory, Baltic amber, Mesopotamian bitumen and Cypriot pottery, the products of at least nine or ten cultures.

Two things about that hull matter more than the inventory. The first is when it sank: the ship is easily filed under the collapse it actually precedes by roughly 120 years, and it dates the system at its height. The second is what kind of voyage it was, and here the chapter should not get ahead of the evidence. The wreck carried gold cups of a type kings sent one another as diplomatic gifts, and a cargo whose components are the very goods the Amarna letters list as royal presents; its 354 oxhide ingots amount to something like 325 talents of copper, against shipments of 100 and 200 talents that those letters record passing between courts. Cemal Pulak, who excavated it, reads the ship as an official dispatch rather than a merchant venture. But it also carried several sets of pan-balance weights on the Syrian shekel of about 9.3 grams, the working tools of perhaps three merchants, and copper ingots of varying weight that had to be weighed at each transaction. His own conclusion is that an official carried the king's goods and did some private trading on the side. The economically interesting point is that administered court-to-court exchange and weighed, priced, transaction-by-transaction commerce were running in the same hold.³⁷

Cycle three ended with the Collapse around 1177 BCE, when the densely interconnected palace economies of the eastern Mediterranean failed together, and failed as a system. Their dependence on long-distance metal shows the exposure plainly: bronze needed tin, tin sat almost nowhere near the cities that used it, so it had to travel, and its role, as Carol Bell put it, was probably not far different from crude oil's today. When the web tore, the cost was systemic. Mainland Greece may have lost forty to sixty per cent of its people, falling from perhaps six hundred thousand to around three hundred and thirty thousand, and Babylonia possibly as much as three-quarters before any recovery; Ugarit, destroyed around 1190 BCE, saw some eight thousand inhabitants flee and was not reoccupied for roughly six hundred and fifty years. The collapse was over-determined — earthquake swarm, drought, the movements conventionally called the Sea Peoples, internal rebellion, the cutting of the trade routes — and that is precisely Cline's argument, that once a system is this tightly coupled no single cause is needed. The Sea Peoples in particular need handling with care: Cline calls them a convenient scapegoat, a construct assembled by nineteenth-century scholarship, and the tide has turned against reading them as the agent of the collapse rather than one of its symptoms. This was integration's first systemic failure, and its lesson, inherited by the whole module, is that a tightly connected system can break in a way a loose one cannot.³⁸

³⁶ **Sources:** Cline (2014), prologue on Alashiya and Cypriot copper, chs. 1-2 on Akkadian as the diplomatic language and on the "Golden Age" of internationalism and globalization, (Loc 2260) on three centuries of globalised economy, (Loc 3810) on the first great international era. **Read more:** Cline, 1177 B.C. (2014).

³⁷ **Sources:** Pulak, "The Uluburun shipwreck: an overview", *International Journal of Nautical Archaeology* 27 (1998), pp. 188, 193, 199, 201-2, 204, 209-10, 214-15, 219-20. The date rests on a tree-ring reading giving a last observable ring at 1305 BCE, with the ship sinking after that but probably not much later; note that the absolute anchoring of the Aegean dendrochronology was explicitly provisional in 1998 and has been revisited since, so the year should be treated as approximate. Cline (2014), ch. 2, on the ship as a microcosm of international contact. **Read more:** Pulak (1998).

³⁸ **Sources:** Cline, 1177 B.C.: *The Year Civilization Collapsed* (2014), ch. 4 for the destruction of Ugarit and the flight of some eight thousand inhabitants, and ch. 5 for the over-determined, systems-collapse reading of the causes. Cited by chapter: the held edition is an EPUB without pagination. **Read more:** Cline (2014).

1.3. After the collapse: the Phoenician centuries, c. 1177–500 BCE

The wreckage took a long time to clear, and what replaced the palaces did not look like them. Iron changed the metals arithmetic, because iron ore is nearly everywhere and the tin bottleneck that had made bronze a strategic problem simply dissolved. But iron did not rebuild the trading world. That was done by a handful of small coastal cities on the Levantine shore — Byblos, Sidon, Arwad and above all Tyre — whose Canaanite inhabitants the Greeks called *Phoinikes*, and by the Cypriots alongside them. Why those two and not others is a question with an answer. The Canaanites of the Levantine coast were not newcomers but the survivors of the Bronze Age cities, carrying on in the same places under a new name. What they were not was imperial. Carol Bell states it directly: Cyprus and Phoenicia “were able to seize the day when crisis struck the region”, being “unencumbered by imperial agendas, and already familiar with operating within a decentralised trading environment”, so that their traders “remained open for business, with their primary objective being to generate sufficient returns to continue to trade”.

Eric Cline goes further and calls both societies anti-fragile, flourishing in the chaos rather than merely surviving it, the Phoenicians taking the sack of Ugarit and the other port cities as their opportunity to win control of the Mediterranean routes and the Cypriots doing the same with iron goods and iron technology. The economies that failed in 1177 were centralised, redistributive and tightly coupled. What replaced them at the centre of Mediterranean exchange was a set of merchant cities with no empire and no great king, which is the chapter’s lesson about coupling, restated from the other side.³⁹ One document shows the change better than any argument about it. Around 1075 BCE an Egyptian temple official named Wenamun was sent from Karnak to Byblos to buy cedar for a ceremonial barge. In the Amarna world of two centuries earlier this would have been a matter between courts. It was not any more. Wenamun was robbed at Dor on the way; the prince of Byblos, Tjekkerbaal, received him with open hostility; and the timber was not cut until payment had been shipped from Egypt and inspected. The payment is itemised, and reads like an invoice: four gold bowls and another gold vessel, five bowls of silver, ten pieces of clothing, five hundred mats of smooth linen, five hundred ox hides, five hundred ropes, twenty sacks of lentils and five baskets of fish.

Only then did the prince detail three hundred men and three hundred oxen to fell the trees, which lay through the winter before being hauled to the shore. An Egyptian priest could no longer requisition Lebanese cedar on the strength of who he served. He had to buy it, in advance, from a man who was free to be rude to him. That is the shift from the gift economy the Uluburun ship was carrying to a price economy, recorded in one bad-tempered voyage.⁴⁰ The most famous thing they made was a colour, and it is perhaps the clearest illustration the ancient world offers of a price set by scarce supply meeting status-driven demand rather than by the cost of the raw material. Purple dye comes from a gland that has to be cut from a living sea snail, with the shell broken to reach it, which is why the archaeological signature of a dye-works is a bed of crushed shells rather than whole ones. The quantities are brutal: experimental archaeology puts about two hundred snails into a square of wool 125 centimetres on a side to reach a deep purple, and at least five thousand into a whole cloak, while older estimates ran to twenty thousand animals for a garment’s trim alone. A batch took four days at minimum, and the work stank badly enough that Strabo, centuries later, thought it made Tyre unpleasant to live in. It would be wrong to conclude that the labour made the value, because if effort made things valuable every laborious process would be lucrative.

³⁹**Sources:** Cline, *After 1177 B.C.: The Survival of Civilizations* (2021), ch. 3, “The Mediterranean Became a Phoenician Lake”, for the continuity of the Canaanite population, the anti-fragile reading of Phoenicia and Cyprus, and the Carol Bell quotation; Pilkington, *The Carthaginian Empire, 550-202 BCE*, on the Levantine cities after the collapse. Cited by chapter: the held edition is an EPUB without pagination. **Read more:** Cline (2021).

⁴⁰**Sources:** Cline, *After 1177 B.C.* (2021), ch. 3, quoting and discussing the Tale of Wenamun, with the itemised payment and the felling of the timber; the voyage is dated to about 1075 BCE. The Tale is a literary text of uncertain historicity, and should be taught as a story that reflects commercial conditions rather than as a transaction record. **Read more:** Cline (2021), ch. 3.



Figure 1.9. *Bolinus brandaris*, the murex sea-snail from which Tyrian purple was won at enormous labour — a processed luxury whose value lay in the work locked into it. Photo: H. Zell, Wikimedia Commons, CC BY-SA 3.0.

What made purple dear was that an inelastic, labour-bound supply met courts that would pay almost anything for a colour only they were permitted to wear. States knew exactly where that left them. In a Hittite letter of the early thirteenth century a vassal complains that a raid has cost him his purple-dyers — three groups of them, one his own, one the king's, one belonging to a god — and the diplomatic effort to recover them is the substance of the letter. Skilled dyers were capital, and were treated as such.⁴¹

From the tenth century these cities founded settlements westward — Cyprus first, then North Africa, Sardinia and Sicily, and by about 800 to 780 BCE, on radiocarbon dates, southern Spain — and among them was *Qart Hadasht*, New City, which is Carthage. The traditional foundation date of 814 BCE is a Greek chronographer's arithmetic rather than an excavated one; the archaeology says late ninth or early eighth century and declines to say more. Carthage was nobody yet: through this chapter's period it was a modest, peripheral, largely mud-brick town, one of the least archaeologically visible of the western foundations, and the empire it later built belongs to the next chapter. What drove them west is the interesting question, and it is genuinely open. The traditional answer is metals. Silver was becoming a proto-currency across the Iron Age Near East, with thirty-four silver hoards recovered from Cisjordan alone, and demand for it was sharpened by three things at once: the collapse of the Bronze Age exchange networks, the Levant's own lack of silver, and the rise of the Neo-Assyrian empire with its appetite for tribute. On this reading, first argued at length by Susan Frankenstein and developed by Maria Eugenia Aubet, the western settlements were commercial factories and the whole diaspora was a response to Assyrian pressure on Tyre, and the mining frontiers do open when the model predicts, in south-western Spain and at Laurion in the Aegean.

The case against is archaeological, and it is not that the metals trade did not happen but that it does not explain the settlements, since the excavated colonies look like farms, agricultural towns from their foundation. Appendix E sets out the argument. This chapter takes the position the evidence supports and no more: the metals demand was real, the settlements were more than trading posts, and the westward silver flow that the tidiest version of the story wants remains an inference rather than a demonstrated fact.⁴² A word about the people already there, because this section has so far

⁴¹**Sources:** Silver, *The Purpled World* (2023), reporting Ruscillo's experimental work on murex yields and the Hittite correspondence on captured dyers. Note that the held copy of Silver lacks book pagination, so these are cited by section; the underlying experimental figures are Ruscillo (2005). Silver also reads the Aegean Bronze Age textile and purple trade as evidence of a far more commercialised world than the orthodox picture allows, even casting the Trojan War as a trade war; that bolder reading is a contested hypothesis rather than settled fact. **Read more:** Silver (2023).

⁴²**Sources:** Pilkington, *The Carthaginian Empire*, on the western foundation sequence, the radiocarbon dating of the Iberian

been told from the ships. The society the Phoenicians met in south-western Iberia is known as Tartessos, and the temptation is to file it as the native partner in a colonial encounter. Sebastián Celestino and Carolina López-Ruiz argue that this gets the thing backwards. Tartessos, on their account, cannot be understood as an impermeable entity separate from the contact: it *is* the cultural horizon that the contact produced, formed from an indigenous substratum meeting several waves of Phoenician settlement, with contributions from elsewhere in the Mediterranean at different stages. There was no fixed Tartessian culture that the Phoenicians then traded with; the two made a third thing together. It is also, they note, a society known almost entirely through other people's writing, having left no historical narrative of its own, which is the same evidential predicament as the Phoenicians themselves and a reason to be careful about who gets described as the active party.⁴³

Two other things travelled with the pots, and the first was a script. Short Phoenician inscriptions, mostly personal names and ownership marks, turn up scratched on ordinary locally made jars in the harbour houses of Castillo de Dona Blanca near Gadir — thirty-eight of them from the eighth and seventh centuries — and by the seventh century the Iberians around Gadir were writing too, on objects of trade. This is writing as a merchant's tool rather than a bureaucracy's, and it is the counter-case to Uruk. Cuneiform was expensive: it took hundreds of signs, a scribal career to learn them, and it grew out of running a large institution's accounts. The Phoenician script was a consonantal alphabet of twenty-two letters, one for each consonantal phoneme in the language, and a trader could carry it in his head. A consonantal alphabet had been adapted to the cuneiform technique at Ugarit in the fourteenth and thirteenth centuries BCE, but the schematised, reduced form that became the Phoenician script came into common use around the twelfth century or a little later, in the specialists' own phrasing with the reopening of trade links after the collapse.

From there it went everywhere: to the Israelites, Moabites, Ammonites, Edomites and Aramaeans, whose national scripts are all essentially its children; westward with the settlements; and into the Aegean, where the Greeks adapted it, adding vowels, with the oldest Greek alphabetic inscriptions dating to about the second quarter of the eighth century BCE. The economic point is the one to carry forward. When the cost of learning to record a transaction falls from a career to an afternoon, record-keeping stops being a function of the state. Uruk's accounting produced a scribal class; Tyre's produced a merchant class that could write. Both are answers to the same problem, and the second is very much cheaper.⁴⁴ The other thing that travelled was competition. Within two centuries the neighbours had learned the writing, the farming and the politics, and Greek settlement in the western Mediterranean came to outnumber Phoenician. Settler counts measure presence rather than commercial advantage, so read that as a sign and not a measurement; but what it points at, advantage decaying by imitation once the technique is cheap to copy, is a pattern this book will meet again in Venice, in Amsterdam and in Lancashire. By 500 BCE the pieces of the next order were in place. Persia had reunited the Near East, iron had loosened the tin bottleneck, the Lydians had struck the

settlements, the status of the 814 BCE tradition, Carthage's archaeological invisibility before about 650 BCE, and at p. 61 for the negative case and for the Huelva sherd counts, sailing distances and agricultural character of the settlements; López-Ruiz and Doak (eds), *The Oxford Handbook of the Phoenician and Punic Mediterranean* (2019), for the Iron Age silver economy and the Cisjordan hoards, for the Frankenstein thesis and Aubet's commercial-diaspora development of it, and for the shift away from exclusively commercial explanations of colonization. **Read more:** López-Ruiz and Doak (2019).

⁴³**Sources:** Celestino Pérez and López-Ruiz, *Tartessos and the Phoenicians in Iberia* (Oxford University Press), for the argument that Tartessos is the product of contact rather than a pre-existing entity encountered by it, and for the absence of self-generated written narratives. **Read more:** Celestino Pérez and López-Ruiz.

⁴⁴**Sources:** López-Ruiz and Doak (eds), *The Oxford Handbook of the Phoenician and Punic Mediterranean* (2019), on the twenty-two-letter consonantal alphabet and its match to Phoenician phonology, the Ugaritic cuneiform antecedent, the schematised script emerging around the twelfth century with the reopening of trade links, its spread through the Levantine national scripts, and the dating of the Greek adaptation by grapheme comparison; Pilkington, *The Carthaginian Empire*, on the Castillo de Dona Blanca inscriptions and indigenous Iberian writing at Gadir. The claim that the script's cheapness had economic consequences is this book's inference from those facts, not a finding the handbook states. **Read more:** López-Ruiz and Doak (2019).

first coins and the Greeks were already minting small change, and Carthage, still a modest city, had founded its own first colony, Ebusus, in 654 BCE. The board was reset.⁴⁵

1.4. Reading the period: the four questions

The three thousand years from about 3500 to 500 BCE can be put through the same four questions that drive the rest of this book, with the warning that the answers here are softer than any later chapter's, because the evidence is thinner and the system looser. They are worth running all the same, because this is where the questions themselves are forged. The first question was about direction, and the answer is that the system did both, repeatedly, with no single arc running through the period. The three cycles narrated above — the Uruk sphere, the Gulf trade, the Late Bronze Age palace economies — each integrated and then came apart, which is the module's rhythm in miniature. The lesson the sequence bequeathed to every later chapter is that integration bred fragility: the routes that carried trade also carried shocks, and a more connected system was a more exposed one. The Collapse of 1177 BCE is the clearest instance in the book, because the palace economies that failed together were the most tightly coupled thing the ancient world had built.⁴⁶

The second question was about channels, and this period's answer is that all three of the module's corridors were born here at once: the maritime Indian Ocean, the desert and oasis route that would later be called the Silk Road, and the great Eurasian steppe. Of the three, the steppe was the era's dynamic engine rather than its periphery, a band of unbroken grass running some seven thousand kilometres from eastern Europe to the Great Wall of China. The ocean, the corridor this book centres on, was still largely waiting, its monsoon geography laid down but its traffic thin. To see how the channels actually worked it helps to follow one good, and the most revealing is tin. Bronze, the period's defining metal, needed it, yet tin lay almost nowhere near the cities that used it, so it had to travel long distances to reach them — and its price marked the route, in barter ratios that the cuneiform records preserve centuries before any coin existed to express them. The volumes were not trivial either: a single cargo such as the one lost off Uluburun around 1300 BCE carried perhaps a tonne of tin in one hull.⁴⁷

! Follow the money

With no coins yet, the period's bullion tracer was the relative price of metals, and tin is the clearest one to track. It had to travel because bronze needed it and it lay far from the cities that consumed it. The gap between its price at Assur, about 6.7 shekels of silver per 100 shekels of tin, and in Anatolia, about 14.3, moved the metal from where it was cheap toward where it was dear, centuries before coinage. The gap is also a measurement: roughly a doubling over 950 kilometres is what carriage cost in this technology, and later chapters watch that wedge narrow. On the Introduction's test — a gap bounded by the cost of carriage — that makes Assur and Kanesh about as integrated as the technology of the day allowed, and the measure of later integration is not that such gaps disappear but that they shrink toward the physical cost of carriage. This is the same "follow the metal" logic the module will use for millennia to locate the centre, except that here it shows up in barter ratios scratched onto clay rather than in the

⁴⁵ **Sources:** Pilkington, *The Carthaginian Empire*, on the relative numbers of Greek and Phoenician settlers by 600 BCE and on Ebusus. **Note on a gap:** the held corpus supports the archaeology of merchant literacy but not the history of the alphabet itself — its letter count, its date, or the mechanism of its transmission to Greek. **Read more:** Pilkington.

⁴⁶ **Sources:** Scott (Page 1199); Kriwaczek (Page 72); Cline (Loc 507). **Read more:** Cline, *1177 B.C.: The Year Civilization Collapsed* (2014).

⁴⁷ **Sources:** Cunliffe (Loc 158, 909); Frahm (Page 47); Anthony (Loc 6829); Cline (Loc 1646). **Read more:** Cunliffe, *By Steppe, Desert, and Ocean* (2015).

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Figure 1.10. The price of distance: 100 shekels of tin cost about 6.7 shekels of silver at Assur but about 14.3 in Anatolia, where it was scarcer. The gap is not arbitrage — it is the cost of carriage, quoted, centuries before any coin existed to express it, and integration is that wedge narrowing (stylised; values from the Old Assyrian records). Source: Frahm.

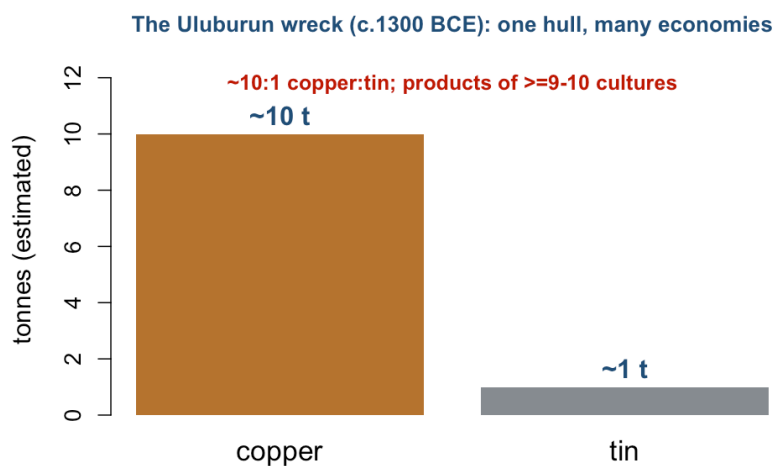


Figure 1.11. A single Bronze Age cargo as a dataset of one: the Uluburun wreck (c.1300 BCE) carried ~10 tonnes of copper and ~1 tonne of tin — the ~10:1 bronze recipe — plus glass, resin, ivory and amber, with products of at least nine or ten cultures. Bulk metal moved by sea long before coins (estimated tonnages). Source: Cline.

output of any mint.

The third question was about modes: what crossed the connection? Two kinds of thing did, and the chapter is clearer if they are kept apart. On one side are real flows, and the production function names them: output itself moves as goods, useful knowledge as technique, people as labour, produced assets as capital. On the other side run monetary and financial flows, things that settle a transaction and claims that carry purchasing power across time. Silver sits on both sides at once, which is the complication this period is best at teaching. The evidence for each is a different instrument. Goods are somebody else's land, arriving in usable form. A cargo of tin is a mineral endowment that Mesopotamia did not have, and that is what makes tin and not grain the story: a small flow of an input nothing can substitute for has more structural consequence than a large flow of something anybody can grow. High-value processed goods travelled too, and mattered out of proportion to their bulk, because a supply that could not easily be expanded met a demand that would pay almost anything — Ugarit paid its yearly tribute to the Hittites not only in gold but in dyed wool and finished garments, cloth being among the most concentrated stores of value a Bronze Age economy could ship, and the Tyrian purple described above is the vivid later case.

There is also one piece of evidence here for the other side of the equation, and the archaeology of trade almost never reaches it: demand. Ancient proteins recovered from dental calculus at Bronze Age Megiddo, around 1650 to 1320 BCE, record what individual people actually put in their mouths — cereals, sesame and dates as expected, and then soybean in one individual, turmeric in one, probable banana in one. At least one person in the second-millennium Levant ate a South Asian spice and a South Asian pulse. What that licenses is narrow. It establishes contact and is silent on commerce; it cannot say how many, how often, or whether a market existed, and every exotic in the study rests on a single person. The study's authors read it less cautiously than this and write of a pattern of established trade, and the gap between what evidence licenses and what is claimed from it, including by the people who produced it, is worth more to an economist than the finding.⁴⁸

Technology moved, and it moved slowly. The chariot took something like eight hundred years to cross a continent, from Sintashta graves east of the Urals before 2000 BCE to China around 1200, and that is the diffusion speed of this world. People moved, and two studies measure it in a way no document can. The Harappan cemeteries are the first. Strontium and lead isotopes in tooth enamel, seventy-seven teeth from fifty-five individuals at Harappa and Farmana, and the design is what makes the result strong: the three molars form at different ages, so one skeleton yields an ordered sequence of childhood residences rather than an average. First molars fall outside the local range; second and third converge on it, at Farmana around three to five years of age. These were first-generation immigrants. The economics is in where they came from. The migrants point north-west to the upper Indus and north-east to the Punjab plains, and the lead in the Farmana burials points at the Khetri copper belt about a hundred and fifty kilometres south, so migrants and metals travelled the same corridors. Resource peripheries exported the material and the people who knew how to get it, which is a relationship this chapter meets again at its far end, in Cornwall.

The sample will not carry a migration rate, and the reason is a selection rule rather than a small sample size: fewer than six hundred Indus skeletons exist from a century of excavation across some 680,000 square kilometres, and formal cemetery burial is rare within that. A cemetery here is a complete sample of a small institution rather than a thin random sample of Indus society.⁴⁹ The second study is in Britain, and it measures a flow of people with no price attached to it at all. Five isotope systems run on 131 pigs from four Late Neolithic henge complexes in southern Britain, around 2800 to 2400 BCE, and the strontium range at Durrington Walls is wider than at any other published British site, wider than a Roman legionary fortress sitting on far more varied geology, while three kilometres away Middle Neolithic animals from West Amesbury Farm show the narrowest range of any British assemblage. So the wide catchment is not a property of the landscape; it is new, and it is institutional. The economics turns on the animal, because pigs drive badly over distance and there is no reason to walk one for hundreds of kilometres rather than carry the meat, unless the animal and the people who raised it are meant to arrive together. The isotopes are therefore measuring the geographic reach of a mobilisation organised without money, writing, prices or a market. Hold it in mind for the fourth question: peripheral in a network is not the same thing as simple.⁵⁰

⁴⁸**Sources:** Scott et al. (2021) on the Megiddo and Tel Erani dental calculus; the identifications rest on two and three peptides respectively, at or barely above the standard authentication threshold; Cline (Loc 2308) on Ugarit's tribute in cloth. **Note:** there is no published rebuttal of these identifications — the caution here is this book's reading of a general methodological standard, not a critique in the literature. **Read more:** Scott et al., "Exotic foods reveal contact between South Asia and the Near East", *PNAS* 118 (2021).

⁴⁹**Sources:** Valentine et al. (2015) on strontium and lead isotopes at Harappa and Farmana. **Caveats the authors state:** the local range is defined from faunal samples whose spread suggests some of the animals were themselves imported, which makes the non-local verdict conservative but is an inference; breastfeeding by immigrant mothers and childhood exposure to imported lead-bearing cosmetics are argued against rather than eliminated; the South Asian ore and slag reference data are in the authors' own word equivocal. **Read more:** Valentine et al., *PLOS ONE* 10 (2015).

⁵⁰**Sources:** Madgwick et al. (2019), the largest fully published multi-isotope faunal dataset in archaeology. **Caveat:** provenance here is by exclusion — the animals are shown not to be local, not traced to farms — and with thin British comparative data for sulfur and oxygen, most of the geographic weight rests on strontium. **Read more:** Madgwick et al., *Science Advances* 5 (2019).

Capital was present and thin. Ships, pack animals, irrigation works, workshops and stores are the capital of this world, and there is nothing in it that could be called a capital market. That point needs care, because finance in this period was sophisticated without being broad, and the two are easy to confuse. Money and credit are not capital; they are what let capital be assembled and what bridges the delay between committing resources and getting them back. That delay is the thing to picture: perhaps a hundred days between committing goods at Assur and seeing any silver return, with the donkeys, fodder, drivers and tolls all paid for in advance. A trade with that shape cannot run on cash in hand. So it ran on credit, and we know it did because the Kanesh archive is a merchant archive: some twelve thousand private letters and account tablets from the trading families themselves rather than from a palace.

For once in this period we are reading the traders, and the claims and obligations in those letters are denominated in silver, across households and across a political frontier, with no coin anywhere in the system. The institutional depth goes back further. Interest-bearing debt is securely attested from the later third millennium and may reach to about 3000 BCE, and rulers periodically wiped the slate: the earliest recorded is Enmetena of Lagash, around 2400 BCE, and the Sumerian word for what he granted was *amargi*, literally a return to the mother, debtors going home. It is the earliest attested word for freedom in any written language, and the first freedom in the written record is freedom from debt.

One last flow belongs here, and it complicates this book's own spine before the spine has started. Before coinage, silver in the Levant circulated as hacksilver, cut and weighed, and because it was not endlessly remelted its lead-isotope signature reads onto the ore field that was worked. Seventy-five items from three dated Phoenician hoards show the source moving: Anatolian silver is prominent in the tenth and ninth centuries, but by the ninth-century hoard at Ein Hofez only two of twenty-nine items are consistent with Anatolian ore and most point to Iberia. The metal is running west to east. The expectation set up by Chapter 5 and Chapter 6 is that silver flows toward Asia. The first traceable flow in the record does the opposite. Direction is a function of where the ore sits and where the demand sits, and both of those move.⁵¹ The fourth question was about Europe, and this is the answer that has changed most. The old line was that Bronze Age Europe sat outside the world system whose cores were in West Asia and the Indian Ocean. That is not right, and the thing that changed it is tin. The obstacle had never been the artefacts but the absence of a reference base: nobody had characterised the ores, so there was nothing to compare an ingot against. A Durham-led programme built one, and the discrimination runs in two steps.

Cornish and Devon cassiterites carry up to 176 parts per million of indium where Iberian and French ores are under ten, and indium survives smelting, which is what makes it readable on an ingot rather than on a rock; that removes Iberia and France. Lead-isotope model age then separates the two survivors, the Erzgebirge coming out at 318 to 323 million years against Cornwall and Devon at 274 to 293, and tin from three radiocarbon-dated Israeli wrecks gives 290. One sentence keeps that honest in a seminar: tin isotopes alone cannot do this job, because the tin provinces overlap too much, and the work is done by indium and by lead-isotope model age. What is settled is the Cornish case for those wrecks. What is not settled is Uluburun, where two laboratories read the same cargo and reach incompatible answers, one programme putting the ingots inside Cornish and Devon ranges and another attributing roughly two-thirds to the Taurus and a third to Central Asia, three thousand kilometres east. Concede the strongest point against the Cornish reading rather than smoothing it:

⁵¹ **Sources:** Eshel et al. (2019) on lead isotopes in hacksilver from Dor, Akko and Ein Hofez; Frahm (Page 47) and Goetzmann (loc 592, 836) on Old Assyrian credit; Graeber (Loc 8075, 4560) on the *amargi* cancellations; Barjamovic, Chaney, Cosar and Hortacsu (2019), *QJE* 134(3), on the Kanesh corpus. **Caveats:** many hacksilver items are mixtures rather than single-source, and copper-alloyed hoards were excluded from the study, which is a selection rule of unknown direction. Graeber's wider debt-before-barter thesis is contested; the specific Mesopotamian facts here are not, and Goetzmann is the safer citation for the finance-history framing. **Read more:** Goetzmann, *Money Changes Everything* (2016).

the Uluburun ingots measure 2 to 7 parts per million of indium against up to 176 in Cornish ore, and indium is the discriminator. The verdict here rests on the Israeli wrecks.⁵²

So the answer is not “peripheral” in the sense students expect. Britain supplied tin, without which there is no bronze, to a network whose demand, capital, weights, writing and organisation all sat in West Asia. That is a resource periphery *inside* the system, which is a different thing from being outside it, and the production model is what makes “periphery” precise rather than dismissive. The tin came from decentralised small alluvial workings, worked part-time alongside farming, handed to coastal trading places and passed down the line. There was no direct contact of any kind between Britain and the Levant: nobody in Cornwall knew where the metal ended up and nobody in Ugarit knew where it started. Britain supplied an input; it did not organise, price or capture the trade in it. Which is this chapter’s sharpest instance of N as power, and the pattern recurs at Potosi, at Banda and at every commodity exporter in the twentieth century: the holder of the scarce resource was not the holder of the rents.

The scale needs care. Perhaps 100 to 200 tonnes of tin a year across all of Europe and the Mediterranean in the mid-second millennium BCE — and say the caveat in the same breath, because the authors call the figure speculative. It is scaled from copper output at three mines on an assumed ten per cent tin-to-copper ratio, then extrapolated across more than a hundred known mines, and a number built by multiplying an uncertain quantity by an assumed coefficient inherits the uncertainty of both. For proportion: Cornwall and Devon produced about 400 tonnes a year in 1200 CE on substantially the same technology with some six thousand mainly part-time workers. A strategic flow, then, and not a large one. None of which makes Europe simple. Varna on the Black Sea, around 4500 BCE, yielded some three thousand gold objects weighing roughly six kilograms, more gold than every other fifth-millennium find in the world put together, and the Durrington pigs above show the same island mobilising animals across most of its length to one place. Organisational sophistication and peripheral position are independent facts, and this period has both. Behind all four questions stood the forces this module tracks: geography came first, with the Himalayan uplift, the monsoon and the great rivers setting the board; climate was the first dynamic driver, its swings bearing on the rise and fall of cores; and technology — the sail married to monsoon knowledge, the horse, the wheel, bronze — turned geographic possibility into movement.⁵³

1.5. Which input was binding?

Every chapter closes its analysis on the second half of the ledger: which of A, K, L and N was the binding constraint? N was binding everywhere, and it bound in two different ways. As a level, there was no tin in Mesopotamia and Uruk hauled its stone eighty kilometres, so a bronze-using city had to be connected to a continent whether it wanted to be or not. As variance, the river under the Indus cities was not short of water so much as unreliable about it. Two kinds of scarcity in one input, and the second is the one most often missed. A moved, and moved slowest, at eight hundred years for the chariot to cross a continent. L was the residual claimant, since a better technique or a windfall of land was captured as more people rather than as higher income per head, which is Malthus written in this book’s notation and why the period’s thriving is counted in cities and populations. K was real but thin, and finance was sophisticated without being broad: credit crossed a political frontier

⁵²**Sources:** Williams et al. (2025) for the Durham ore programme, the indium discrimination and the model ages; Powell et al. (2022) for the Taurus and Musiston attribution of 105 Uluburun ingots; Berger et al. (2023) for the indium measurements that cut against the Cornish reading of that cargo. **Worth knowing:** Berger is a co-author on two of the three, so the lines of evidence are less independent than they look, and lead-isotope provenancing at Uluburun is compromised by lead contamination running from 22 ppm to 11.5 per cent. **Read more:** Williams et al. (2025).

⁵³**Sources:** on scale and the production model, Williams et al. (2025); Anthony (Loc 3941) on Varna; Madgwick et al. (2019) on the Durrington catchment. **Read more:** Anthony, *The Horse, the Wheel, and Language* (2007).

at Assur and Kanesh centuries before anyone struck a coin, and it was relationship-based, running between known households and named agents, with nothing in this world resembling an impersonal capital market. And control of N is where the economic power sat, with Dilmun taking a cut at a chokepoint without producing anything, Cyprus with its copper and Cornwall with its tin. Britain had the tin; West Asia had the trade.

1.6. The verdict: where was the centre?

The verdict for this period is that there was no single centre, and it is delivered with low confidence by design. What the evidence shows is several loosely linked riverine cores rather than one integrated world economy with a weight that can be located. The Near East, and Uruk above all, was the most central node, a large urban demand pole and the supplier of metals into the long-distance trade. The Indus led the Gulf trade through its mature phase, and it survives as a bright spot in a thin record rather than as a documented centre. Which way that thinness cuts matters. Because the humid tropics destroy organic evidence, what survives from the Indus is a lower bound on its activity rather than an estimate of it, which licenses a claim about direction and nothing about magnitude. What it does not license is treating the sparseness of the record as though it were evidence for the argument, a move this chapter made in an earlier draft and should not make. Egypt and the forming East Asian core sat largely apart from the western-Eurasian system that the evidence actually documents.⁵⁴

The centre-of-gravity question that runs through the rest of the book is therefore not yet applicable. This period neither supports nor contradicts the claim that Asia was central, because it predates a connected system in which a centre of gravity is a meaningful idea. The first chapter at which the question becomes answerable is the next one, when Alexander and the Achaemenid inheritance fuse the channels into something like a single trans-Eurasian system. What this chapter contributes is structural rather than evidential: it lays down the monsoon and the channel geography that will later make the Indian Ocean the natural integrator. Two cautions govern how the period's own thriving should be read. The denominator has to be kept honest, since these were overwhelmingly agrarian subsistence economies in which the great majority farmed and never handled a trade good, and the long-distance traffic described here was a thin and lucrative network laid over that mass rather than the bulk of output. And whatever thriving the period shows was measured in numbers rather than income per head: more food meant more people rather than richer people, the Malthusian condition humanity shared with every other species. The switch from measuring an economy by its scale to measuring it by the living standards of the people in it lies far ahead, at the Great Divergence of Chapter 7.⁵⁵

The debate: how much weight should geography carry?

Cunliffe, and Diamond before him, lean hard on geography as the deciding force — the mountains and monsoon and grass corridors setting what was possible. The counter stresses human agency and contingency, the choices that geography permitted but did not dictate. The climate-collapse story is contested in the same way. The 4.2 ka weakening of the monsoon is invoked behind the Harappan decline, and drought behind the Late Bronze Age Collapse, yet the palaeo-record does not cleanly show a sharp, sustained drought at the right moment, so mono-causal climate accounts overreach. The sensible posture is to hold both at once: deep structures shaped the board on which the period was played, but people still chose the moves. Name the positions, weigh them, and leave the question genuinely open.

⁵⁴ Sources: Scott (Page 805); Frahm (Page 46). Read more: Frahm, *Assyria* (2023).

⁵⁵ Sources: Pearson (Ch.1) on the monsoon deep structure and the channels; Scott (Page 631) on the agrarian denominator. Read more: Pearson, *The Indian Ocean* (2003).



Figure 1.12. The Alexander Mosaic (Pompeii) — the turn into the next chapter, where Alexander and the Achaemenid inheritance fuse the three channels into the first trans-Eurasian world system. Public domain, Wikimedia Commons.

Data exhibit — the Uluburun cargo, a dataset of one. With almost no quantitative records, a single well-excavated shipwreck does the work that a price series does in later chapters. The cargo is itemised above, and Figure 1.11 plots the two metals. *What you could do with this:* infer how many separate economies one hull implicates, and ask what the copper-to-tin ratio implies about where the metals were sourced and where the bronze was to be made — reasoning from one rich object to a whole system, with the uncertainty stated.

1.7. Threads forward

What this chapter hands on is a period fully described rather than a verdict: a world of loosely linked riverine cores, three cycles of integration and collapse, and the first evidence that connection carries shocks as well as goods. That last lesson runs from the Late Bronze Age Collapse through every later disintegration the book meets. The hand-off to the next chapter is geographic and institutional at once. By about 500 BCE the Achaemenid empire had reunited the Near East, iron had loosened the tin bottleneck, and coinage had begun to standardise money. Alexander is where it turns, and the three channels born here would fuse into the first genuinely trans-Eurasian world system — Rome, Parthia, the Kushans and Han China strung along the routes — with the monsoon road this chapter laid down as its maritime spine and India at the fulcrum. At that point, for the first time, “where did the weight sit?” becomes a question the evidence can answer.⁵⁶

⁵⁶**Sources:** Scott (Page 631); Cline (Loc 507) on the fragility lesson. **Citation withdrawn:** the Achaemenid reunification, iron and early coinage claim was attributed to Cline, 1177 B.C., but that book does not discuss the Achaemenid reunification and the word “coinage” does not appear in it; the claim needs a source that carries it. Schaps, *The Invention of Coinage and the Monetization of Ancient Greece* (2004), covers the coinage half. **Read more:** Findlay & O’Rourke, *Power and Plenty* (2007).

Classic research: the foundations

The questions this chapter raises — was Bronze Age trade a market or a redistribution; how much weight should geography carry — descend from a bench of historians, archaeologists and anthropologists.

- **Polanyi (ed., 1957)**, *Trade and Market in the Early Empires* — the substantivist thesis that early “trade” was administered and redistributive rather than market exchange. The reference point for this chapter’s central debate; the recent cliometric work below is the market-side rebuttal. Pair them.
- **Algaze (1993; 2008)**, *The Uruk World System and Ancient Mesopotamia at the Dawn of Civilization* — the “Uruk world system” and the early peak-and-contraction that predates the Gulf trade.
- **Adams (1981)**, *Heartland of Cities* — the classic survey-archaeology of Mesopotamian urbanisation and settlement.
- **Sherratt (1981)**, “Plough and Pastoralism: Aspects of the Secondary Products Revolution,” in *Pattern of the Past* — the traction/wool/dairy thesis; teach as contested (discrete event vs gradual adoption).
- **Possehl (2002)**, *The Indus Civilization: A Contemporary Perspective* — the standard synthesis and the “transformation, not collapse” reading of the Harappan decline.
- **Anthony (2007)**, *The Horse, the Wheel, and Language* — horse domestication, the Sintashta chariot, and the Indo-European expansions; the steppe-as-engine story.
- **Cline (2014)**, *1177 B.C.: The Year Civilization Collapsed* — the systemic-fragility reading of the Late Bronze Age Collapse; this chapter’s first “integration creates fragility” case.
- **Graeber (2011)**, *Debt: The First 5,000 Years* — credit, debt and the *amargi* cancellations; the case that finance preceded coinage.

At the research frontier: recent cliometric work

A striking amount of recent economics reaches back into deep prehistory — fitting trade models to cuneiform tablets, and testing the origins of states, farming and the Malthusian regime. Every entry verified.

- **Barjamovic, Chaney, Coşar & Hortaçsu (2019)**, “Trade, Merchants, and the Lost Cities of the Bronze Age,” *Quarterly Journal of Economics* 134(3): 1455–1503 — a structural gravity model fit to ~12,000 Old Assyrian merchant tablets from Kanesh that recovers trade costs and even triangulates the locations of lost cities. The chapter’s proof that Bronze Age long-distance trade was a genuine, estimable market. [DOI 10.1093/qje/qjz009]
- **Mayshar, Moav & Pascali (2022)**, “The Origin of the State: Land Productivity or Appropriability?” *Journal of Political Economy* 130(4): 1091–1144 — hierarchy and taxation arose from the *appropriability* of storable cereals, not from higher farm productivity; the institutional micro-foundation under the riverine cores. [DOI 10.1086/718372]
- **Matranga (2024)**, “The Ant and the Grasshopper: Seasonality and the Invention of Agriculture,” *Quarterly Journal of Economics* 139(3): 1467–1504 — rising climatic seasonality pushed foragers into storage and sedentism, the precondition for farming and surplus; grounds the chapter’s “geography and climate as master force” in recent data. [DOI 10.1093/qje/qjae012]
- **Ashraf & Galor (2011)**, “Dynamics and Stagnation in the Malthusian Epoch,” *American Economic Review* 101(5): 2003–2041 — the canonical empirical statement that, before 1800, technology and land raised population density rather than income per head. The frontier anchor for this chapter’s quantity-vs-quality yardstick, and the thread it hands to Chapter 7. [DOI 10.1257/aer.101.5.2003]

i Research in focus — Barjamovic, Chaney, Coşar & Hortaçsu (2019): a gravity model on cuneiform

- **Aim** — to show that Bronze Age long-distance trade obeyed the same economic regularities as modern trade, and to use them to locate cities now lost.
- **Question** — can a structural gravity model, fit to ancient merchant records, recover trade costs and even the geographic position of unknown trading posts?
- **Data & method** — roughly 12,000 clay tablets from the Old Assyrian merchant colony at Kanesh (early second millennium BCE), recording shipments between named cities; a structural gravity model in which trade falls with distance is fit to the recorded flows, and the locations of lost cities are estimated as the coordinates that best rationalise the trade pattern.
- **Findings** — trade fell with distance just as the modern gravity equation predicts; the model recovers sensible trade costs and places several lost cities close to where independent historical guesses had put them, with a few testable new predictions.
- **Caveats** — the data are one merchant network in one region and period, heavily weighted to tin and textiles; the gravity structure is imposed, not proven, and the city-location estimates are probabilistic.

Sources: Barjamovic, Chaney, Coşar & Hortaçsu, "Trade, Merchants, and the Lost Cities of the Bronze Age," *Quarterly Journal of Economics* 134(3), 2019 [DOI 10.1093/qje/qjz009].

i Research in focus — Ashraf & Galor (2011): the Malthusian regime, and why "success" meant numbers

- **Aim** — to test the central prediction of the Malthusian model for the pre-industrial world: that productivity gains raised population, not living standards.
- **Question** — across the long pre-1800 era, did better technology and land translate into higher income per head, or simply into more people?
- **Data & method** — assembled cross-country data on population density and income for the pre-industrial period and related them to measures of technological and land productivity, testing the Malthusian mechanism.
- **Findings** — places with better technology and more productive land supported denser populations but not higher income per head; prosperity was converted into numbers, exactly as Malthus predicted, until the regime broke around 1800.
- **Caveats** — deep-history income and population figures are reconstructions with wide error bands; the result characterises the regime on average, not every local episode.

Sources: Ashraf & Galor, "Dynamics and Stagnation in the Malthusian Epoch," *American Economic Review* 101(5), 2011 [DOI 10.1257/aer.101.5.2003]. The quantity-vs-quality distinction it formalises is the yardstick paid off at Chapter 7.

Questions for consideration

Essay / exam style — reward the four-questions toolkit and the live debate, not recall.

1. "Before prices, how can we know a 'world economy' existed?" Answer with Bronze Age evidence — the diffusion of goods, shared standards, and the direction of the ships.

1. Deep structures: a world without a centre

2. “Integration creates fragility.” Assess this claim using the Late Bronze Age Collapse and the contraction of the Uruk world system.
3. The Harappans ran the Gulf trade, yet the Indus survives as “a bright spot in a thin record.” Explain how that thinness can *support* rather than undermine the claim that the record underdocuments the East.
4. How much explanatory weight should climate carry in the end of the Harappan urban phase? Distinguish abrupt-forcing, gradual-hydrological, and multi-causal accounts.
5. “Was Bronze Age trade a market, or redistribution?” Use Mesopotamia (Polanyi) and the Kanesh tablet evidence (Barjamovic et al.) to frame both sides.

💡 Cross-cutting questions (collected at the end of the book)

The book closes with a bank of questions spanning several chapters. Those this chapter opens:

- **Quantity vs quality.** This chapter introduces the distinction between aggregate (“more people”) and per-capita (“richer people”) success. Trace it through Chapter 3, Chapter 4 and Chapter 7: when is a big economy legitimately a *rich* one?
- **Measuring integration.** Compare measuring integration by the diffusion of goods (this chapter) with measuring it by price convergence (Chapter 8): what does each method see that the other misses?

Data exercise

```
# The Uluburun wreck as a dataset of one, and population as a prosperity proxy.
# 1. Itemise the Uluburun cargo (~10 t copper as 354 oxhide + 121 bun ingots; ~1 t tin,
#   estimated; ~175 glass ingots; >=149 Canaanite jars holding ~1 t terebinth resin;
#   135 Cypriot vessels; ebony, ivory, Baltic amber, Mesopotamian bitumen; products of
#   at least nine or ten cultures). How many economies does one hull imply?
# 1b. The 135 Cypriot vessels aboard are twice the number known from the whole Aegean for
#   the entire Late Bronze Age (Pulak 1998: 204). If one wreck doubles the record, what
#   does the surviving record measure – trade, or the accidents of preservation?
# 2. What does the ~10:1 copper:tin ratio say about the intended bronze recipe?
# 3. Load a long-run population series (HYDE 3.3 or Our World in Data) for one region.
# 4. Discuss why "more people" did NOT mean "richer people" – the quantity-vs-quality
#   yardstick – and state the uncertainty in deep-history population estimates.
```

Key data

Figure	Value	Source
Uruk population, c.3300 BCE	~40,000–50,000 over ~250 ha; not walled at this date (the wall is Early Dynastic)	Scott (2017); Frahm (2023)
Uruk state textile workshops	up to ~9,000 women (~a fifth of the city)	Scott (2017)
Badakhshan lapis, distance to Mesopotamia	~2,500 km	Kriwaczek (2010)
Price of tin, in silver	~6.7 shekels silver per 100 shekels tin at Assur vs ~14.3 in Anatolia	Frahm (2023)

Figure	Value	Source
Assur–Kanesh caravan	50+ donkeys, ~950 km / ~50 days; ~200% gross mark-up on cloth (not profit — costs come out first)	Frahm (2023)
Chariot diffusion	Sintashta (before 2000 BCE) → China (~1200 BCE)	Anthony (2007)
Interest-bearing debt / Hammurabi rates	by c.3000 BCE; 20% silver, 33 1/3% barley	Graeber (2011)
Uluburun cargo, c.1300 BCE	~10 t copper (354 oxhide + 121 bun ingots), ~1 t tin (est.); products of >=9-10 cultures	Pulak (1998)
LBA collapse, mainland Greece	-40–60% (~600,000 → ~330,000)	Cline, <i>After 1177 B.C.</i> (2021)
Varna gold, c.4500 BCE	~3,000 objects, ~6 kg (> all other 5th-millennium finds)	Anthony (2007)

This is the data-poorest chapter; the transferable skill is reasoning from proxies and single rich objects with the uncertainty stated, not from continuous series.

Further reading

- **Core:** Cunliffe, *By Steppe, Desert, and Ocean* (2015); Pearson, *The Indian Ocean* (2003), ch. 1; Scott, *Against the Grain* (2017).
- **Supplementary:** Anthony, *The Horse, the Wheel, and Language* (2007); Cline, *1177 B.C.* (2014); Possehl, *The Indus Civilization* (2002); Graeber, *Debt* (2011); Frahm, *Assyria* (2023); Kriwaczek, *Babylon* (2010).
- **The debate:** Polanyi (1957) and Barjamovic, Chaney, Coşar & Hortaçsu (2019) on market vs redistribution; on geography and climate as master forces, Mayshar, Moav & Pascali (2022), Matranga (2024) and Ashraf & Galor (2011); on the Harappan decline, the palaeoclimate work of Staubwasser, Giosan and Gupta and colleagues, with Possehl's caution.

2. The first world system: Rome was the customer, Asia the workshop

i Text to follow

This chapter is being revised and is not yet available. The Preface, the Introduction and Chapter 1 are published now; the remaining chapters will follow as each is finished.

Part II: 500 to 1400 CE

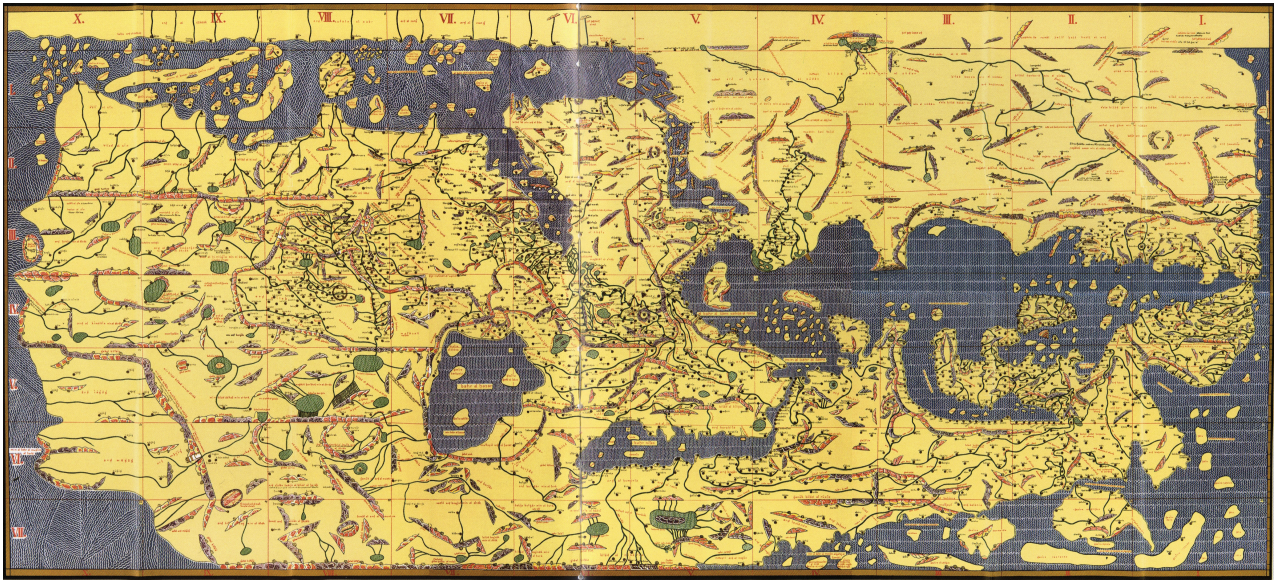


Figure 2.1. Al-Idrisi's *Tabula Rogeriana* (1154), reassembled by Konrad Miller (1928) and drawn south-up — the medieval Indian-Ocean world placed on top, at the height of its reach. Wikimedia Commons, public domain.

3. Maritime apogee: Baghdad and Guangzhou, not Aachen

i Text to follow

This chapter is being revised and is not yet available. The Preface, the Introduction and Chapter 1 are published now; the remaining chapters will follow as each is finished.

4. The high-medieval boom and the Mongol climax

i Text to follow

This chapter is being revised and is not yet available. The Preface, the Introduction and Chapter 1 are published now; the remaining chapters will follow as each is finished.

Part III: 1400 to 1760



Figure 4.1. A silver rupee of the Mughal emperor Akbar (Lahore mint) — the coin into which the world's new silver was minted as it drained, through Europe, toward the Asian demand core that priced it highest. Photo: Drnsreedhar1959, Wikimedia Commons, CC BY-SA 3.0.

5. The world becomes one — and stays Asian

i Text to follow

This chapter is being revised and is not yet available. The Preface, the Introduction and Chapter 1 are published now; the remaining chapters will follow as each is finished.

6. Companies, coercion and the contest of the oceans

i Text to follow

This chapter is being revised and is not yet available. The Preface, the Introduction and Chapter 1 are published now; the remaining chapters will follow as each is finished.

Part IV: 1760 to 1914



Figure 6.1. Indian painted-and-dyed cotton chintz, c.1700–1800 — the textile whose pull on world demand set off the European import-substitution that became the British cotton industry, and with it the Great Reversal. Science and Industry Museum, Wikimedia Commons, CC0.

7. The Great Divergence: the centre moves west

i Text to follow

This chapter is being revised and is not yet available. The Preface, the Introduction and Chapter 1 are published now; the remaining chapters will follow as each is finished.

8. The first global economy: integration on colonial terms

i Text to follow

This chapter is being revised and is not yet available. The Preface, the Introduction and Chapter 1 are published now; the remaining chapters will follow as each is finished.

Part V: 1914 to the present



Figure 8.1. The stacked containers of the *Edith Maersk*, dwarfing the two crew crossing the walkway — the standard box that made late-twentieth-century globalisation cheap and helped carry the workshop back to Asia. Photo: Maersk Line, Wikimedia Commons, CC BY-SA 2.0.

9. Disintegration: war, depression, and the colonial world

i Text to follow

This chapter is being revised and is not yet available. The Preface, the Introduction and Chapter 1 are published now; the remaining chapters will follow as each is finished.

10. The Asian rebalancing: the arc comes home

i Text to follow

This chapter is being revised and is not yet available. The Preface, the Introduction and Chapter 1 are published now; the remaining chapters will follow as each is finished.

Conclusion: How strong is the spine?

i Text to follow

This chapter is being revised and is not yet available. The Preface, the Introduction and Chapter 1 are published now; the remaining chapters will follow as each is finished.

A. Datasets catalogue

i Draft pending

From `teaching/ecu4413_world_econ/module_design/datasets_by_topic.md` (near camera-ready): per-topic datasets with student-ready / illustrative / specialist flags, the four-questions mapping, and what a student could do with each.

B. A guide to the sources and their biases

i Draft pending

The 'data survival and bias' thread, consolidated: record-keeping conventions, climate/ preservation, 'absence is not absence', and 'through whose eyes'. From the module's data-survival note and each chapter's 'How we know'.

C. The centre of gravity, chapter by chapter

! **Preliminary draft — under review.** Published for review; content, figures and citations may still change.

This appendix collects, in one place, the running verdict each chapter entered into evidence: where the centre of economic gravity sat in that period, how confident the reading is on direction and on scale separately, and the evidence that carries it. Read down the table and the spine of the book appears as one movement — Asia central for most of the arc, a European interruption of roughly two centuries, and a partial return. The Conclusion walks these verdicts and weighs how strong the “European dominance as anomaly” reading can be made.

C.1. The table

Each row is the evidence the chapter argued for. “Where the centre sat” summarises the verdict. Direction and scale are graded separately, because they are separate claims: you can be sure which way the weight lay and quite unsure how much of it there was, and saying both is the point of the exercise. “Strongest evidence” names what carries the reading, and how much there is to carry it.

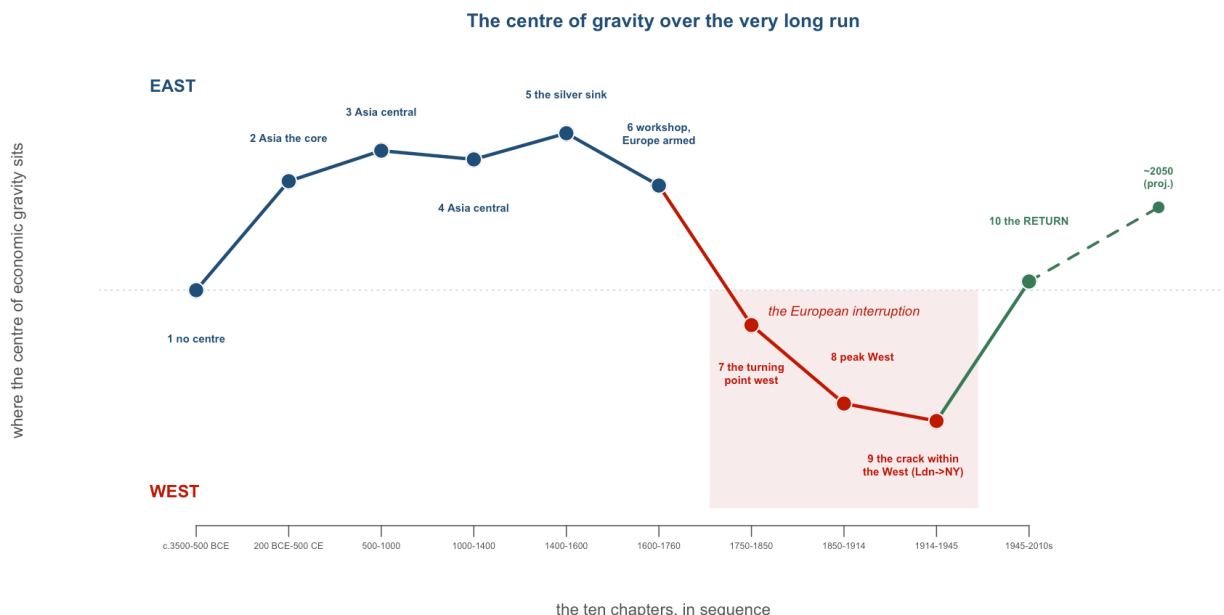


Figure C.1. The centre of economic gravity across all ten chapters. The reading stays in the east for the first six periods, swings west through the great turning point and the colonial centuries (the European interruption), and begins a partial return after 1945. Vertical position is a stylised East–West placement of each period’s centre, not a measured coordinate — only the modern decades rest on a computed series (Quah). Author’s schematic.

The four levels mean this:

- **high** — several instruments, different biases, one answer
- **moderate** — supported, but one instrument carries it
- **low** — a reasonable reading; another would not be unreasonable
- **not applicable** — the question has no meaning here yet

One cell carries a fifth entry. Chapter 7's scale is marked *deferred*, because the series that would settle it is being rebuilt; the reason is set out below.

Table C.1. The centre of gravity, chapter by chapter.

Ch	Period	Where the centre sat	Direction	Scale	Strongest evidence
1	c.3500–500 BCE	No single centre — a multipolar world of riverine cores and steppe corridors, none of them central to the others.	not applicable	not applicable	No one dominant productive core; trade and tribute ran between several states, none central. This is not a low grade with a missing measurement behind it: there was no connected system whose weight could be located.
2	200 BCE–500 CE	Asia the production core of a multipolar relay; Rome the consumer, India the fulcrum, China the maker. Measured by demand rather than production, the peak was Rome.	high	low	Three independent traces pointed east: Pliny's complaint of a drain, the South-Indian coin hoards that made it physical, the Muziris papyrus that priced one cargo of it. The bullion indicated the direction without measuring it, and the magnitudes are contested — Pliny's figures were rhetorical, and the <i>Hermapollon</i> cargo has been reconstructed several ways.
3	500–1000	Asia and the Islamic world central, Europe peripheral; the Abbasid–Tang high tide, with a distinctive northward silver pull to the slave-and-fur periphery.	high	low	Four instruments agreed: city size, where money and manufactures sat, the direction of the bullion, and the direction of the trade in slaves and furs. All four fixed a region and none fixed a quantity. Baghdad at perhaps half a million was a reported estimate, and some accounts ran higher.
4	1000–1400	Asia central and measurably ahead; the Mongol overland climax re-weighted the routes without moving the centre. The first northwest-European seed showed in post-plague wages.	high	low	Song iron output, paper money, the depth of the coin economy and cities approaching a million: four instruments with different biases, one answer. How far ahead is unmeasured — there is no output figure for the period, and the size of the Little Divergence seed was itself disputed.

Ch	Period	Where the centre sat	Direction	Scale	Strongest evidence
5	1400–1600	Asia the silver sink; the oceans joined into one market, and New World silver ran east to buy Asian goods.	moderate	low	The silver arbitrage: metal cheap in the Americas and Europe, dear in China, drawn east to the workshop. One instrument carries the verdict, and the module's own rule is that bullion indicates the centre without defining it. The reading would reach high if Chinese fiscal monetisation and the price differential were argued as separate instruments; the chapter treats them as one story. The headline share — something like half of all New World silver, 1550–1645 — is a broad estimate.
6	1600–1760	Asia still the workshop on the production measure, as late as about 1750; Europe now held the armed and organised levers. On the welfare measure the two ends had already parted by about 1700.	moderate	low	India around a quarter of world manufacturing c.1750, on the Bairoch-type estimates: one much-criticised series, with the continuing eastward silver flow as corroboration rather than as a second measure. The grade attaches to the production measure and not to the period; the welfare ratios of c.1700 returned a different answer.
7	1750–1850	The great turning point: the centre moved west, and growth switched from extensive to intensive. When it crossed is not settled — the chapter's text says 1850, its own shares figure puts the crossing in the middle 1860s.	high	<i>deferred</i>	The Great Divergence in incomes per head. That the move happened is uncontested and the direction grade reflects that. Its date and its placement both rest on the module's centre-of-gravity locator series, which is being rebuilt, so no magnitude and no crossing date are entered here. The row is provisional.
8	1850–1914	Peak Western dominance — resting on the Indian-Ocean periphery it governed.	high	moderate	Europe and its offshoots about 89.8 per cent of world manufactures in 1913 and Western Europe about 60.1 per cent of world exports, against India's 1.4 per cent. The trade shares were counted; the manufacturing shares came from the same Bairoch-type family as Chapter 6, and the drain figure — above 40 per cent of Britain's payments deficits, 1870–1915 — is a reconstruction rather than a measurement.

Ch	Period	Where the centre sat	Direction	Scale	Strongest evidence
9	1914–1945	The order cracked, and the decisive shift ran within the West. Financial leadership passed London → New York; the centre did not move to Asia.	high	moderate	Western Europe's export share fell from 60.1 per cent (1913) to 41.1 per cent (1950) as American creditor power rose, and Asia in aggregate stayed subordinate. The shares were counted; the transfer of financial leadership is read from the record rather than measured, and the interior figures run wide — twelve to eighteen million influenza deaths in India.
10	1945–2010s	The return to Asia in aggregate weight: partial, recent, and inside a US-anchored order. Not a return in income per head.	high	high	Quah's computed centre of gravity drifts from the mid-Atlantic toward India and China; Asia's share of world GDP rose from about 21 to 39 per cent in a single generation; China became the world's largest manufacturer. The one period whose magnitude is computed rather than inferred — and computed on aggregate output, which is heavy in part because the populations are large.

C.2. Reading the table

Four things the table shows, and two of them are gaps.

First, the plateau and the dip. For six chapters the reading stays in the east; for three it swings hard to the west; in the last it begins to return. The European interruption was real and large, but it was an interruption in a longer pattern rather than the pattern itself.

Second, the book knows direction far better than it knows scale. Read the two grades as columns and the difference is plain. Direction is high in seven of the nine periods that return a verdict at all; scale is low in five of them and reaches high only in the final chapter. That is what the surviving evidence allows. Bullion movements, city sizes, the direction of a trade in slaves or cottons, the way a price gradient pulled: each of these tells you which way the weight lay, because each is directional by nature. None of them tells you how much. The pre-modern chapters therefore make a strong claim about where and a weak one about how large, and keeping the two apart is what this table is for.

Confidence does not rise across the book. It opens at not applicable, runs high on direction for three chapters, falls to moderate for the two silver chapters, goes unresolved at the turning point, and returns to high for the last three. What does improve, and only at the very end, is the scale column: Chapter 10 is the single row resting on a computed coordinate rather than an inference.

Third, two moves that were not a simple east–west pendulum. The centre's pull in Chapter 3 ran north, from the Abbasid core to its slave-and-fur periphery, not east; and the decisive shift of Chapter 9 ran within the West, from London to New York, not toward Asia. A centre of gravity can move

inside a region as well as between regions, and the table only reads as a clean arc if those two moves are kept in view.

Fourth, where the table is weak or silent, and why.

- **Chapters 5 and 6 each rest on one instrument.** The silver-sink verdict of Chapter 5 is carried by the bullion flow alone, and the module's own rule is that bullion indicates the centre without defining it. Chapter 6's production verdict is carried by the Bairoch-type manufacturing estimates, and it holds for the production measure only: on welfare ratios the two ends of Eurasia had already parted by about 1700. Both chapters claim high confidence in their own text. On the scale used here, one instrument carrying a reading is moderate, and moderate is what the table records.
- **Chapter 7 is unresolved, and is left so deliberately.** That the centre moved west is not in dispute. When it crossed is. The chapter's text dates the move to 1850, its own shares figure puts the crossing in the middle 1860s, and the locator series that places the marker is being rebuilt. Until that work is finished no date and no magnitude can be entered without guessing, so none is. Read the figure's placement of the turning point as provisional too.
- **The early and late rows are not marks on the same ruler.** Chapters 1 to 6 scored aggregate weight; from Chapter 7 the measure that mattered became income per head. That switch was the Great Divergence itself, and the book announces it rather than hiding it, but it does mean a high grade in Chapter 4 and a high grade in Chapter 10 were awarded on different questions.
- **Chapter 1 is silent, and the silence is the finding.** Its pair of not-applicables is not a gap in the evidence but a statement about the period: there was no connected system whose weight could be located, so the question had no meaning yet. Recording that is more useful than forcing a low grade onto a question nobody could ask.

This is a summary instrument, not a measured series. The vertical placements in the figure are stylised judgments drawn from each chapter's verdict for Chapters 1 to 9, and the placements around the turning point are under revision; only Chapter 10's modern decades rest on a computed centre of gravity (Quah). Treat the table as the book's argument made auditable — every entry traceable to a chapter, and every grade traceable to the instruments behind it — rather than as a quantitative dataset.

D. How the numbers were made

Several of this book's most quotable figures are not measurements. They are constructions: a coefficient applied to an area, a ratio inferred from a handful of texts, a total extrapolated from one surviving document. Where a chapter needs the finding but not the derivation, the derivation lives here, so that a reader who wants to know how confident to be can see the working.

The rule this appendix applies throughout: a number built by multiplying an uncertain quantity by an assumed coefficient inherits the uncertainty of both, and the range it spans is usually wider than the debate about it suggests.

D.1. How urbanisation rates get made, and why two experts differ by a factor of three

The share of a population living in cities is one of the most-used numbers in economic history, and in the deep past it is manufactured rather than counted. Mesopotamia offers an unusually clean demonstration, because two respected reconstructions of the same society disagree by a factor of three and the disagreement is entirely methodological.

Robert McC. Adams, working from archaeological survey, estimated that by about 2400 BCE some 78 per cent of the Sumerian population lived in settlements larger than 40 hectares — an extraordinarily urbanised society by any pre-modern standard. Piotr Steinkeller, working from the place-names recorded in administrative texts, reached the opposite conclusion for Ur III Umma: 70 to 80 per cent of its people lived outside the city, where Adams's survey figures implied about 25 per cent.

Neither is careless. Survey finds sites by walking the ground and reading potsherds, and in an alluvial floodplain small settlements are buried, scoured away or hidden under later silt, so survey systematically under-counts the countryside. Texts record the places an administration had reason to name, which is not the same as the places people lived. Emberling attributes part of the gap to the two methods covering different time spans and part to sites the survey simply missed.

The lesson generalises, and this book will need it repeatedly. An urbanisation rate is a ratio of two constructed quantities, and the method that built each one determines the answer. When Chapter 2 takes up the population of Pataliputra, the same problem appears in a different guise: there the disagreement is not between survey and texts but between two lines drawn on the same site plan.

One further figure from the same chapter is worth holding beside these. Tell Brak, in northern Syria some 750 kilometres from Uruk, had reached about 130 hectares of settled area before 3500 BCE, perhaps 10,000 to 20,000 people, and it was contemporary with the southern cities rather than derived from them. Any account that makes southern irrigation the cause of Mesopotamian urbanism has to explain the north as well.

Sources: Emberling, "Mesopotamian cities and urban process, 3500-1600 BCE", in The Cambridge World History, vol. 3 (2015), pp. 253-78, at pp. 271-72 for the Adams and Steinkeller figures and his account of the discrepancy, p. 275 for Ur at not much more than 50 hectares while the Wool Office employed some 13,000 workers, p. 262 for Tell Brak, and p. 255 for dry farming constraining northern cities to roughly 100 hectares until Neo-Assyrian irrigation.

Chapter 1 keeps the finding — that urbanisation rates are constructions rather than observations, and that two careful scholars can differ by a factor of three — and sends the derivation here.

D.2. How big was Pataliputra? A number and its pedigree

Figures between 40,000 and 400,000 circulate for Mauryan Pataliputra. Following one of them back is a short course in how ancient quantities are made.

The chain begins with Megasthenes, the Seleucid ambassador to Chandragupta, who described the city in his *Indica*. That book is lost. What survives are passages quoted by Strabo and Arrian centuries later, and the version most often cited is McCrindle's translation of 1877. Through them Megasthenes gives dimensions: a parallelogram some eighty stadia long and fifteen broad, girded by a timber wall with 570 towers and sixty-four gates, and a ditch that doubled as the sewer.

Converted, that is roughly nine miles by two. But converted into what *area*? Here the estimates part company, and by more than is usually admitted. Allchin offers two: about 2,200 hectares for the outer rampart circuit, or about 340 hectares for the ground enclosed by the city moat. Erdosy gives a rampart-based range topping out at 1,350 hectares — often quoted alongside the others as if it were a third independent survey, which it is not. And Allchin reports Erdosy again, at 4,500 hectares, for the area Megasthenes's description actually implies.

The last step is a density assumption. The standard coefficient is 120 persons per hectare — a figure Robert McC. Adams derived for Mesopotamia, and which Erdosy then imported into South Asia. Multiply it out and the arithmetic is unforgiving:

Area used	× 120 per hectare	Population
340 ha (inside the moat)		~41,000
1,200–1,350 ha (rampart circuit, Erdosy)		~144,000–162,000
2,200 ha (outer ramparts, Allchin)		~264,000
4,500 ha (the area Megasthenes implies)		~540,000

Nothing separates these answers except a decision about which line on the ground counts as the city. Same lost text, same translation, same coefficient, same modern authorities cited at more than one point on the range — and a spread of about thirteen to one. It is worth being clear what that does to the figures in general circulation: the familiar “about 200,000” needs no unaudited source to explain it, because it falls inside a range the held literature generates on its own. Shonaleeka Kaul, writing the *Cambridge World History* study of the city, takes the moat figure and concludes that “well over 40,000” people lived there, calling even that conservative.

Set against this, what has actually been dug is a thin record. Pataliputra lies beneath modern Patna; dense settlement and a high water table have kept excavation to fragments. The wooden palisade at Bulandibagh — plausibly Megasthenes' wall — has been traced for about 350 feet. There is a burnt pillared hall at Kumrahar whose function is disputed three ways. That is the physical record for the greatest city in India.

None of this means the city was small. Every comparison in the table below survives the uncertainty, because the same conventions were applied to all of them, and on any reading Pataliputra was far larger than any other South Asian city of its day. It means that the number is an artefact of method, and that a book which quotes one figure without its boundary assumption has told you less than it appears to.

City	Enclosed area
Pataliputra	340 ha (moat) to 2,200 ha (ramparts)
Rajagriha, the old Magadhan capital	up to 240 ha
Toshali (Sisupalgarh)	up to 180 ha
Ujjayini	120 ha
Taxila	60 ha
<i>Harappa and Mohenjo-daro, for comparison</i>	<i>c. 85 ha each</i>

The Harappan row needs a warning, and it is the same warning as the rest of this section. That c. 85 hectares is Kaul and Allchin's own comparator, on their own convention, which is what makes the column internally consistent. It is not the figure Chapter 1 uses: there Mohenjo-daro is above 200 hectares, in the same class as Uruk rather than a third of it. The two numbers measure different things — an occupied mound against a wider site extent — so the cross-period comparison inherits a boundary problem at *both* ends, and cannot carry a precise multiple.

Sources: Kaul, "Regional study: Pataliputra", in The Cambridge World History, vol. 4 (2015), pp. 514–36; Allchin, The Archaeology of Early Historic South Asia (1995), pp. 109, 202, and ch. 3 for the comparative areas; Adams, Heartland of Cities (1981), for the density coefficient.

Chapter 2 keeps the finding — that the answer moves by a factor of about six and a half on a single decision about which line on the ground counts as the city, and that the hierarchy is secure where the number is not — and sends the derivation here.

E. The disputes in full

Where a chapter stages a live scholarly disagreement, it keeps the finding and sends the argument here. Nothing in this appendix is settled; that is the point of it. A reader who wants only the state of play can stay in the chapters, and a reader who wants to know why the state of play is unsettled will find the reasons below.

The appendix is organised by chapter, so that each chapter's live disagreements sit together and the apparatus grows with the book rather than being retro-fitted to it.

F. Chapter 1 — Deep structures

F.1. Indus governance, three ways

The three positions run as a spectrum rather than a quarrel. Jonathan Mark Kenoyer argues for rulers of a kind, reading the weight system as established by those rulers, plausibly for taxation — but he writes of the “so-called priest-king” himself and argues for several competing elite classes rather than a single sovereign. Gregory Possehl denies a state in the archaic sense, with no king and no royal apparatus, but affirms elites, ranks and a social pyramid, with authority perhaps sitting in councils. Adam Green goes further than either, arguing for collective governance in which many intersecting institutions rather than any ruling class coordinated labour and produced the public goods: the water infrastructure, the large non-residential buildings, the planning itself.

The evidence Green leans on hardest is the distribution of the seals. More than a thousand were recovered from the excavated areas of Mohenjo-daro alone, and on Franke-Vogt’s analysis the great majority came from ordinary households rather than large public structures. If seals recorded claims on resources, the right to make such claims was spread very widely.

That inference is where the dispute actually sits. The step from where seals were found to how the society was governed is an inference rather than a finding, and Possehl reads the same distribution as a concentration in particular quarters rather than a random spread. Weights may have been used in taxation, and at Harappa some were found by a neighbourhood gateway, though that has only been reported preliminarily and does not hold across sites. What was taxed, and by whom, is unknown.

What the three agree on is enough to carry the economic argument without settling the dispute, and that is why Chapter 1 does not try to settle it.

F.2. Why the Indus cities emptied

Four readings are in play, and the chapter takes none of them as settled.

Abrupt climate forcing. Staubwasser and colleagues, and Gupta and colleagues, treat the 4.2 ka event — a weakening of the monsoon around 2200 BCE — as the trigger. The difficulty is arithmetic: the de-urbanisation is dated around 1900 BCE, three centuries later, and a mono-causal climate account has to explain that lag away.

Gradual hydrological change. Giosan and colleagues emphasise rivers shifting and drying over generations rather than a single sharp drought. This fits the palaeoclimate record better, which shows no drought of the right timing and depth to topple a civilisation.

Internal, ideological failure. Gregory Possehl rejects environmental causation outright and reaches for a failure of whatever held the civilisation together. His is an ideological explanation, not a meteorological one.

Flooding routed through politics. Jonathan Mark Kenoyer keeps the water in the story but makes the causal link political: the rivers moved, the disruption was real, and what broke was the legitimacy of the people who had claimed to manage such things.

Two observations cut across all four. Both standard authorities attribute the drying of the Sarasvati or Ghaggar-Hakra system to **tectonic change rather than monsoon failure**, which detaches the most-cited piece of hydrological evidence from the climate story altogether. And on every reading, “de-urbanisation” fits the evidence better than “collapse”: the cities were given up, the population shifted east toward the Ganges, and the people did not disappear.

1

F.3. Redistribution versus market at Uruk

Polanyi placed a redistributive pattern at the heart of the early Near East, and Uruk’s state textile workshops — credited with as many as 9,000 women — are the exhibit usually offered for it. Three things complicate that.

The chronology. The archives documenting administered redistribution most fully are Ur III, around 2100 BCE, more than a millennium after Uruk. Emberling treats the great collection-and-redistribution centres as an innovation of that empire rather than a feature of the first cities — and Scott, the source of the 9,000 figure, writes in the same passage that reading such evidence back to the Uruk period “is highly questionable”.

The strongest version has been walked back. The “temple-city” model, built on the archive of the temple of Ba’u at Lagash, held that temples owned essentially everything. Continued study of that same archive showed it had dramatically overstated how much land one temple held, and the Ba’u estate turns out to have been supervised by the queens of Lagash rather than by a priesthood standing apart from the palace.

Emberling is more deflationary still: markets, walls and temples cannot be used to explain why cities arose, because none of them developed fully until later.

What survives into the chapter is the position it takes: redistribution and market dealing coexisted, and the balance between them is contested rather than settled. The Uluburun cargo, a thousand years later, shows both running in the same hold.

F.4. The Phoenician metals question

The commercial thesis, argued at length by Susan Frankenstein and developed by Maria Eugenia Aubet, holds that the western settlements were commercial factories and that the whole diaspora was a response to Assyrian pressure on Tyre. The mining frontiers do open when the model predicts, in south-western Spain and at Laurion in the Aegean.

The case against is archaeological, and it is not that the metals trade did not happen but that it does not explain the settlements. The excavated colonies look like farms. They were agricultural towns from their foundation — wine and salted fish at Huelva by the eighth century, Gadir buying farmland across the bay around 800 BCE — and at ninth- and eighth-century Huelva, of some eight thousand potsherds, 99.9 per cent are local or Phoenician and fewer than fifty are Greek. Nathan Pilkington puts the negative sharply: the implication that Iberian metals were destined for Near Eastern markets cannot, he argues, be corroborated from the archaeological record. The Phoenicians

¹ **Sources:** Possehl, *The Indus Civilization* (2002), for the rejection of environmental causation and the ideological reading; Kenoyer, *Ancient Cities of the Indus Valley Civilization* (1998), for flooding mediated by a crisis of legitimacy, and both for the tectonic account of the Sarasvati; Staubwasser et al. and Gupta et al. for abrupt forcing; Giosan et al. for gradual hydrological change.

never permanently settled the Atlantic tin coasts and abandoned the stations they did place there, and a direct run from Tyre to Gadir was some four and a half thousand kilometres and a month's sailing, which allows one voyage a season.

Where the field has moved is easy to overstate in either direction. The older scholarship explained the colonies *exclusively* by trade in metals. The newer work, with specialised settlements, survey, archaeobotany and archaeometallurgy, has opened the explanation to territorial, productive and social motives alongside the commercial one. That is a broadening, not a refutation.

What would settle it: provenance work tying the silver in Levantine hoards to Iberian ores. This book has not found that evidence either way.

2

²**Sources:** on the Indus, Kenoyer, *Ancient Cities of the Indus Valley Civilization* (1998); Possehl, *The Indus Civilization* (2002); Green, "Killing the priest-king" (2021) and his 2022 paper on collective action; Franke-Vogt on the seal distribution. On the Phoenician question, Lopez-Ruiz and Doak (eds), *The Oxford Handbook of the Phoenician and Punic Mediterranean* (2019), and Pilkington, *The Carthaginian Empire*, p. 61.

References

